

SPECIFIC CONTRACT No 2022 P5 02

based on framework Contract No HADEA/2022/OP/0012

Studies and Meetings on Health Emergency Preparedness and Response for HERA

Scientific background document

AI supported supply and demand monitoring

Written by: Alessia Cottone, Chiara Mauro, Irene Mumeni Urbani, Marina Rodés Sánchez and Francisco Lupiáñez-Villanueva

Predi[t *By

intellera Part of Accenture Group

Disclaimer

"This report was produced under the EU4Health Programme under a service contract with the European Health and Digital Executive Agency (HaDEA) acting under the mandate from the European Commission. The information and views set out in this report are those of the author(s) and do not necessarily reflect the official opinion of the European Commission/Executive Agency. The European Commission/Executive Agency do not guarantee the accuracy of the data included in this report. Neither the European Commission/Executive Agency nor any person acting on the European Commission's/Executive Agency's behalf may be held responsible for the use which may be made of the information contained therein."

«Les informations et points de vue exposés dans le présent (ou la présente) rapport n'engagent que leur auteur (ou leurs auteurs) et ne peuvent être assimilés à une position officielle de la Commission/Agence Exécutive. La Commission/Agence Exécutive ne garantissent pas l'exactitude des données figurant dans le présent (ou la présente) rapport. Ni la Commission/Agence Exécutive ni aucune personne agissant au nom de la Commission/Agence Exécutive n'est responsable de l'usage qui pourrait être fait des informations contenues dans le présent texte»

Manuscript completed in May 2025

First edition

LEGAL NOTICE

This document has been prepared for the European Commission however it reflects the views only of the authors, and the European Commission is not liable for any consequence stemming from the reuse of this publication. More information on the European Union is available on the Internet (http://www.europa.eu).

PDF ISBN 978-92-95239-54-8 doi: 10.2925/7812706 HW-01-25-029-EN-N

Luxembourg: Publications Office of the European Union, 2025

© European Union, 2025



The reuse policy of European Commission documents is implemented by the Commission Decision 2011/833/EU of 12 December 2011 on the reuse of Commission documents (OJ L 330, 14.12.2011, p. 39). Except otherwise noted, the reuse of this document is authorised under a Creative Commons Attribution 4.0 International (CC-BY 4.0) licence (https://creativecommons.org/licenses/by/4.0/). This means that reuse is allowed provided appropriate credit is given and any changes are indicated.

For any use or reproduction of elements that are not owned by the European Union, permission may need to be sought directly from the respective right holders.

Executive Summary

The **Health Emergency Preparedness and Response Authority**'s (HERA) mission is to improve the European Union's (EU) ability to prevent, detect, and rapidly respond to health emergencies, ensuring **availability and accessibility** to key **medical countermeasures** (MCM) (HERA, 2023). An essential part of this mission is to gather intelligence on cutting-edge solutions that can boost the efficiency and resilience of **MCM supply chains**.

Over the years, the **complexity** of the medical supply chains has sharply increased, exposing the need to adopt **innovative solutions** to achieve a high level of visibility and improve the capacity to **track and manage** supply chains. Access to quality data is also critical to accurately **anticipate demand** fluctuations and inform evidence-based supply planning or risk mitigation interventions.

Recently, the COVID-19 pandemic further contributed to increasing the focus on **MCM supply chain resilience** and the awareness of the potential impact that **supply disruptions** may have on patients and healthcare services. This led to the use of advanced statistical methods and Big Data to improve supply performance, as well as the implementation of strategies to promote domestic supply chain resilience such as onshoring and strategic stockpiling. Although these mechanisms have been effective in fostering supply chain management, they rely on the **availability of comprehensive information and historical data** to make accurate predictions and planning. **Artificial Intelligence (AI) and simulation methods (SMs)** offer promising solutions to these obstacles, providing quick responses for very complex problems in situations characterised by a high degree of uncertainty.

This scientific background document explores the **potential of AI and SMs in improving demand forecasting and supply chain monitoring of MCM**. The study maps existing innovative methodologies and solutions that enhance demand forecasting and supply chain monitoring and planning, improving MCM supply chain resilience and ultimately strengthening health emergency preparedness and response.

An extensive literature search was conducted, using the academic database PubMed combined with additional desk research of relevant documents developed by experts, competent authorities, or international organisations. Additionally, key experts from academia, governments and industry have been interviewed to collect further insights on the topic.

The study is organised into three chapters as follows.

The *Introduction (Chapter 1)* provides more detailed information on the study's objectives and structure, as well as an overview of the current state of the art. Specifically, this study has **three main objectives**:

1. Analyse the **AI-driven methodologies for demand forecasting**, including epidemics predictions or estimates of products´ needs, to understand:
   - what **methodologies** are available and could be effectively implemented in the EU
   - what **data** characteristics are relevant to implementing AI-driven forecasting
2. Analyse the **AI-driven methodologies for supply chain monitoring** (e.g., supply chain mapping, monitoring and management, identification of actual/potential bottlenecks) to understand what **methodologies** are needed and could be effectively implemented in the EU MCM sector.
3. Analyse the **simulation methods** that can be used by HERA to support decision-making on MCM supply chains while tackling vulnerabilities, to understand:
   - what **methodologies** could be effectively implemented in the EU MCM sector
   - the most appropriate **context** in which they should be used

Chapter 2 includes the main *Results* of the study, which are organised into four subchapters:

The first subchapter, 2.1, **AI-driven methodologies for demand forecasting**, presents the main steps that should be followed to implement a predictive AI model to anticipate future demand of MCM, which include i) Definition of the forecasting´s scope and data availability ii) Selection of the most appropriate methodology and iii) Data collection and processing. This field is rapidly evolving, and several different methodologies have been explored by researchers. Evidence shows that there is no unique correct approach to MCM demand forecasting since different methodologies may be selected based on specific forecasting goals, data availability (quantity quality and dimensionality) and implementation setting (including level of expertise and resources). The machine learning methodologies more frequently applied to demand forecasting are deep learning networks, such as Long short-term memory (LSTM) or Gated Recurrent Unit Networks (GRU), and ensemble models, such as random forests or boosting algorithms.

The second subchapter, 2.2 **AI-driven methodologies for supply chain monitoring**, presents an overview of the main methodologies employed throughout the entire supply chain to continuously monitor and track inventory levels, lead times, supplier performance metrics (e.g., quality), transportation data (e.g., shipping routes), demand forecast, accuracy rates as well as external factors that may impact the flow of goods. Evidence shows that AI methodologies must be employed in coordination with other digital technologies such as IoT, blockchain and Big Data.

The third subchapter 2.3 **Simulation methodologies for supply chain related decision-making** illustrates how simulation can complement AI and Big Data by allowing to test different scenarios and relative outcomes, allowing to determine causality and answering the “what if” question. In a healthcare supply chain context, Discrete Event Simulations (DES) have been found to be better suited for operational problems such as hospital´s resource allocations, while System Dynamic Simulations (SD) are believed to be more appropriate for the study of systemic and dynamic issues, such as to model the spread of diseases and evaluate public health policies. Hybrid simulation methods (such as system modelling and RealOpt) which bring together elements of the different methodologies, have been already applied in health emergencies with promising results.

Finally, the fourth subchapter highlights the major 2.4 **Limitations and challenges to implementation** of the discussed methods. Indeed, although the literature reviewed and the consultations with key experts highlighted the potential benefits of integrating AI-tools and SMs into existing demand forecasting and supply chain monitoring systems, various barriers were identified in the areas of data visibility, integrity and quality; privacy, security and anonymisation; human and financial resources; interoperability and applicability and cultural and organisational resistance.

These main findings of the study are further discussed in the *Conclusion (Chapter 3),* which also includes key recommendations for policymakers to successfully implement the methodologies and increase the control of MCM supply chains. These include the following:

- Enhance **supply chain visibility** by **mapping of the end-to-end supply chain**. Data visibility and quality can be aided by the use of 4.0 technology enablers such as Internet of Things and AI.
- Enhance **collaboration** and promote **capacity building**, for instance by creating spaces for information sharing and tackling industry concerns surrounding privacy and security.
- Boost **investment in AI research and development**, providing stakeholders with the right knowledge and infrastructure to develop and apply the relevant technologies.

Overall, this report finds that the adoption of AI-based tools and simulation methodologies can enhance EU´s crisis preparedness and response capabilities by supporting effective demand forecasting and supply chain monitoring of MCM.

By addressing the current barriers that hinder the application of AI in MCM supply chain, the EU can harness the potential of AI and simulation methods to ensure a resilient MCM supply chain and ultimately ensure that every European citizen receives the healthcare they need.

Contents

Executive Summary **1**

1. Introduction **1**

1.1 Study scope *1*

1.2 Study objectives *1*

1.3 Structure of the study *2*

1.4 Overview of the state of the art *2*

1.5 Overview The role of AI and simulation methods in demand forecasting and supply monitoring *6*

2. Results **10**

2.1 AI-driven methodologies for demand forecasting *10*

2.2 AI-driven methodologies for supply chain monitoring *23*

2.3 Simulation methods for supply chain decision-making *30*

2.4 Limitations and challenges to implementation *38*

3. Discussion and Conclusion **42**

Bibliography **47**

Annex 1: Methodology **56**

Scientific literature review *56*

Consultation activities *57*

List of Figures

Figure 1 Link between monitoring population health and MCM demand and supply chain management 4
Figure 2. IoT, blockchain and AI methodologies in a medical product supply chain.....24
Figure 3. PRISMA flow diagram summarising the records of the literature review 57

List of Tables

Table 1. Definition of the AI techniques considered in this study 8
Table 2. Summary of methodologies used for demand forecasting in the relevant scientific literature 14
Table 3. Summary of the relevant AI-driven demand forecasting methodologies from literature review findings 19
Table 4. Overview of simulation methodologies 31
Table 5. Overview of simulation methodologies in different healthcare supply chain areas...34
Table 6. DES simulation applications with other methodologies.....35
Table 7. Consultation activities. 58

List of Boxes

Box 1. Overview of the benefits for the main stakeholders involved in the MCM supply chain6
Box 2. Working definition of “AI system¨ 7
Box 3. Working definition of “Simulation method” 9
Box 4. Considerations from the experts: AI-driven methodologies for demand forecasting ...22
Box 5. Considerations from the experts: AI-driven methodologies for supply chain monitoring 29
Box 6. Considerations from the experts: simulation models for supply chain 37
Box 7. Considerations from the experts: limitations and challenges to implementation.....41
Box 8. Focus on practical applications of AI-supported methodologies and simulation methods for supply chain monitoring and demand forecasting in real-world scenarios.....43
Box 9. Research questions and query strings.....56

Acronyms and abbreviations

AI	Artificial Intelligence
ARIMA	Autoregressive integrated moving average
ANN	Artificial Neural Networks
API	Active Pharmaceutical Ingredients
ATHINA	Advanced Technology for Health Intelligence and action IT system
ABS	Agent based simulation
ANFIS	Adaptive neuro-fuzzy inference system
ANN	Artificial Neural Networks
DES	Discrete Event Simulation
DIN	Drug Identification Number
EU	European Union
FL	Federated Learning
GRU	Gated Recurrent Unit Networks
HERA	Health Emergency Preparedness and Response Authority
IoT	Internet of Things
ICUs	Intensive Care Units
MCM	Medical Countermeasures
ML	Machine Learning
NLP	Natural Language Processing
LASSO	Least Absolute Shrinkage and Selection Operator regression
LSTM	Long-short-term memory
PPE	Personal Protective Equipment
RF	Random Forest
RFID	Radio-frequency identification
RNN	Recurrent Neural Networks
SD	System dynamic (simulation)
SCM	Supply chain management
SVR	Support vector regressor

Glossary

Term	Description
Artificial Intelligence	Technology that enables computers and machines to simulate human learning, comprehension, problem-solving, decision-making, creativity and autonomy.
Artificial Intelligence system	A machine-based system that, for explicit or implicit objectives, infers, from the input it receives, how to generate outputs such as predictions, content, recommendations, or decisions that can influence physical or virtual environments.
Blockchain	The blockchain is a shared, immutable ledger that facilitates the process of recording transactions and tracking assets in a given network.
Bullwhip effect	Amplification of variability in demand caused by the retailer incorrectly forecasting demand and leading to discrepancies between inventory produced and demand.
Data visibility	Access to high-quality data on both demand and supply factors which aids successful decision-making.
Demand Forecasting	Demand forecasting is a scientific and methodical estimation and/or prediction assessment of the future need for a specific product or service
Deep Learning	Subset of ML that uses multi-layered neural networks, called deep neural networks, to simulate the complex decision-making power of the human brain.
Extreme Gradient Boosting	A scalable, distributed gradient-boosted decision tree (GBDT) machine learning library. It provides parallel tree boosting and is the leading machine learning library for regression, classification, and ranking problems.
Friendshoring	Trade practices where supply chain practices are focused on partners regarded as political and economic allies.
Internet of Things (IoT)	The Internet of Things describes the network of physical objects—“things”—that are embedded with sensors, software, and other technologies to connect and exchange data with other devices, systems and the cloud over the Internet.
Machine Learning	Branch of AI and computer science that focuses on using data and algorithms to enable AI to imitate the way that humans learn, gradually improving its accuracy.
Medical Countermeasures	Medicinal products, medical devices and other goods or services that are necessary for preparedness, protection and response to serious cross-border threats to health.
Natural Language Processing	The systematic approach means a computer that gathers knowledge on how humans use, apply, and understand language.
Onshoring	Process of sourcing or relocating business operations within domestic national borders.
Public health surveillance	Continuous, systematic collection, analysis and interpretation of health-related data.
Radio-frequency Identification	A wireless system utilising radio-frequency electromagnetic fields to obtain data for tracking and identifying items.
Reshoring	Process of moving a business or part of a business that was based in a different country back to its original country.
Simulation methods	A simulation is a research technique that replicates events and processes under test conditions. It is often a highly complex mathematical process. In the context of supply chain management, simulation methods create realistic scenarios and allow testing different strategies in a controlled environment and across a range of different indicators, actors, and parameters.

Term	Description
Synthetic Data	Artificially generated data that mimics real-world data. It is created by computing algorithms and simulations based on generative artificial intelligence technologies. A synthetic data set has the same mathematical properties as the actual data it is based on, but it does not contain any of the same information.
Supply chain	A supply chain is a network of suppliers, manufacturing facilities, warehouses, distribution centres, retailers, producers, vendors and customers that deals with the procurement of raw materials, the manufacture of products, and the distribution of products to the customers. They are composed of multiple stages carried out by different entities interconnected through physical, information, and financial flows.
Supply chain monitoring	Supply chain monitoring involves using a set of tools, with analytical capabilities, to track, trace and assess the performance of supply chain operations. It is a critical part of effective supply chain management (SCM) as it helps identify and remedy issues in global supply chains that could impact the operations.
Supply chain mapping	A simplified model of a supply chain which presents appropriate and accurate information in a manner that can be easily understood and at the same time be sufficiently informative to aid supply chain visibility, analysis and integration

1. Introduction

1.1 Study scope

Over the years, increasing attention has been dedicated to healthcare **supply chain efficiency and resilience** due to its critical role in ensuring stable, fast and equitable access to medical countermeasures (MCM) (National Academies of Sciences, Engineering, and Medicine, 2022). Increasing **visibility**, fostering **monitoring** capacity and improving **forecasting** accuracy are recognised as some of the key success factors in building robust supply systems.

In this context, innovative data analytics, including **artificial intelligence (AI) solutions and advanced simulation methods**, can pave the way to actively predict, prevent, plan, act against potential MCM supply chain disruptions and react towards them.

By collecting relevant information on the most cutting-edge methodologies and tools to address supply chain issues, this study contributes to **Health Emergency Preparedness and Response Authority's (HERA) intelligence-gathering activities**, providing instruments to guide **evidence-based decision-making** and, therefore, strengthening European Union (EU) healthcare systems.

Specifically, this study focuses on analysing the **AI-driven methodologies and simulation methods** that can support MCM **demand forecasting and supply chain monitoring** activities. The final aim is to advise **HERA** on what can be considered the **most suitable methodologies** to be applied to MCM supply chains in the EU context, what data fields are needed to effectively implement them and what challenges and barriers may be encountered.

The methodologies and simulation methods analysed in this report may include solutions to obtain more accurate demand forecasting of MCM, for example, based on an estimate of disease outbreak trajectories, or tools to map the supply chain to increase visibility. Simulation methods, together with other analytical methods, can also be adopted to test various scenarios in a cost-effective way, helping decision makers to understand the potential impacts of disruptive events on MCM and choose the most effective course of action or public health intervention.

1.2 Study objectives

Overall, this report aims to support **HERA's mission of improving the EU's** ability to prepare, prevent, detect, and rapidly respond to health emergencies, ensuring availability and accessibility to key MCM (HERA, 2023). Specifically, the study pursues three main objectives:

1. Analyse the **AI-driven methodologies for demand forecasting**, including epidemics predictions or estimates of product needs, to understand:
   - what **methodologies** are available and could be effectively implemented in the EU.
   - what **data** characteristics are relevant to implementing AI-driven forecasting.
2. Analyse the **AI-driven methodologies for supply chain monitoring** (e.g., supply chain management, monitoring of the supply chain, identification of actual/potential bottlenecks) to understand what **methodologies** are needed and could be effectively implemented in the EU MCM sector.
3. Analyse the **simulation methods** that can be used by HERA to support decision-making and optimise MCM supply chains, to understand:
   - what **methodologies** could be effectively implemented in the EU MCM sector
   - the most appropriate **context** in which they should be used

The study brings together literature review and experts' consultations on both peace time and emergency scenarios to focus on the so-called **preparedness phase** of health emergency situations. This includes studies on AI applications for inventory management, MCM demand forecasting and regular supply monitoring that aim at increasing overall resilience of the supply

chain. Nevertheless, the report also includes evidence that can inform the **response-phase** of emergency situations, such as models that utilise disease trajectories to forecast MCM demand and build simulations of possible scenarios according to different health interventions.

1.3 Structure of the study

This study is divided into three main sections:

- **Section 1 - Introduction** presents the purpose of the study together with the background and overview of the state of the art of supply chain monitoring and demand forecasting and the role of AI tools and simulation methods.
- **Section 2 - Results** presents key findings that emerged from the literature review and from the consultation with key experts, according to research questions. These are organised into four main parts:
    - AI-driven methodologies for demand forecasting
    - AI-driven methodologies for supply chain monitoring
    - Simulation methods for supply chain related decision making
    - Limitations and challenges to implementation
- **Section 3 - Discussion and Conclusion** synthesises key takeaways from the analysis of key findings, including possible directions for future research and policy actions

In addition**, Annex A -** outlines the methodology used in this report, detailing the key research questions, literature research strategy, and data extraction methodology. A PRISMA diagram is presented to give a clear snapshot of the information considered in the literature review. Additionally, information on the consultation of key experts is provided.

1.4 Overview of the state of the art

The **healthcare supply chain** can be defined as all the systems, components processes, stakeholders and resources involved in the manufacturing, sourcing and management of medical goods and services from suppliers to the end-users, i.e. patients (Arora & Grigas, 2018). Supply chains play a crucial role in health systems, as directly impact accessibility, cost and quality of healthcare services, including the availability of pharmaceutical products and medical devices (Long, et al., 2023). Over the last decades, the complexity and extension of supply chains have sharply increased, driven by major trends such as globalisation, digitalisation, and technological innovation (Panzitta, et al., 2017). In today's globalised economy, medical supply chains have become highly vulnerable to disasters (both natural and human-made), pandemics and epidemics (Zheng, et al., 2022). A supply chain is typically divided into upstream and downstream operations. **Upstream operations** involve sourcing, procuring, and transporting raw materials (for example active pharmaceutical ingredients) to manufacturing facilities, focusing on supplier management, procurement, and inventory control (GEP, 2024). Key challenges include disruptions due to external factors like natural disasters or political unrest, as well as maintaining **visibility** (defined as “companies having access to timely, high-quality and useful supply chain information” (Williams, et al., 2013) and managing inventory levels efficiently. **Downstream operations** handle the distribution of finished products (for example vaccines, medicines or medical devices) to customers, including warehousing, transportation, and order fulfilment. Challenges in this stage are often centred around meeting customer demands for faster delivery and managing reverse logistics, such as product returns and related costs (GEP, 2024).

In 2020, the COVID-19 pandemic posed unprecedented challenges for healthcare systems worldwide and dramatically exposed the vulnerability of the supply networks of medical products (Belgodère, et al., 2023). Recent events such as increasing geopolitical tensions, adverse effects caused by climate change and antibiotic resistance, pose additional challenges to health security. Consequently, there is a growing recognition of the need to improve access

to Medical Countermeasures (MCM), defined as all the medicinal products, devices, vaccines, personal protective equipment (PPE) and other goods or services that are necessary to be prepared and respond to serious cross-border threats (Brintrup, et al., 2023). MCM supply chain encompasses the sourcing of raw materials and biological products needed for their production as well as their storage, transportation, and delivery to healthcare facilities or populations in need (Ergun et al., 2023). In the context of health emergencies, **MCM** must be produced and **distributed timely, equitably and in adequate volume**. This is because their smooth and timely provision is essential to a successful response as allowing to prevent, protect against, treat and diagnosis diseases (Lal, 2022).

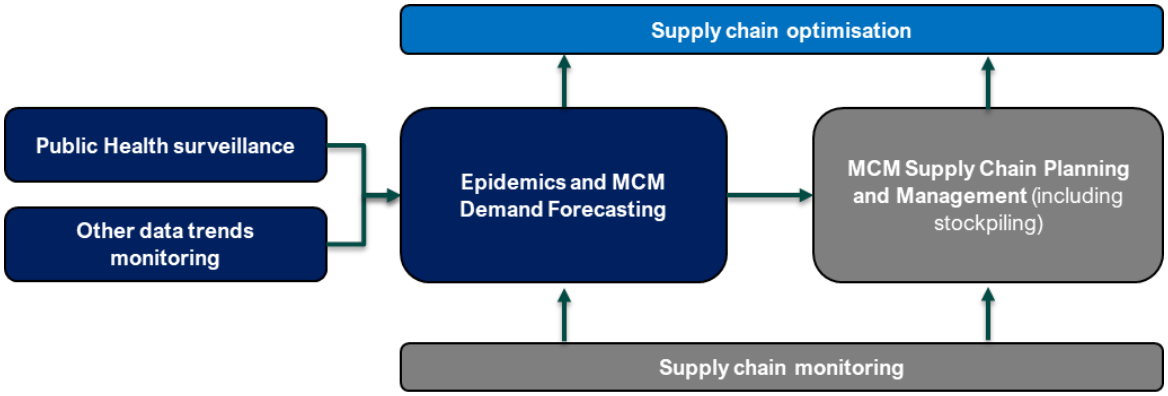
To ensure the resilience of MCM supply chains in future health threats and emergencies, governments and international organisations must be able to **proactively anticipate, plan,** and **put in place effective actions** to prevent or mitigate shortages caused by both operational (inherent uncertainties such as mismatch between supply and demand) and disruption risks (e.g. major events such as natural disasters and health emergencies) (Tang, 2005). This process, however, is possible only through the continuous **monitoring** of a wide range of data in the supply chain (Christopher & Lee, 2004). This can be defined as **the regular analysis and detection of all the activities** essential to effective risk planning and management. This includes both upstream and downstream operations, from sourcing, manufacturing and stockpiling to product delivery and distribution. Monitoring allows to have a supply chain that is traceable and transparent, for instance by tracking the origin and destination of the products as well as their consumption. It is also strictly correlated to **data visibility** on the end-to-end supply chain, defined as access to high-quality data on both demand and supply factors, which aids successful decision-making (Barratt & Oke, 2007). Accurate mapping of the supply chain, rendered possible by data visibility, is essential to effective supply chain management, improvement and digitalisation (MacCarthy, Ahmed, & Demirel, 2022). According to (Rathipriya, et al., 2023), another mechanism to effectively anticipate fluctuations in supply chains is **demand forecasting**, a scientific and methodical assessment of future demand which provides information on the resources needed to ensure adequate services, allowing to match supply accordingly. Demand forecasting involves predicting the healthcare needs of a population, both in terms of health services as well as medical products (Huang, 2020). Although this has been considered one of the most effective methods to predict and anticipate shocks in supply, demand forecasting is also one of the **weakest links in health supply chains** and inaccurate estimates of the demands have often caused inadequate planning, eventually leading to shortages (Blanchard, 2021). This is especially true in a context characterised by frequent fluctuations in demand due to demographic trends such as shortages of healthcare workforce in EU countries and ageing population (Looi, 2024), changes in disease patterns, new disease outbreaks or pandemics.

Demand forecasting and **supply chain monitoring** are two highly **interconnected activities** that together can inform supply chain management and decision-making to help guarantee efficient and stable delivery of health products (**Error! Reference source not found.**). Robust forecasting of future epidemic outbreaks or MCM demand needs a high amount of quality and historical data. These may be collected through **public health surveillance** (i.e. continuous, systematic collection, analysis and interpretation of health-related data (WHO, 2024) but also **monitoring other relevant trends**, which include, for instance, environmental changes or social and market dynamics. **Forecasting methods** have been extensively developed over the years, especially after the recent pandemic, to predict factors such as infected cases, the number of deaths and demand for medical products, such as vaccines, ventilators and ICU beds (Petropoulos, 2022). Relevant data to forecast demand for MCM are also provided through **supply chain monitoring** activities that increase the visibility on products, processes and resources. Indeed, as outlined in Section 2.1, demand forecasting methodologies are mainly based on the level of consumption of the previous year, which can be measured by monitoring the downstream supply chain. In addition to the identification of consumption patterns, tracking lead times and inventory levels at the different stages can provide additional insights. More advanced demand modelling usually integrates additional variables, including

epidemiological ones, to anticipate factors and trends that will impact future demand. The combination of insights from predictive models and supply chain tracking can support **evidence-based planning and management** of the entire supply chain from sourcing to stockpiling. All these activities, together along with **supply chain optimisation,** defined as the process of improving performance, reliability and efficiency of supply chain network (IBM, 2024) can lead to optimising the supply chain, avoiding wastage of resources, preventing critical shortages and increasing the resilience to future health emergencies.

Figure 1 below provides a visual representation of the link between epidemics and demand forecasting, supply chain monitoring and supply chain management.

Figure 1 Link between monitoring population health and MCM demand and supply chain management



***Source**: author's elaboration based on literature review findings.*

The nature of MCM supply chain is inherently complex, given the products (for example varying materials and perishability), the impact of supply disruptions on citizen's health, high levels of internationalisation, regulatory and privacy concerns and the range of different stakeholders involved (Wollek, et al., 2022) see Box 1 for detailed explanation).

To make predictions and optimise such a complex supply chain network**, traditional statistical and epidemiological forecasting** methods have been widely used to solve MCM supply chain issues (Petropoulos, 2022). **Statistical methods** for health forecasting often consist of using time series historical data together with additional parameters. They include generalised regression, moving averages, multilevel time-series models (Solamain & Cohen, 2024) and exponential smoothing (Tseng & Shih, 2020). One of the most common methodologies applied to demand forecasting is **ARIMA** (Auto-Regressive Integrated Moving Average) (Li, et al., 2023). This is a traditional statistical time series forecasting method that estimates future needs based on historical data, assuming stability in patterns. The model includes three components: an autoregressive (AR) component that uses as a predictor a linear combination of lagged values, a moving average (MA) component based on past forecast errors (white noise), and an integrated component where differencing is applied on the data to make it stationary (Motamedi, et al., 2024). Nevertheless, other methods have also been used, such as Simple Moving Averages (SMA) or Single Exponential Smoothing (SES) (Harris-Wilson, 2023). Limitations of traditional statistical models include using established patterns with the assumptions of stability and not considering external factors (e.g. government interventions, human behaviour) that might significantly influence results. An example of this is (Petropoulos, 2022) model that used a time-series approach to predict short-term COVID-19 cases, without factoring actions that could influence the spread such as lockdown policies.

In addition to statistical methods, **epidemiological forecasting methods** have also been used to predict how biological factors and population variables can further influence MCM demand. For instance, model predictions on the spreading of pandemics can then help inform demand on vaccines and help shape policies on vaccines production and stockpiling (Hyder, Buckeridge, & Leung, 2013). With their use increasing as a result of the COVID-19 pandemic

to obtain insights into disease dynamics, **compartmental models** (modelling techniques used to study the evolution of pandemic) have been extensively applied to understand how disease development can influence factors such as demand for medical products by Goic et al (2021). These include different methods among which one of the most common is the SEIR (Susceptible, Exposed, Infected, Recovered) (Piccolomini & Zama, 2020). However, compartmental models have limited flexibility to accommodate dynamic variations in the environment, which are particularly frequent during health emergencies. For instance, during the COVID-19 epidemic, many critical parameters have frequently changed over time (e.g. the delay between the identification of a new case and the requirement of an ICU bed, the duration of mechanical ventilation and the likelihood of requiring urgent care) and most compartmental models are not able to properly capture them. According to Lee et al. (2021), **systems modelling,** which combines biological epidemiological model, simulations and AI on human behaviour and processes, optimisation and simulation of resources and interventions; can provide powerful predictions and address the limitations of compartmental models.

Regarding **monitoring of MCM supply chain**, mechanisms to improve its effectiveness through data and visibility have mainly focused on **international and domestic policies**, including onshoring and friendshoring, partnerships to increase data-sharing across countries and across government and industry and strategic stockpiling (Mulopulos, 2023). Specifically for medical products such as PPE, there have been efforts in reducing dependencies from countries with low traceability through increase in domestic manufacturing or friendshoring (Dallas, Horner, & Li, 2021). Indeed, the dependency on China and India for API supply has been highlighted as one of the main reasons contributing to sustained shortages of antibiotics in Europe in 2022, and 64% of European Medicines Agency (EMA) compliance notices between 2018 and 2019 were related to facilities located in those countries (European Commission, 2023); (Miljković, 2024). Worldwide governments have also started to mandate companies to share data on their end-to-end medical supply chain to be able to monitor factors such as human rights practices and environmental violations in the supply chain. Some countries, like New Zealand, have made significant efforts to tackle the issue of transparency by disclosing information about the names and locations of active ingredient producers, finished product manufacturers, product sponsors, and marketers (Boston Consulting Group, 2022).

Regarding the use of **mathematical modelling** for **supply chain monitoring** however, experts have found big gaps in the literature (Beldek, 2020). As the need to quickly adapt to demand and supply shocks arose globally due to the COVID-19 pandemic, several methodologies using novel technologies have started to be applied to help predict outbreaks, improve target treatments and contain the spread of the virus. Between those, technologies called **Industry 4.0 enablers** which include blockchain, big data and artificial intelligence (AI) have been one of the fastest growing areas of interest (Brintrup, et al., 2023). The focus of this paper will be on artificial intelligence and used in conjunction to other methodologies specifically for the use of MCM supply chain monitoring and demand forecasting.

Box 1. Overview of the benefits for the main stakeholders involved in the MCM supply chain

Benefits of improving MCM SC monitoring and demand forecasting

Adding to the complexity of the supply chain are the different stakeholders involved, each one with different interests (Senna, et al., 2023). These stakeholders can benefit from improved demand forecasting and supply chain monitoring:

- **Industry.** This includes suppliers, manufacturers, providers, wholesalers, distributors and procurement agencies. Given the globalisation and complexity of MCM supply chains, a current challenge for the industry is to achieve full end-to-end visibility, starting from the origin of their raw materials. Adulyasak et al., (2023) claimed that configuration and management of supply chains should also be improved to better identify shocks such as sales anomalies, panic buying and pre-hoarding.
- **Healthcare providers & patients.** Monitoring the supply level and matching it with accurate estimates of the demand contribute to anticipating or preventing supply disruptions, which, in case of an insufficient level of inventory, may lead to shortages. In turn, shortages may significantly impact the quality of care and availability of medical products (delayed care, sub-optimal treatments, increased out-of-pocket expenses) therefore affecting patients' health outcomes. In addition, shortages also generate uncertainty, additional burdens, and higher costs for public institutions and healthcare providers (Wong, 2023); (Phuong, 2019). Health service institutions are increasingly struggling to cope with changes in demand and manage services and resources accordingly, such as Intensive Care Units (ICU) beds. Comprehensive supply monitoring and demand forecasting are needed to broaden the information health services, thus improving their accessibility, coverage and quality, allowing for better organisation and management (Cao, 2023). Supply monitoring can also prevent counterfeit medicines from entering the pharmaceutical supply chain by improving medicine and pharmaceutical traceability (Kumar & Tripathi, 2019).
- **Government agencies responsible for crisis preparedness**. To be able to design effective policies, government and healthcare institutions require data to map the national supply chain. With this aim, they have been increasingly relying on health surveillance systems to allow the identification of early, accurate and reliable signals of anomalies and disease outbreaks, and consequently to predict demand for a specific medical product. However, traditional statistical forecasting models have had limited accuracy and predicting power, particularly during health emergencies, which often rely on predictions with restricted historical data and standardisation (Petropoulos, 2022). In addition, further complexity derives from the fact that pharmaceutical companies often keep information on their suppliers confidential as this is typically considered commercially sensitive (National Academies of Sciences, Engineering, and Medicine, 2022).

1.5 Overview The role of AI and simulation methods in demand forecasting and supply monitoring

A. Artificial Intelligence

Artificial intelligence (AI) is a rapidly evolving tool revolutionising many aspects of healthcare. It is a vast field that includes a variety of methodologies such as computer vision, natural language processing, and machine learning (ML) used for pattern recognition, prediction, and large-scale data analysis by Cannas et al., (2024).

In 2023, OECD member countries approved a revised version of the organisation's definition of an AI system (OECD, Artificial intelligence, 2023). The updated definition reflects developments over the last five years, improves technical accuracy, and makes it more 'future-proof'. The EU's AI Act has aligned the definition of AI to the updated OECD definition in its final text. In this study, we will consider the updated OECD definition of 'AI systems' as a baseline to ensure technical accuracy and international alignment (

Box 2).

Box 2. Working definition of "AI system¨

> "An AI system is a machine-based system that, for explicit or implicit objectives, infers, from the input it receives, how to generate outputs such as predictions, content, recommendations, or decisions that can influence physical or virtual environments. Different AI systems vary in their levels of autonomy and adaptiveness after deployment."

Prior to 2019, AI has been predominantly employed in medicine and healthcare administration. The COVID-19 pandemic shifted its focus from medicine to **public health** (Olawade, 2023) and accelerated the use of AI-based technology to improve the timeliness and accuracy of outbreaks' detection, improve target treatment and contain the spread of the virus. An example is the application of a dynamic neural network model to predict outbreak risk of ZIKV in Americas by Akhtar et al., (2019). AI-based solutions have also been used to broaden the information health services have access to thus improving their accessibility, coverage and quality, allowing for better organisation and management (Cannas et al., 2024).

The application of **AI in supply chain management** has been defined as a game changer, including in the medical supply chain (IBM, 2019). AI can improve the visibility and flexibility of MCM supply chains, and it has already been used to estimate real-time demand forecasting, increase the quality and safety of MCM by tracking counterfeits and authenticate medical devices, improve inventory management, stockpiling, distribution and supply chain risk management (Kumar & Tripathi, 2019). An example is the use of AI to improve vaccine supply chain management and estimate real-time demand (Gao, 2023). Consistently, as outlined in the EU Commission Communication on "Addressing medicine shortages in the EU" (European Commission, 2023) and HERA 2023 work plan (European Commission, 2023); AI may also be used as a critical asset for supporting demand forecasting and supply chain monitoring activities. This is being reflected in the increase in AI-based function supply chain applications (Riahi, 2021), with the Global AI market expected to grow at a CAGR (compound annual growth rate) of 36.6% between 2024 to 2030 (Grand View Research, 2024).

To strengthen data collection, analysis and monitoring of MCM supply chain, the EU has launched the Advanced Technology for Health Intelligence and Action IT System (**ATHINA),** which will collect data related to health threats and provide an initial assessment to ensure effective response to public health emergencies. The EU-HIP project (the EU interoperability with HERA's IT platform) will help Member States strengthen their reporting systems to create interoperability within ATHINA (Norwegian Institute of Public Health, 2024). In parallel, the European Medicines Agency (EMA) is creating the European Shortages Monitoring Platform (ESPM) will support the monitoring of supply, demand and availabilities of critical medicines throughout normal times, emergency situations and by request of the EMA´s Executive Steering Group on Shortages and Safety of Medicinal Products (MSSG), (EMA, 2024).

Bathula et al., (2024)claim that although the potential applications of AIin healthcare supply chains are numerous, the challenges to its application are still substantial. This is partly due to the nature of healthcare supply chains as, they rely on a wider range of inputs from different stakeholders, causing data storage and security and system interoperability issues. . Particularly regarding security and privacy, healthcare supply chain is made particularly vulnerable by several factors. Firstly, because of the high interconnection across all the different systems, devices and stakeholders that are included in the healthcare supply chain (Stanley, 2019). Secondly, because of the complexity of data included, which increases the risks of and exposure to cybersecurity threats (Christopher & Peck, 2004). Thirdly, as attackers aim to obtain patients' sensitive data, they can threaten services to release or inappropriately use patients' data. This can have severe impacts on patients trust and the organisation's

regulatory compliance (Wright, 2023). According to the European Agency for cybersecurity (ENISA), the healthcare sector has experienced a 400% increase of cyberattacks in the supply chain (ENISA, 2021). In their latest report published in July 2023, ransomware attacks (a type of malware that prevents the owner from accessing their device and related data) were claimed to represent more than half of the main threats in the health sector. Nearly half of cyberattacks aimed stealing patient´s data (ENISA, 2024).

Furthermore, according to Balasubramanian et al. (2023), research has been scattered, with most studies focusing on specific applications of AI such as medicine development.

The OECD working definition provided the overarching theoretical framework for AI systems, but for the purposes of this study, the scope was narrowed to specific AI techniques and algorithms. Particularly, the OECD Framework for the Classification of AI Systems provides a comprehensive and structured approach that helps characterise AI systems based on key dimensions such as description, examples, data and main tasks (OECD, 2022). Although not covering all types of AI, the OECD framework has been used to categorise in a systematic way AI systems based on their characteristics and the produced outputs. As a result, the study team considered the following AI techniques, namely **Symbolic AI, Statistical AI, and Hybrid AI**. **Error! Reference source not found.** provides an overview of the three categories considered with their main characteristics and tasks performed. **Symbolic AI models** allow to make interferences and perform classification (e.g., Knowledge-based and expert systems). **Statistical AI models** facilitate trend and pattern prediction based on data (e.g., Linear regression, Logistical regression, Decision trees, Neural networks, Deep learning). Lastly, **Hybrid AI models** combine the previous approaches to extract complex features and enhance pattern recognition and learning abilities.

Table 1. Definition of the AI techniques considered in this study

	Symbolic AI model	Statistical AI model	Hybrid AI model
Description	Symbolic or **knowledge-based** AI uses human-generated logical representations to infer a conclusion from a set of constraints (variables). These constraints include **rules**, **ontologies** and **search algorithms** that rely on explicit descriptions of variables and descriptions of the interrelations between these variables.	**Statistical AI models** (e.g., neural networks and deep learning) identify patterns based on data rather than expert human knowledge. Primarily used for recognition, more recently, these models are also being used for tasks such as generation, synthesising and generating images or audio.	Many applied AI systems **combine symbolic and statistical** models into “hybrid” models. For example, NLP algorithms often combine statistical approaches building on large amounts of data and symbolic approaches that consider issues such as grammar rules.
Examples	Knowledge-based and expert systems	Linear regression, Logistical regression, Decision trees, Neural networks, Deep learning	NLP
Data	Labelled and categorised	Unlabelled and uncategorised	Can be labelled or unlabelled
Task	Making inferences and performing classification	Trend/ pattern prediction	Complex feature extraction and pattern recognition/ learning abilities

***Source**: author’s elaboration based on literature review findings.*

Among the AI tools the most widely used for supply monitoring and demand forecasting are **ML** methodologies, a specific category of AI which works with self-learning algorithms able to leverage on the experience gained from available historical data to improve results (Mediavilla, Dietrich, & Palm, 2022). Other characteristics of this type of AI have contributed to its vast use

for demand forecasting, such as its capacity to deal with large amounts of noisy data, which are typical of historical demand data (Mediavilla, Dietrich, & Palm, 2022).

Further developments of ML led to the diffusion of Deep Learning methodologies, a subset of ML that uses multi-layered neural networks, called deep neural networks, to simulate the complex decision-making power of the human brain.

To conclude, AI methodologies can potentially offer several key benefits to supply chain management, including enabling predictive analytics and automating and optimising processes through ML algorithms. Cannas et al (2024) argue that AI also enhances strategic sourcing and supplier evaluation and strengthens risk management, contract management, and compliance efforts while minimising the need for manual intervention. The following section (2. Results) will delve deeper into this.

B. Simulation methods

Simulations are one of the most used research methods in supply chain management (Shafer & Smunt, 2004) as well as in healthcare systems, according to Melnyk et al (2024). They are considered an effective tool to study and forecast supply chain behaviours over time (Ivanov, 2020).

Simulation methods can provide solutions where analytic calculations and linear programming cannot (Grigoryev, 2018). Furthermore, they have been put forward to address some of the limitations that characterise AI, Big Data and ML, particularly regarding the dependency on historical data. As AI and analytics rely on data to predict future trends, Melnyk et al (2024) define them by what has already happened in the past and do not perform well when dealing with low-probability, uncertain scenarios. Furthermore, AI tools are defined as ´black box´ approaches because they are unable to provide insights on the relationship between input and output, and thus determine causality (Nas & Koyuncu, 2019). Simulation methods are believed by some to overcome these limitations and, in the context of supply chain management, aid the analysis of time-dependent responses of key actors and their actions in the supply chain and in a system that might be destabilised. However, by nature, simulation does not identify critical features as in AI/ML, but rather allows users to run trial-and-error simulation experiments to pinpoint bottlenecks (Rudin, 2022).

Box 3. Working definition of "Simulation method"

"A simulation is a research technique that replicates events and processes under test conditions. It is often a highly complex mathematical process. In the context of supply chain management, simulation methods create realistic scenarios and allow to test different strategies in a controlled environment and across a range of different indicators, actors, and parameters¨

Discrete event simulation, one of the most common type of simulations, has been wildly used to model supply chain since 1950s (Wang & Ingham, 2008). As they allow for flexibility and can deal with uncertain and variable scenarios, they are often included when modelling healthcare systems. Furthermore, the increased uncertainty faced by medical supply chains, together with the use of Industry 4.0 enablers in supply chain management, have prompted experts to call for an increased application of simulation for supply chain, as allowing to carry out the '**what if' analysis,** observe the potential impacts, and provide insights on how to prevent them (Saisridhar, Thuerer, & Avittathur, 2024). Simulation methods are run with the use of commercial **software**, such as Arena, Simul8, AnyLogic and Vensim (the latter two also for hybrid simulations). Simulations can be used to measure the performance of solutions obtained from different techniques, such as meta-heuristic algorithms (Shirazi, 2021) and, according to Ordu et al., (2021), optimisation methods. For instance, Discrete Event Simulation (DES) could be used to decide how many beds to be assigned to a specific hospital region in an emergency scenario. This solution could be tested against different scenarios and probability distribution with simulation techniques (Ortiz-Barrios, et al., 2024). In terms of data

needs, simulations use historical data for initialisation and execution (Mustafee & Fakhimi, 2024).

Simulation methodologies can be implemented at a policy-level (government institutions) as well as organisational (e.g. hospitals, pharmaceutical supply chains) to test scenarios according to different health interventions and disruptions such as disease outbreaks and supply chain disruptions and foster better decision-making. Historically, they have mainly been used to model spreading of diseases such as AIDS (Harper & Shahani, 2003), influenza (Savachkin & Uribe, 2012) and more recently, COVID-19 (Currie, 2020). Simulation has also been used and implemented successfully (coupled with machine learning and optimisation) for patient treatment outcome prediction, hospital patient workflow, clinical operations, and resource allocation (Lee, et al., 2015). Compared to traditional statistical and mathematical epidemiological models, which predict the number of cases and identify appropriate measures to reduce transmissions, simulations have been shown to give more accurate and detailed predictions and allow to consider the **impact** that **individual´s behaviours** with their different characteristics have on the whole system (Savachkin & Uribe, 2012). Modelling COVID-19 cases through simulation has permitted **evidence-based decisions** on **resource management** in hospitals and MCM supply chains (Currie, 2020). For instance, simulation of COVID-19 cases was conducted to inform simulations of plasma demand (Shirazi, 2021) and vaccine demand (Kamran, 2023).

2. Results

The results are organised into four subchapters. The first three subchapters provide a detailed explanation of the processes, specific methodologies, and data that could be used for increasing the efficiency and resilience of supply chains through **i) demand forecasting**, **ii) supply monitoring** and **iii) simulation methodologies for supply optimisation and decision making**. Additionally, the last subchapter outlines the **main barriers to the implementation** of the identified AI-driven methodologies and simulation methods.

The comprehensive literature review pointed out that there is extensive adoption of AI-driven methodologies and tools in the context of **supply chain management**. According to the Supply Chain Agility Index (2024), out of 1700 supply chain management leaders surveyed across the US, UK & Ireland and APAC; more than half the respondents are using AI, automation (including smart robotics), or ML for at least one supply chain management application. The primary motivation behind the adoption of these technologies is **to enhance operational efficiency and optimise resource allocation**. The benefits that these technologies bring, however, extend beyond mere operational/ financial efficiency: automation can in fact facilitate better compliance management and quality assurance through consistent monitoring and analysis, reducing the risk of non-compliance penalties (Hamway, 2024).

Across the management of the entire supply chain, evidence shows that AI is mostly used in demand forecasting (Section 2.1). Regarding supply chain monitoring, AI is often used through data collected and transmitted through blockchain and Internet of Things (IoT), (Section 2.2). Data related to demand forecasting and monitoring are increasingly used to test and inform possible scenarios through simulation methodologies methods (Section 2.3), driving effective decision-making. Major barriers persist to the actual implementation of these methodologies (Section 2.4).

2.1 AI-driven methodologies for demand forecasting

Demand forecasting is an essential step in supply chain management since it is an informative tool to drive strategic decisions on the allocation of resources, selection of mitigation measures, production and logistics planning and inventory management

This **research field is rapidly evolving**, and several **different approaches** for forecasting have been adopted by stakeholders or explored in the literature, making it complex to identify

the most suitable methods for all different contexts and objectives (Mediavilla, Dietrich, & Palm, 2022) (Soyiri & Reidpath, 2013).

Forecasting methods can be mainly classified into **statistical**, **AI-based**, or **hybrid methods**, which combine the characteristics of a statistical forecasting model with the ones from the AI domain. In addition, these methods can be **univariate analysis** (traditional time series methods such as ARIMA or Prophet models) and **multivariable analysis** (e.g., LSTM networks, ANFIS, random forest), depending on whether they include single or multiple predictors in the model.

Specifically, for demand forecasting, **predictive AI** (mainly **ML methods**) has been applied for its ability to facilitate and improve statistical modelling using historical data, combined with other variables, to predict future trends.

Many methodologies have been developed for epidemic prediction especially during **extreme events,** when ensuring stable supply, avoiding overstocking and having an adequate allocation of resources is even more critical (Nguyen, Ding, & Pathirana, 2021). Establishing the form of estimates (e.g. point, distributional) is also important for successful application of forecasting models to inform their complexity and how they might be used for planning. However, most of the forecasting models still provide inaccurate estimates, partly due to lack of incorporation of disease spread and social behaviour within most forecasting models.

In addition, ML tools are already used by the majority of international pharmaceutical corporations during **normal times** to **improve efficiency** in the **management of the supply chain,** by matching the supply with demand fluctuation and reducing inventory levels (Rathipriya, et al., 2023). In this way, firms not only manage the complexity of their supply chains but also pursue a competitive advantage.

Similarly, ensuring accurate demand forecasting is essential for governments and **public healthcare services providers** for planning and budgeting (Subramanian, 2021). Underestimation of demand can in fact lead to reduced **coverage and healthcare level** such as low inventory level among manufacturers (lean approach), partly contributing to the inability to respond to sudden demand surge. Oppositely, overestimation leads to **wastage** of public resources.

The literature review allowed to identify three main steps that need to be addressed to effectively apply an AI-driven methodology to forecasting:

1. Definition of the scope of the forecasting and data availability
2. Selection of the most appropriate methodology
3. Data collection and processing

1. Definition of the scope of the forecasting and data availability

The first step is to clarify the specific aim that forecasting methods should meet, as well as understand the amount and type of data available for the analysis. Considering MCM supply chains, forecasting methodologies could be mainly used to anticipating **epidemics and disease outbreaks** or Forecasting **demand of MCM** for planning production, optimise inventory and stockpiling and healthcare resource management.

The first field of research has become increasingly relevant after the COVID-19 pandemic since its use, in combination with traditional **surveillance** methods; proved to be extremely valuable in getting rapid and accurate insights into disease dynamics (e.g., estimating the **number of cases, number of deaths, expected trajectories**), especially in situations of rapidly changing variables (Zhao, et al., 2024). Through advanced **epidemiological models,** Lee et al (2022) showed it is possible to **quantify the number of patients** who require treatments over time, taking also into account **uncertainty**.

These models then can be used to build accurate **product demand forecasting**, for instance, to quantify the medicines needed during the epidemic, based on future trends beyond historical patterns (Zhao, et al., 2024); (EMA, 2021). For instance, many authors applied AI-driven methodologies to forecast the demand for products and services during the COVID-19 pandemic, including the number of **hospital** ICUs (Zhang & Li, 2024); (Borges & Nascimento, 2022), **vaccines** (Hu, Xu, & Liu, 2023) and **equipment** (Koç & Türkoğlu, 2022).

Other authors focused on **normal times** mainly forecasting the need for **blood and platelet** products (Motamedi, et al., 2024); (Kwon, et al., 2024) (Li, et al., 2022) (Li, et al., 2023); (Schilling, et al., 2022), **vaccine** utilization (Gao, 2023); (Hariharan, et al., 2020), **medicines** quantity or **shortages** (Pall et al., 2022).

The above supply chains investigated in the literature are often characterised by highly **uncertain demand patterns**, due for instance to extremely short shelf life or unplanned supply disruption, making demand estimates even more crucial. Considering **the condition** under which the model will be run, different methodologies and data may be required **during crises** with respect to **normal times.** Emergency situations are often accompanied by a drastic increase in demand, sudden supply interruptions and strong changes in consumer behaviour that can also lead to excessive overstocking (Nguyen, Ding, & Pathirana, 2021). Under this high level of uncertainty and lack of essential data, standard epidemiological and statistical models have shown low accuracy for long-term prediction (Ardabili, 2020). Complex AI-supported epidemiological models can provide significant benefits in anticipating potential strong fluctuations in demand and are increasingly used (Zhao, et al., 2024).

In addition**, different aims and contexts** may require the use of different models as well as different data availability. For instance, the literature highlights two main types of models used for **demand forecasting of products** based on whether this is considered directly linked to supporting **inventory management decisions** or separately (Li, et al., 2023). Indeed, a so-called **1-stage** model is applied when demand prediction and ordering are considered simultaneously as part of a unique, constrained optimisation model for making inventory decisions. Although this approach facilitates implementation and resource savings, it is only effective when demand is relatively stable and predictable with a low error rate, and production lead times are short (Li, et al., 2023). On the contrary, in a **2-stage** model, first, the demand forecast is estimated, and then, the inventory decisions are planned, as separate independent steps. This type of model minimizes errors and increases the clarity of the result of demand forecasting and, consequently, of inventory decisions. Usually, a 2-stage approach is used by larger organizations with wide portfolios. However, the choice between these models also depends on the level of uncertainty in demand and the production lead time. It is important however to outline that these models have the limitations of not considering lead times of raw materials and their supply patterns, limiting their application in real-world situations (Li, et al., 2023).

Time horizon, meaning the horizon forecasted by the AI method, also needs to be defined based on the organisation's needs and other contextual factors. Forecasting can target **short-term** (hourly, daily, weekly), **medium-term** (monthly), or **long-term** (yearly) horizons (Mediavilla, Dietrich, & Palm, 2022). Hariharan et al (2020), for instance, decided to forecast vaccine utilisation in health facilities in Tanzania with biweekly frequency based on discussions with healthcare providers, the need to have sufficient data size to make robust forecasts and considerations on the frequency of power outages in the regions. The EU Executive Steering Group on Shortages of Medicines Caused by Major Events, suggested that considering the goal of **informing supply planning**, demand forecasting should cover at least a period of **6 months** (EMA, 2021)**.**

Finally, **data availability** (i.e. quantity and quality) is one of the main constraints driving the choice among different models and influencing prediction accuracy (Schilling, et al., 2022), as further discussed below (in 3. Data collection and processing).

2. Selection of the most appropriate methodology

Once the scope and the data availability have been defined, a specific methodology should be selected. As the literature highlights numerous AI-driven forecasting methodologies, this section provides a brief **description of the most frequently mentioned** ones, specifically focusing on studies relevant to MCM supply chains which are summarised in **Error! Reference source not found.**.

Even though the focus of this study is on MCM supply chain, both the literature review and expert consultations indicate that there are no strong differences in the methodologies applied across sectors, meaning the findings from healthcare could be generalised to other sectors and vice versa. The literature search conducted identified studies focusing on energy and water management, transportation (e.g. estimate traffic, demand for electric vehicles or air passengers), retail, food and agriculture. Most of these studies applied the models discussed in this section. For example, **electricity demand** forecasting has been a widely studied area for many years, and Artificial Neural Networks (ANN) have been frequently used in this field to handle nonlinearities and deal with multiple types of input (Román-Portabales A, 2021). Among the several ANN-based models compared by Román-Portabales et al., (2021), the ones that can capture recurrent patterns, such as Long Short-Term Memory (LSTM), proved a higher predicting performance. A recent study by Aldarraji et al (2024) applied and compared cutting edge models for energy forecasting in Iraq, including XGBoost, Random Forest and LSTM, Temporal Convolutional Networks, and Multi-Layer Perceptron.

Similarly, in the **retail sector**, there is growing evidence that ML methods offer significant improvements in forecasting accuracy. Models such as recurrent neural networks (RNN), gradient boosted and regression trees (discussed below) are mentioned as techniques proving a superior performance than traditional time series model or linear regression, especially when forecasting multiple related sales series and are trained in a global learning manner (i.e. with parameters estimated collectively from a group of sales time series), (Fildes, 2022).

The literature review conducted by Mediavilla et al. (2022) included studies from sectors such as automotive, logistics, and defence and showed that the most recurrent AI models applied are among the ones discussed in Table 2. The choice between different models is mainly driven by the **specific context, purpose and data characteristics**, as well as the level of **expertise** and **knowledge** inside the organisation, including the capacity to manage system complexity and data. Fildes (2022) also referenced Wolpert and Macready's *no-free-lunch theorem*, to underscore that no single method, even the most complex one, consistently outperforms others across all series or time periods. Thus, no single forecasting method can dominate in every scenario.

Different classifications of AI methodologies have been proposed in the literature to support stakeholders in the selection of the most suitable one. For instance, Mediavilla et al. (2022) classified AI methods into traditional ML ones, deep learning and ensemble methods. Additionally, they underlined that the parameters to be considered in the selection are **data dimensionality** (univariate, multivariable with less than 10 attributes and with more), **data volume** (methods processing up to 500, from 501-1000 or more than 1001 input data) and forecasting **time horizon**.

Finally, other factors may drive the choice between models in real-world implementation, such as the cost of implementation and its interpretability, since this is an important aspect in ML forecasting models (Fildes, 2022).

Table 2. Summary of methodologies used for demand forecasting in the relevant scientific literature

Machine learning		
Traditional	***Deep Learning***	***Ensemble***
Prophet Least Absolute Shrinkage and Selection Operator regression (LASSO) Support vector regressor (SVR)	Artificial Neural Networks (ANN) Recurrent Neural Networks (RNN) Long-short-term memory (LSTM) Gated Recurrent Unit Networks (GRU) Adaptive neuro-fuzzy inference system (ANFIS)	XGBoost Random Forest (RF)

***Source**: Authors' own elaboration based on literature review findings*

ML - Prophet Model

Similar to traditional statistic methodologies such as ARIMA, the Prophet model is a **univariate time series model** that forecasts demand based on **historical data**. However, unlike ARIMA, the Prophet model is a **ML based tool** that also includes considerations on **trend, seasonality, holiday effects and outliers** (Motamedi, et al., 2024). According to Motamedi et al. (2024), accounting for the impact of weekends and holidays on the platelet demand variation, the Prophet model can ensure a 10% improvement in accuracy concerning ARIMA. Satrio et al., 2021 also compared the two univariate models, and ARIMA proved to be highly inaccurate in predicting COVID-19 cases concerning Prophet, which had a 91% accuracy. This model was originally developed by Facebook to forecast events, and it is now available as an open-source software package in Python and R. Its scalability makes it ideal for handling large datasets, and its interpretability makes it accessible to a diverse range of users (Motamedi, et al., 2024). In the literature relevant to MCM supply chains, it has been used in comparison with other models to **predict daily platelet demand** (Motamedi et al., 2024) or in combination with the LSTM method to increase accuracy in **forecasting COVID-19 ICU demand** (Borges & Nascimento, 2022). To note however, Motamedi et al. (2024), extensively cited above, applied the model to a normal time scenario and consequently has a certain limited applicability to emergency scenarios, characterised by more uncertain demand patterns. A modified Holt-Winters, also a univariate model, provides accurate and robust forecasts for multi-seasonal data with a damped trend, and can handle both normal time and emergency scenarios.

Univariate models such as Prophet and Holt-Winters are particularly valuable **in resource-limited settings** as they are more accessible and **do not require extensive expertise** for training and usage (Motamedi, et al., 2024). Consistently, the World Health Organization (WHO) underlined the need for robust but simple models for demand forecasting that could be easily employed in different settings. In addition, the Prophet model could be helpful in cases where there is solely access to data on historical demand but in adequate quantity. Indeed, Prophet models require an **extensive amount of data but for a few variables, i.e. historical demand,** including specification on the **seasonality** to capture the periodic fluctuations as weekly, monthly or yearly patterns, and the non-periodic holiday effect (Motamedi, et al., 2024).

ML - Least Absolute Shrinkage and Selection Operator regression (LASSO)

The Least Absolute Shrinkage and Selection Operator method, also called LASSO regression, is a **ML model** that can be implemented to forecast demand based on **historical data** while also considering **multiple predictors** in the process. It is often used as a feature selection tool, and it can be used as a variable selection method to **identify the most relevant predictors** for multivariable models. By accurately selecting appropriate variables, it reduces

the impact of predictors that would be irrelevant or less important. Consequently, it can help **reduce model complexity** and **improve the accuracy** of prediction.

According to the literature, in the current state of the art, the LASSO model, together with the LSTM networks (explained below), are among the most promising prediction models in the context of **platelet demand forecasting**. For instance, (Schilling, et al., 2022) predicted daily platelet demand by comparing LASSO and LSTM models and showing that both provide **similar performance and sufficient accuracy** to reduce waste and shortage in a retrospective simulation study. In the comparison between models developed by Motamedi et al., (2024) the LASSO model ensured **better performance concerning univariate time series** (ARIMA and Prophet), especially on the shorter training sets, and even marginally better than LSTM.

ML - Support vector regressor (SVR)

Support Vector Machine (SVM) is a ML tool initially used for **classification issues** and then extended to regression problems. Support vector regressor (SVR) is a regression model theoretically based on statistical learning that ensures a relatively high learning performance. This makes the model significantly suitable for predictions in small-sample, non-linear, and multi-dimensional fields (Zhang & Li, 2024).

The SVR model has been used to forecast the **number of COVID-19 confirmed cases**, and related **deaths,** overall ensuring an acceptable level of performance. (Parbat & Chakraborty, 2020) claimed that the estimates based on this model reached a 97% accuracy. Several studies compared SVR to different forecasting methods to evaluate its performance. For instance, SVR was compared to other frequently used models to forecast **medical equipment** demand in Turkey (Koç & Türkoğlu, 2022), as well as to predict the trend of **COVID-19 new cases and ICU bed occupancy** in Iran (Shafiekhani, Namdar, & Rafiei, 2022). Guo et al., (2021) combined SVR to Prophet model, using SVR to capture nonlinearities in the residuals generated by Prophet and to predict demand in manufacturing. Recently, the model has also been applied by Zhang et al., (2024) to estimate **demand for ICU healthcare resources** based on a prediction of several critically ill patients in the short term, to support decision-making at hospital level, improve rescue efficiency and reduce costs. The authors implemented a model combining bidirectional LSTM with SVR model optimised by a genetic algorithm (SVRGA), building a more robust estimation methodology.

Deep Learning - Artificial Neural Networks (ANN)

Neural Networks (NNs) are deep learning methods that simultaneously integrate multiple algorithms by replicating the way the human brain processes. NNs have proved to be highly effective in analysing disease transmission, spread, and causation (Stergiou, et al., 2022). Among NNs, Artificial Neural Networks are one of the simplest variants and have been increasingly used for demand prediction due to their superior prediction quality (Rathipriya, et al., 2023). These models overcome some limitations of traditional linear forecasting and thus improve the estimation of demand by modelling data with nonlinear behaviours.

Rathipriya et al., (2023) investigated several methods and techniques for demand forecasting of **pharmaceutical products** and concluded that the ANN-based model was the most appropriate for time series dataset. Specifically, they conducted an analysis considering **weekly forecasts** of the demand for Anatomical Therapeutic Chemical (ATC) thematic medicines based on their sales dataset.

ANN models have also been applied to forecast **blood demand**. (Fanoodi, 2019) did it by considering **daily demands** for eight types of blood platelets, while (Khaldi, 2017) implemented it to forecast the monthly demand for three blood components (red blood cells, platelets, and plasma)

Deep Learning - Recurrent neural networks (RNN) - Long-short-term memory (LSTM)

Recurrent neural networks (RNN) are more complex NNs compared to ANN since they can store the output of processing nodes and reintroduce this information into the model, and in this way **learn and improve prediction**. Specifically, every node in the RNN model functions as a memory cell, continuing the computation and implementation of operations. In case of incorrect predictions, the system can autonomously learn and refine its prediction during backpropagation. However, RNNs have some **difficulty in memorising information on long sequences** due to a common limitation called the vanishing gradient problem.

The **Long–Short Term Memory Neural Networks** (LSTM) are a class of RNN designed to solve this issue thanks to their capacity to learn long-term dependencies in sequential data, by storing time-dependent information and mapping non-linear relationships between inputs and outputs (Motamedi, et al., 2024). LSTM networks map a set of input neurons (also called units) to a set of output neurons through hidden layers, named memory cells (Motamedi, et al., 2024). LSTM links hidden layers across time steps, in a way that allows the output of one time point to arbitrarily enter the next (i.e. output terminal or the hidden layer). The LSTM architecture includes layers of forget, input and output gates, which determine if an entry is relevant and what information should be retained or discarded.

For its characteristics (i.e. long-term information remembering, excellent capacity for self-learning and non-linear fitting), the LSTM has become the **most widely adopted model for demand forecasting** among the deep learning models for time series forecasting, especially after the development of open-source libraries and improvements in computational power (Borges & Nascimento, 2022) (Zhang & Li, 2024). This model has been frequently implemented to predict complex dynamic sequences such as ultra-short-term industrial power demand, highly fluctuating product demand, or hourly natural gas demand (Borges & Nascimento, 2022).

In the healthcare sector, it has been adopted by many authors to model data regarding the **COVID-19 pandemic**, for example, to forecast number of new cases (Yan, et al., 2020), predict trends and prevent the outbreak in Canada (Chimmula, 2020) and forecasting the confirmed cases at national and province levels in Iran (Azarafza, 2020). In a study by (Koç & Türkoğlu, 2022) LSTM was applied for the first time to forecast the need for respiratory equipment and ICU beds during the COVID-19 pandemic. Specifically, the authors developed a model in three main steps: data normalization, multilayer LSTM network and dropout-dense-regression layers. To prove the effectiveness of the model, an experimental study based on COVID-19 data collected in Turkey was conducted. The timeframe covered 77 days, of which 68 days were used to feed the input layer of the system and train the model and the last 9 days of data as a test database.

Despite the vast benefits of LSTM networks (limited amount of data needed, low sensitivity to retraining, capture of nonlinear relationships and long-term correlations), these methodologies have substantial drawbacks such as complexity and lack of interpretability, which require a **high level of expertise** in the ML area since **poor training can lead to incorrect results**.

This model has also been used in **combination** with others to further **improve its accuracy**. Zandavi et al., (2020) for instance, combined LSTM with dynamic behavioural models to predict the number of COVID-19 cases and deaths. The performance of this hybrid model resulted higher than the LSTM alone in case of limited availability of data.

Zhang and Li (2024) combined a support vector regression (SVR) prediction model optimised by a genetic algorithm (GA) with bidirectional long-short-term memory (BILSTM) to estimate the **demand for ICU resources** based on the prediction of the increase in the number of critically ill patients in the short term.

Finally, in the context of vaccine demand prediction, Hu et al. (2023) showed that LSTM and RNN were less accurate than GRU. However, the authors explored also ML tools for conducting sentiment analysis to understand the attitudes of online vaccine reviews and, thus,

improve vaccine quality. Results showed around 80% accuracy of trainable BILSTM deep learning model for sentiment analysis of vaccine reviews, indicating that this can be an effective supporting tool.

Deep Learning - Gated Recurrent Unit Networks (GRU)

The Gated Recurrent Unit (GRU) networks are a further evolution of RNN and can be considered a simplified version of LSTM. Despite the structure being similar, they are characterised by having two gates, one to reset information and one to update it (Dey & Salem, 2017). This simplified structure contributes to making the model easier to be trained and used with respect to LSTM. However, the comparison of their performance is currently under discussion.

Hu et al., (2023) proposed a **vaccine** supply chain management system that combines blockchain, the Internet of Things (IoT), and ML to improve performance by increasing transparency, product quality and demand prediction, contributing to reducing expiration. Specifically, RNN, LSTM, and GRU ML models were explored to train data to predict demand. Their performance has been evaluated through the ability to forecast the U.S. influenza vaccination volume from 1980 to 2020. Comparing the errors of the different models analysed, GRU estimates had a higher level of accuracy than LSTM and RNN (average annual prediction error of influenza vaccine within 3%). As a result, the model combined with blockchain and IoT technologies has been used in situations of high uncertainty such as the COVID-19 pandemic, also providing considerable cost savings to the vaccine supply chains.

Borges & Nascimento (2022) included GRU among ML models to be analysed for **ICU forecasting** during COVID-19. Results indicated that the performance of the model was higher (lower Mean Average Errors) with respect to many other well-established methodologies such as ARIMA, RNN or RFR. However, in this study, Prophet and LSTM models, as well as their combination, proved a higher level of accuracy.

Deep Learning - Adaptive neuro-fuzzy inference system (ANFIS)

Adaptive neuro-fuzzy inference system (ANFIS) integrates nonlinear artificial neural networks (ANN) and fuzzy logic systems to estimate nonlinear functions under uncertainty, thus improving learning and prediction capability (Shafiekhani, Namdar, & Rafiei, 2022). The ANFIS model has been extensively utilised for forecasting due to its superior ability to maintain the power and accuracy of its predictions despite uncertainty and environmental changes, making it especially valuable during crises (Shafiekhani, Namdar, & Rafiei, 2022) (Zivkovic M, 2021).

For instance, Shafiekhani et al., (2022) used the ANFIS model to predict the **number of COVID-19 new cases** over time in Iran. ANFIS model had better predictive power with fewer prediction errors compared to other models considered. The study used data on daily COVID-19 cases admitted in hospitals, including the emergency departments and IC, over a year. (Ardabili, 2020) also investigated several outbreak prediction models used during the **COVID-19 pandemic** to make informed decisions and adopt relevant mitigation measures. Among the ML approaches analysed in the study, ANFIS was identified as the most effective methodology together with multi-layered perceptron (MLP).

Ensemble models

Ensemble forecasting models are techniques that **combine and integrate different prediction models** to overcome specific limitations and enhance the accuracy of output prediction. For instance, Goic et al. (2021) used an ensemble approach to predict the number of ICU beds during the COVID-19 pandemic for different regions and consider a time horizon of 14 days ahead. Specifically, they integrated several autoregressive and ML models to capture different components of the evolution during the outbreak. The forecast outputs of the different models were combined using a trimmed mean ensemble. This ensemble model proved a good accuracy in forecasts, achieving average prediction errors respectively of 4% and 9% for one and two-week horizons.

Ensemble methods also include some famous prediction methodologies such as Random Forest Regressor or Boosting algorithms, as explained below.

Ensemble model - Random Forest Regressor (RFR)

The Random Forest Regressor or RFR is a **ML ensemble model** based on the use of decision trees (Motamedi, et al., 2024) (Hariharan, et al., 2020) (Iwendi, 2020). Indeed, random forests are classified as an ensemble method since it is a model composed of several decision trees that combined can mitigate overfitting issues. In this way, the model becomes more flexible and powerful with lower variance while retaining the simplicity and interpretability of decision trees. The benefit of this model is that it can **capture nonlinear relationships between predictors** while remaining easily interpretable.

In the literature, Hariharan et al. (2020) trained and applied this model to **forecast biweekly vaccine utilization** at health facilities in Arusha and Tanga. The study highlighted that RFR proved a better performance than the other five frequently used ML algorithms (i.e., regularised linear regression, SVM, k-nearest neighbours, autoregression-based univariate time series models). Indeed, before selecting RFR, the authors considered a wide range of commonly used ML algorithms, including RNN, building exploratory models for comparison. However, these failed to yield comparable performances and gave largely uninterpretable predictions. The data used in this study included a total of 32 features in the ML model, including key vaccine utilization determinants based on vaccine nature, health facility features and catchment population around each facility. According to the comparison between models performed by Motamedi et al. (2024), if several data variables are available to implement a model, the RFR methodology (as LASSO and LSTM) can forecast the demand of platelets with high accuracy, even if a small amount of data is available and without frequent retraining. In addition, RFR models maintain interpretability while also having the ability to capture nonlinear relationships. Even though RFR has many advantages over other models, just like the others, according to Lee et al (2023), it cannot handle mixed inseparable data or imbalanced data.

Ensemble model - Boosting algorithms - XGBoost

Similarly to RFR, boosting algorithms are ML ensemble methods that combine several weak prediction models (as decision trees) to create an improved prediction model. The model builds complexity by iteratively adding models, with errors that are corrected by the subsequent one, until the training data is accurately predicted or reproduced by the model. This type of algorithm is the one used by several methodologies such as Extreme Gradient Boosting (XGBoost), Light Gradient Boosting Machine (LGBM), and Category Boosting (CatBoost) (Kwon, et al., 2024). Specifically, XGBoost is frequently used in demand forecasting due to its simplicity in implementation, high accuracy, speed, and scalability (Pall, et al., 2023).

For these characteristics, XGBoost has been used to forecast demand for **red blood cell transfusions**, which is subject to long-time dependencies, changing patterns in seasonality, and nonlinear effects of clinical factors. Specifically, (Li, et al., 2022) combined XGBoost in a hybrid model together with Seasonal and Trend decomposition using Loess (STL), a widely used efficient univariate time series model, to enhance the accuracy of the prediction. The study leveraged on TRUST database, including all patient data across four hospitals in Hamilton, Ontario, for a total of 227,944 transfusions for 40,787 patients from 2012 to 2018. The aim was to combine daily forecasting with inventory optimisation to develop a comprehensive strategy for hospitals. Similarly, Kwon et al. (2024) used XGBoost in an ensemble model, combining it with LGBM and CatBoost. **blood demand** prediction in Korea. This study was based on open public data related to the national blood supply and information related to blood transfusion in a medical institution on a 2-year basis.

XGBoost model has also been applied to predict **medicine shortages** in Canada by Pall et al (2023). The authors highlighted how the model was accurate in predicting medicine shortage class at 69% (none, low, medium, high), one month in advance. In addition, this was done by

avoiding the need for access to any inventory data from pharmaceutical manufacturers since the estimates were based on sales data from 22 Canadian pharmacies and historical medicine shortage data. It is however important to note that MCM demand prediction, particularly during a pandemic response, poses different and more complex challenges than blood demand prediction since the latter allows for continuous (daily) capturing of demand usage over seasonal and disruptive events fluctuation for the past few decades. MCM demand data instead, is sparse and often non-existing until after the emergency.

Table 3. Summary of the relevant AI-driven demand forecasting methodologies from literature review findings

Object of forecasting	AI-driven methodologies	References
Disease outbreak - COVID-19 cases, hospital admission, ICU	SVR	*Parbat & Chakraborty (2020)* *Shafiekhani et al., 2022*
	LSTM	*Shafiekhani et al., 2022*
	LSTM+ SVR	*Zhang et al., 2024*
	ANFIS	*Zivkovic et al., 2021* *Shafiekhani et al., 2022* *Ardabili et al., 2020*
	Phopet + LSTM	*Borges et al., 2022*
	Ensemble	*Goic et al., 2021*
	Random Forest	*Iwendi et al., 2020*
Blood and platelet	Prophet	*Motamedi et al., 2024*
	LASSO	*Motamedi et al., 2024* *Schilling et al. 2022*
	ANN	*Fanoodi et al. 2019* *Khaldi et al. 2017*
	LSTM	*Motamedi et al., 2024*
	LASSO + RNN + LSTM	*Schilling et al., 2022*
	Random Forest	*Motamedi et al., 2024*
	Ensemble (XGBoost + LGM + CatBoost)	*Kwon et al., 2024*
	XGBoost + STL	*Li et al., 2022*
Vaccination	RNN - LSTM	*Hu et al., 2023*
	GRU	*Hu et al., 2023*
	Random forest	*Hariharan et al., 2020*
Pharmaceutical products	ANN - LSTM	*Rathipriya et al., 2023*
Medical equipment	LSTM	*Koç & Türkoğlu 2022*
Medicine shortages	XGBoost	*Pall et al., 2022*

3. Data collection and processing

After the appropriate methodology has been selected, data need to be collected and processed to ensure the robustness of the results.

As anticipated, the **volume of data available** plays a critical role in model selection since the performance of different methodologies also depends on data availability. Mediavilla et al. (2022) highlighted that most AI methods work with **low-volume data** due to challenges in accessing real data and technical limitations in terms of software and hardware. Authors classified different high-performing AI methods used for demand forecasting based on data dimensionality, time horizon and data volume. Most models considered in this analysis are classified as low-data volume models, including XGBoost and Random Forests. However, the amount of data required depends not only on the type of model selected but also on other characteristics such as **time horizon** and the **number of variables** included. For univariate models, the data need is simple and linear to the length of the time horizon since only one variable is needed for each time stamp. When combining LSTM with random forests in a multivariable model, low data are needed for short-term forecasting while a medium amount is required for the medium-term.

Discussing the quantity of data needed, it is important to underline that the AI model needs to be trained using a relatively high amount of data (i.e. **training dataset**), and additional data needs to be used to evaluate the forecasting performance. (i.e. **test dataset**). Although there are some AI/ML interpretable models that were designed and work well in practice training on only a small set of data (e.g. vaccine immunogenicity prediction that accelerates vaccine design and facilitates target delivery (Querec, et al., 2009); where their AI/ML results were instrumental to the eventual world-wide clinical trial of the Malaria vaccines (Kazmin, et al., 2017), Lee, et al., (2016)a); the majority of the AI models cannot be trained properly without a significant amount of data. It is common practice that around 70% of the dataset is used to train the model and the rest as validation and test data (Rathipriya, et al., 2023); (Hariharan, et al., 2020)Motamedi et al. (2024) provided useful insights into the different performance of models with the same forecasting objective, applying different **training window sizes** (two years and eight years of data) and retraining periods. Overall, the results indicate that while univariate models can benefit from a larger training window size (ARIMA performance improves by around 20 comparing the two, but this tends to be overfitting and hence creates uncertainty regarding its predictability), the performance of the multivariable models is not sensitive to training window (i.e. similar performance with both training windows), meaning that multivariable models have robust performance with different data volumes. This highlights the trade-off between model complexity versus performance robustness.

As regards **retraining frequency**, all the models benefit from retraining more frequently but with different sensitivities (univariate models benefit more from frequent retraining) (Motamedi, et al., 2024).

Data availability is key to effectively train all ML models. However, in real-world scenarios, data are dispersed over different organisations, and **privacy concerns** are a major barrier to data sharing and collaboration, as also discussed in Section 2.4 (Li, et al., 2021). Indeed, as mentioned by Li et al., (2021), data of different organisations, e.g., hospitals, can become “data islands”: isolated and limited sources of data that cannot effectively be used to train a high-quality model and achieve good predictive accuracy. **Collaboration** can provide significant benefits in this regard, but **data sharing** is particularly challenging due to policies and regulations, ethical reasons or issues related to business competition and confidentiality (Li, et al., 2021) (Li, et al., 2020). Finding ways to enhance collaborative research among countries is even more crucial when facing global health emergencies like COVID-19 (Li, et al., 2021). At the same time, also privacy issues are more significant when considering international collaboration (Li, et al., 2021).

For this reason, the literature increasingly highlighted **Federated Learning** (FL) as a key approach to mitigating many of the systemic privacy risks and costs resulting from traditional centralized ML (Kairouz, et al., 2021). Indeed, FL is a learning paradigm where statistical models are trained by multiple parties collaboratively without exchanging raw data but model updates, e.g., gradient information (Li, et al., 2020) (Li, et al., 2021).

Usually, a **central server** (e.g. service provider) coordinates this collaborative effort in training the model, keeping the data decentralized. Large companies or organizations can have this role, but in some scenarios, a powerful central server may not be available or desirable, and it can even constitute a bottleneck. For this reason, also **fully decentralized learning** can be developed to replace communication with the server with peer-to-peer communication between individual clients (Kairouz, et al., 2021).

The literature explores FL paradigms and classifies them according to several criteria.

For instance, based on how data is distributed over the sample and feature spaces, FL systems are typically categorized as **horizontal or vertical**. In horizontal FL, the datasets of different entities are similar in terms of features, but there is a limited intersection on the sample space, while in vertical FL, samples are similar but there are minimal common features.

In general, the literature agrees that FL constitutes a **critical step towards protecting data**, enabling the collaborative training of ML models among different organizations (Li, et al., 2021). With this purpose, **several applications** of FL in the healthcare sector have been studied, including not only ML models for demand forecasting (e.g., in combination with LSTM for blood forecasting (Wei, et al., 2023)) or risk prediction (Zheng, Kong, & Brintrup, 2023), but also in combination with blockchains to overcome some FL limitations and further boost security and privacy (Myrzashova, et al., 2023).

The **quality of data** also plays a critical role in ensuring appropriate forecasting results. Dataset creation always requires data **pre-processing activities** to improve demand forecasting results integrity. Many methodologies may be applied to the data to ensure they are robust, consistent, and ready to be used, such as data cleaning i.e. handling missing values, correcting outliers, standardising scales, and removing duplicates (Kwon, et al., 2024). Among these, handling of missing data remains a major challenge. Some authors applied dimension reduction to eliminate unnecessary dimensions to improve the accuracy of the forecast or clustering to group products with similar properties, although many of these techniques lead to unimplementable solutions (Mediavilla, Dietrich, & Palm, 2022). Hence, feature selection remains critical as it ensures the interpretability of the AI models. Healthcare data are usually considered to be noisier than the ones from other sectors, increasing the difficulty in applying AI and ML technologies, as explained in Section 3 (National Academies of Sciences, 2017).

In terms of **data dimensionality**, i.e. **number of attributes** included in the model, **multivariable models** supported by AI, such as LSTM or RF, tend to have a higher performance compared to univariate ones (Motamedi, et al., 2024).

The **variables** that could be adopted as **predictors** for demand forecasting depend on the type of products and the setting. Specific statistical methodologies exist to select the appropriate predictors that can contribute more to improving the accuracy of the forecasting model (e.g. LASSO or feature importance technique) (Kwon, et al., 2024); (Motamedi, et al., 2024). It is important to underline that the variables included in the models significantly differ based on the **object of the forecasting** and among models with similar objectives. In general, while the adoption of common data standards would facilitate comparisons across multiple models adopted with the same forecasting objectives, this may not be feasible since these models may still have different data requirements.

For **platelet demand forecasting** Motamedi et al., (2024) had access to high dimensional data (more than 100 variables) including data on blood inventory (product name and type, received date, expiry date), patient characteristics (age, gender, patient ABO Rh blood type), transfusion location (intensive care, cardiovascular surgery, haematology) and laboratory test results (platelet count, haemoglobin level, creatinine level, and red cell distribution). Additionally, they also calculated new predictors based on those data. Using LASSO regression, they selected 29 clinical predictors to train the multivariable models. In doing this selection also, correlations among variables need to be considered since they affect the

performance of the model. To note, however, in doing so they created depending on variables not suited for predictive modelling

To forecast **medicine shortages,** Pall et al (2023) used sales data from 22 Canadian pharmacies and publicly available historical data on shortages to build a dataset that included three major sections: medicine information (e.g., drug identification number (DIN), drug classification information, DIN strength and common name) Monthly Dispensed Data (e.g., days of supply, number of patients, prescription counts and sales quantity for each DIN by month and year) and Top DINs and Shortages (info on most-supplied medicines and indicators of shortages and their timeframes).

When considering **epidemics forecasting**, the variables included usually relate to case counts (confirmed, probable, suspected), mobility of individuals and populations, host susceptibility (immunization coverage data), environmental susceptibility (climate temperature, precipitation,.), healthcare capacity, and spatial demographic data as population density (Desai, et al., 2019). Zhao et al., (2024) outlined potential data sources for collecting information on these types of variables including health records travel and mobility data, genomic and pathogen data for disease prediction, satellite imagery, news media and social media trends.

In the EU context, during the COVID-19 pandemic, there had been a **high heterogeneity in the models** adopted by Member States, including the use of different data sources, data fields, level of granularity of outputs and time period, as underlined by the EU Executive Steering Group on Shortages of Medicines Caused by Major Events, in its recommendation on demand forecasting (EMA, 2021). Data differences across countries would lead to more sophisticated models to be built for those countries with more data available. Defining a set of common data elements and identifying variables that are essential for forecasting allows to foster harmonisation in the EEA/EU and, thus, comparability of predictions. Considering this, the selection of common variables may be based not only on consideration of predictors' capacity to contribute to forecasting accuracy but also on the overall level of availability of the information in the different countries. Nevertheless, a common data element should be set to encourage countries to collect as much as they can. Having different countries with different variables allows researchers to objectively compare models built with more variables and evaluate objectively if the extra variables benefit the model. Ultimately, this allows the generation of a set of good variables that are important to collect.

The recommendations outlined that possible common variables to be considered for demand forecasting during the COVID-19 pandemic are the projected number of COVID-19 patients in ICUs, distinguishing between those who require mechanical ventilation and those who do not, the average consumption of the medicines for each patient group, the expected demand for non-COVID-19 patients (from normal time historical data), confirmed deaths as it indicates estimates of infection, and how current inventory levels, whether excess or planned for replenishment, may influence future demand.

Once predictors have been selected and the training dataset has been built, the model must be **trained** according to specific learning algorithms. Finally, the model **forecasting performance must be evaluated** for instance by computing a specific measure (e.g. the mean square error (MSE), the Root Mean Square Error (RMSE), the mean absolute percentage error (MAPE), Symmetric Mean Absolute Percentage Error (SMAPE) or the sum of the squares of the errors).

Box 4. Considerations from the experts: AI-driven methodologies for demand forecasting

Highlights resulting from Experts' consultation

The experts consulted from academia and some from industry confirmed to be familiar with most of the methodologies identified through the literature review and listed above. Experts underlined that traditional statistical models such as ARIMA are well established, while ML models have more recently been explored.

It emerged that there is no best methodology to be applied to demand forecasting of MCM since different models proved to perform accurate estimates and their efficacy depends on many factors. One expert underlined that the selection of the most appropriate model mainly depends on methodological assumptions and data characteristics.
For instance, some models may not apply to time series data, characterised by correlated observations. Deep Learning methodologies require large datasets to be trained effectively, and in the context of healthcare supply chains, these are very often not available.
Among the methods listed above, **ensemble** models may be very effective since they allow the combination of different types of ML models to overcome specific limitations. Random Forest is a foundation of the boosting algorithm but was mentioned as probably the weakest ensemble approach as it only uses decision trees and does not combine different types of ML approaches. It was said to also be characterised by some challenges in terms of overfitting and is not particularly adequate for time series data, especially with a small sample size.
XGBoost was mentioned by some as a state-of-the-art model in ML, especially in combination with traditional time series methods (hybrid model). However**,** another expert outlined XGBoost limitation as not working well on imbalanced and mixed groups.
An expert suggested **Holt-Winters** with novel sparsity and multi-seasonal approach as a very powerful optimisation tool that was used in the context of vaccine supply chain and demonstrated a very high prediction accuracy thanks to its dynamic capability. Indeed, the expert underlined that forecasting demand for MCM differs from other products because, in addition to population distribution and historical data, it requires dynamic knowledge of disease progression (epidemiological modelling), combined with human and societal behaviour (simulation and AI) and logistics/inventory information (ML, optimisation and simulation). This was defined as the essence of system modelling.
Experts also mentioned the importance of **reinforcement learning (RL**) algorithms to predict the demand of MCM under uncertainty. Through RL, AI can run many simulations of potential scenarios and to learn how different resources and variables are related to each other to discover complex interactions.
Currently, research is focusing on applying new approaches such as federated learning methods to demand forecasting and combining ML demand forecasting with inventory policy to optimize its management, thus reducing waste and avoiding shortages.
Finally, there is increasing interest in building more generalizable and transferable models as well as improving standardization techniques able to mitigate the impact of the difference in data collection across different types of settings. The **interpretability** of models is also increasingly discussed as a key driver in model selection for their practical implementation.
Advanced statistical forecasting methodologies have been applied to forecast demand during health emergencies (mainly the COVID-19 pandemic) and normal times (e.g. forecasting of antibiotics demand based on historical data to identify trends and prevent potential shortages). US stakeholders consulted mentioned that AI-driven methodologies have been applied by the US government in the healthcare and defence sector. In the EU, public sector demand forecasting is mainly based on historical data of the previous year's consumption or simple methodologies that very often does not include other additional variables. Quantitative data or comprehensive information on epidemiological projections are not included since are generally unavailable. Only one member state is currently using some form of AI for forecasting.

However, there is growing interest and awareness of the potential benefit of implementing AI for demand forecasting in terms of mitigating shortages and supply issues. Moreover, a procurement authority is working to include AI in forecasting, to predict changes in consumption trends but also avoid legal/financial issues related to underestimating consumption and reduce the environmental impact of wastage caused by an overestimation of the demand.

Some wholesalers and private companies are adopting simple AI tools, and consultancy and service companies are increasingly offering services in this regard. Additional insights on the state of the art of real-world applications of AI for forecasting are provided in Box 8.

2.2 AI-driven methodologies for supply chain monitoring

Supply chains are composed of multiple stages carried out by different entities across the globe, interconnected through physical, information, and financial flows (Serdarasan & Tanyas, 2012). Consequently, reaching a high degree of **control over the supply chain** requires clear

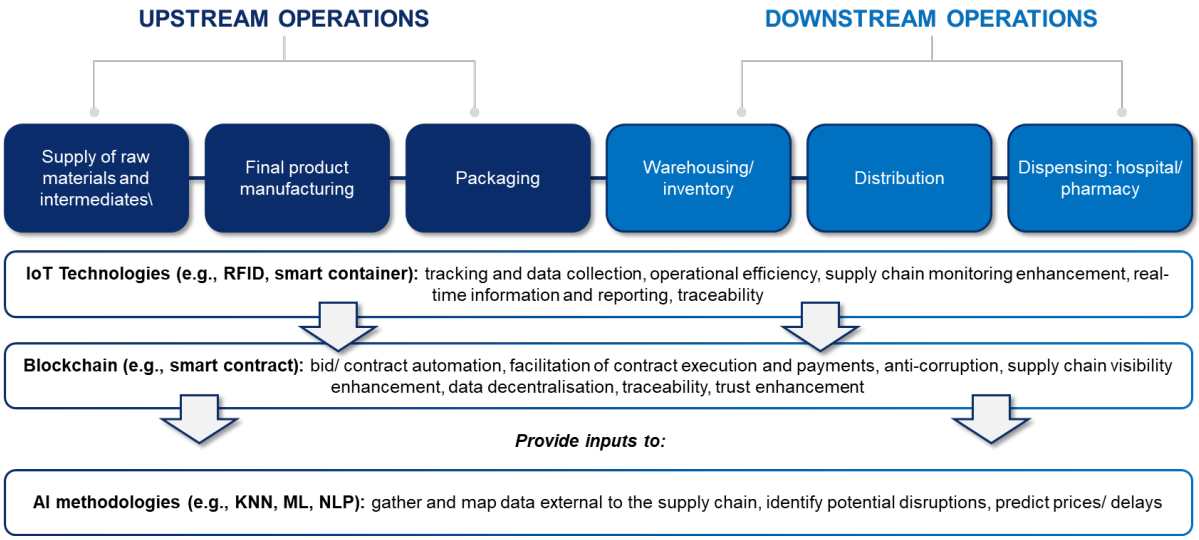
visibility into all the flows of products over these different steps, the level of stocks and of other relevant market dynamics that can influence the supply process.

Supply chain monitoring involves using a set of tools with analytical capabilities that can track, trace and assess the performance of supply chain operations. Moreover, **mapping the supply chain** is essential to better understand all the components and flow of materials and information in the upstream and downstream operations of the supply chain, as well as identify vulnerabilities and potential risks (Norwood Bailey & Peel, 2021). In this sense, monitoring and mapping the supply chain is a critical part of effective supply chain management (SCM) as it helps identify and remedy issues in global supply chains that could impact both the upstream and downstream operations.

Moreover, supply chain monitoring has a bi-directional relationship with **demand forecasting**. On one hand, demand forecasting is at the basis of supply chain monitoring and planning as it influences critical aspects (such as warehouse stocking levels and transportation scheduling) of the logistics and inventory. On the other, improving visibility and data collection through better supply chain monitoring can provide useful inputs to inform more robust demand forecasting systems. For example, real-time data on order fulfilment rates, shipment delays, and customer demand patterns can be used to refine future forecasts, enabling more adaptive and responsive supply chain operations.

Figure 2 provides a simplified visual representation of a medical product supply chain and the main benefits innovative technologies (i.e. IoT, Blockchain and AI) can provide to improve monitoring.

Figure 2 IoT, blockchain and AI methodologies in a medical product supply chain



***Source**: Author's own elaboration based on literature review*

In a supply chain, critical aspects to monitor include **inventory levels, lead times, procurement data, supplier performance metrics** (e.g., on-time delivery rate, cost competitiveness, order accuracy rate), **transportation data** (e.g., available shipping routes, real delivery times), **demand forecast quality (when compared to actual demand), and accuracy rates.** Jackson et al. (2024) argued that external factors, such as adverse **weather or geopolitical events**, that may impact the flow of goods are also important to consider Supply chain monitoring allows identification, tracking, tracing and assessment of data and data patterns in the supply chain. It helps to identify opportunities for cost savings, **demand forecasting** (see Section 2.1) inventory management, and supply chain management process improvements (Mele, Musulin, & Puigjaner, 2005). The main supply chains of MCM considered

in the literature and in this Study are those of blood, vaccines and COVID-19-related products and services. To achieve continuous and accurate monitoring, there must be a continuous stream of high-quality, granular data (e.g. data that comes directly from the supplier or the customer but also adjourned data from external sources such as news items detailing events that could impact the supply chain) throughout the entire supply chain. Many healthcare systems have good hospital **inventory tracking**, especially on medications and critical equipment for patients. However, there is a lack of more granular supply chain data in the upstream operations (e.g. manufacturing), where many raw materials are sourced from international suppliers (Finkenstadt & Handfield, 2021). Good data tracking and the associated data visibility allow for demand estimates and provide a means for re-supply at the right time.

The **digitalisation** of the supply chain offers businesses and governments a set of additional approaches to complement existing methods to monitor and map supply chains (Kosasih & Brintrup, 2022) and attempts to solve problems such as limited **data visibility due to lack of digital infrastructures** along the supply chain, **reluctancy to share data** by the main actors and the data being unstructured (Hu, Xu, & Liu, 2023).This digitalisation is allowing rapid evolution of SCM environment, particularly due to the critical role that industry 4.0 enablers are playing (Younis & Wuni, 2023). These technologies are being employed by researchers and industry practitioners to enable the fast collection and real-time update and integration of data, automation of SC mapping and consequently resolving several pain points in SCM at all stages, improve operational efficiency, manage demand volatility, tackle cost fluctuations, and make evidence-based decisions (MacCarthy, Ahmed, & Demirel, 2022) RFID technology is now extensively used in hospitals to improve SCM effectiveness by increasing the ability to track and locate equipment. For instance, Coustasse et al., (2013) found that the implementation of radio-frequency identification (a technology that uses radio waves to identify objects or people) in the hospital supply chain to track medical equipment and supplies resulted in reduced costs and overall efficiency.

Inventory management, a critical element of the downstream operations of the supply chain, involves tracking inventory from manufacturers to warehouses and from these facilities to the point of sale. It requires inventory visibility, knowing when to order, how much to order and where to store stock (IBM, 2024). To achieve this, it is imperative to have real time monitoring and thus visibility of the products, which can be achieved through the use of blockchain technology (disparate parties can be connected through a fast and immutable record of all transactions) and IoT (data from the sensors provides insight into inventory location and status). AI can then uses tracking data to forecast demand and estimate appropriate levels of inventory make inventory monitoring more accurate) (IBM, 2024).

The dynamic nature of the global market necessitates a sophisticated approach to supply chain management. A study by McKinsey & Company (2017) found that the realisation of the full potential of supply chain digitisation is often hindered by technological and managerial gaps and challenges. Consequently, businesses must adopt innovative strategies to foster resilience and competitiveness.

Emerging technologies are reshaping supply chain operations, with a growing emphasis on **automation, artificial intelligence, blockchain, and the Internet of Things**. These advancements facilitate enhanced transparency, traceability, and predictive analytics, ultimately driving operational efficiency and customer satisfaction.

C. Blockchain, Internet of things (IoT) and Big Data

The first instance of **blockchain technology** can be found in Satoshi Nakamoto's whitepaper (2008). Blockchain, one of the 4.0 technology enablers, is underpinned by cryptographic security, distributed ledger principles and a decentralised architecture. Saberi et al. (2019) claim that its core attributes - information immutability, transparency, security, auditability and decentralisation, make it a powerful tool for SCM (2019). Indeed, evidence shows that SCM makes extensive use of this technology as it brings about numerous benefits, such as safe

and efficient distribution of vaccines (Hu, Xu, & Liu, 2023), automate procurement contracts or enhancing overall visibility (Abeyratne & Monfared, 2016). For instance, blockchain can increase visibility of the supply chain by generating an established network of suppliers that constantly communicate through instant ordering, delivery and payment notifications, thus allowing supply decisions to be made based on daily or hourly updates (Finkenstadt & Handfield, 2021). Blockchain applications in supply chains often work together with other cutting-edge technologies (Wamba, Queiroz, & Trinchera, 2020). For example, blockchain can be combined with the IoT, big data analytics and, more recently, **ML** (Garba, et al., 2021).

From a blockchain-centric perspective, the technology offers potential solutions to challenges posed by AI. Solaiman and Dimitropoulos (2024) highlight how **blockchain can enhance AI's compliance with regulatory frameworks**, particularly in business and corporate transactions.

Conversely, AI can contribute to overcoming blockchain's inherent limitations. AI, with its ability to process vast amounts of data, can optimise blockchain operations and facilitate decision-making (Saberi, et al., 2019). AI-blockchain integration has, in fact, the potential to deliver more robust outcomes as this synergy enables data sharing, analysis, and prediction without intermediaries, streamlining supply chain processes. Zhang et al. (2024) further, demonstrate how AI can extract valuable insights from blockchain data to inform strategic decisions.

Another Industry 4.0 technology is **The Internet of Things (IoT**), a scalable network infrastructure of interconnected devices that facilitates real-time data exchange across the supply chain and enables real-time data exchange both across upstream and downstream supply chain; providing great support in the context of Industry 4.0. Its ability to connect and collect data from various devices and sensors offers unprecedented visibility into complex supply chain operations (Manavalan & Jayakrishna, 2019). As supply chains grow increasingly intricate with multiple stakeholders, IoT becomes indispensable for monitoring and managing end-to-end processes (Pundir, Jagannath, & Ganapathy, 2019). While IoT adoption is gaining momentum, many supply chains still rely on manual data handling. By automating data collection, IoT empowers businesses to make informed decisions and enhance operational efficiency. Technologies like **Radio Frequency Identification (RFID)** exemplify the practical applications of IoT in supply chain visibility. By connecting cloud data centres, edge computing technologies, and industrial control networks, RFID provides sufficient computing capability and in-time manufacturing services and further enables more efficient and flexible production. RFID and IoT can be combined to create solutions that monitor and manage objects in real-time (Ren, Jia, Laili, & Huang, 2023).

The **convergence of blockchain and Internet of Things** (IoT) holds significant potential for biomedical operations within healthcare. IoT devices, such as sensors and gateways, can capture real-time patient data, which is subsequently transmitted to a blockchain for secure storage and sharing (Nguyen, Ding, & Pathirana, 2021). As mentioned above, blockchain technology facilitates decentralised communication, eliminating the need for a centralised authority. In healthcare, decentralised communication happens between IoT devices and IoT users (including medical professionals and patients) and each IoT activity, such as sensor data collection, triggers network operations within the blockchain. This process results in an updated public ledger accessible to all network participants, while simultaneously obviating the role of a central governing entity.

Finally**, Big Data and predictive analytics (BDPA)** encompass a suite of techniques designed to extract value from massive, complex datasets characterised by high volume, velocity, and variety (Duan & Xiong, 2015). This technology addresses critical challenges in data management, including capture, storage, transfer, sharing, and analysis (Erevelles, Fukawa, & Swayne, 2016). When applied to supply chains, BDPA can significantly enhance performance by improving visibility (Barratt & Oke, 2007), building resilience and robustness, and driving overall organizational performance (OP) (Waller & Fawcett, 2013). Alotaibi & Mehmood (2018) highlight that healthcare providers allocate a substantial amount of resources

to high-cost medical equipment and pharmaceuticals. Optimising supply chain processes is therefore imperative to enhance efficiency and reduce expenditures. Moreover, the vast quantities of data generated by medical equipment present a unique opportunity to leverage big data analytics. By harnessing the power of these insights, healthcare providers can refine **forecasting models**, make informed decisions, and ultimately improve the overall performance of their supply chains.

D. Artificial Intelligence and Machine Learning

Blockchain, IoT and BDPA provide the data basis for AI solutions to be developed. As more data become available through their use, the use of AI in SC management increases in parallel. AI is being applied by more and more companies at a global scale (Kosasih & Brintrup, 2022) through a variety of methods including decision tree, natural language processing and artificial neural networks (Nas & Koyuncu, 2019)

Literature findings show that AI has been used in supply chain as early as 2009 (Valluri, North, & Macal). Initially, AI methodologies and tools were utilised to **automate simple tasks**, for example using **Robot Process Automation (RPA).** RPA can perform tasks such as entering data, processing transactions, and managing records with speed and accuracy unmatched by human counterparts. In warehouses, for example, RPA can automate the sorting and packing processes, increasing the speed of operations, and reducing operational costs. As technology advanced and with the advent of COVID-19, **AI's role expanded** to include **mass surveillance** (both of population health data and MCMs production and distribution data) and **decision-making/ strategic planning** (Hu, Xu, & Liu, 2023).

In supply chain management, AI can be used to increase end-to-end visibility, forecast disruptions and identify solutions in a fast and accurate manner often through the creation of monitoring platforms that link production, production, logistics and warehouses (De Giovanni, 2019). The application of AI and ML specifically for supply chain monitoring in MCM is still in the development stage, mainly because it largely depends on the amount and quality of the data produced across the entire supply chain, which, as discussed above, are still limited. Moreover, companies across the SC consider their data as proprietary and are reluctant to share it (Schöpper & Kersten, 2021).

Many digital technology companies claim to be using AI and ML to improve and optimise supply chain mapping, and consequently supply chain monitoring (MacCarthy, Ahmed and Demirel, 2022). The digital company Everstream for example, is utilising AI and ML for incident monitoring in the supply chain (Everstream, 2024). Using data such as news reports, updates on product shipments, materials, locations and weather insights; they help improve the transparency and reliability of the supply chain. Consequently, they claim to be able to provide their clients with 24/7 early warning notifications on disruptions in the SC (Hayden & Shillingford, 2024). There are also cases of these technologies having governments as clients. Altana AI, for instance, is collaborating with the UK, US and Australian governments to map their supply chain and to optimise national policies on national security and border policy (Altana, 2024). To note, the AI applications outlined above, also depend on the existing quality of the supply chain mapping and must rely on the collaboration across stakeholders to share data (MacCarthy, Ahmed, & Demirel, 2022).

There are also some applications of AI coupled with Industry 4.0 enablers, specifically Natural Language Processing (NLP), Neural Networks and Deep Reinforcement Learning. The company OpenSC for instance, claims to be able to test the sustainability and ethical standards of a supply chain utilising deep learning and Geographical Information System (GIS), as well as improving the traceability of products through blockchain, RFID, and QR codes (MacCarthy, Ahmed, & Demirel, 2022). **Natural Language Processing (NLP)** is defined as “the systematic approach means of a computer that gathers knowledge on how humans use, apply, and understand language” (Perazzoli, de Santana Neto, & de Menezes, 2022) and in the context of supply chain mapping, it has proved useful to analyse data external to the supply chain,

such as news on the web or social media about weather, traffic or natural disasters. The vast majority of these sources are written in natural language; hence NLP and ML techniques prove to be useful to analyse this wealth of data (Schöpper & Kersten, 2021) as well as communications happening across the supply chain (Mah Muam, Skalna, & Muzam, 2022). The UK-based company Versed AI[1] provides companies with the possibility to gain full visibility into their supply chain, also using NLP techniques to analyse public data and extract buyer-supplier relationships, thus allowing companies to get visibility from tier 1 to all the tier-n suppliers. It should be noted that MacCarthy, Ahmed & Demirel (2022) point out some possible limitations of NLP, as these methods are still in development and suffer from low recall (percentage of supply relationships truly captured), particularly due to the absence of large, labelled datasets. Furthermore, they potentially suffer from selection bias as not all buyer-supplier relationships are mentioned in public company documents and websites.

Another AI methodology employed in the downstream operations of the supply chain is **Extreme Gradient Boosting** (XGBoost). Literature shows evidence of its use both in logistics (Zhong, et al., 2023) and inventory management (Singh & Shiny, 2024). XGBoost can be defined as "a scalable, distributed gradient-boosted decision tree machine learning library. It provides parallel tree boosting and is the leading machine learning library for regression, classification, and ranking problems" (NVIDIA, s.d.). When combining XGBoost with Q-learning algorithms, decision-making in a supply chain can be optimised. Indeed, among the advantages of this algorithm is flexibility in handling vast volumes of data and fast execution in processing data and can then be used to predict future demand and prediction (Zhong, et al., 2023).

Another example of AI applications is the AI platform offered by IBM (IBM Sterling Supply Chain insights with Watson) that provide companies (including pharmaceutical ones) with a dashboard with smart alerts, a resolution room with machine reasoning capabilities, digital playbooks, and a Control Tower to accelerate data integration, optimize inventory and logistics (IBM, 2019). Similarly, Accenture claimed to offer an N-tier Supply Chain Navigator that leveraged OpenAI GPT to support procurement managers in analysing supplier network data. The tool offers real-time insights and supports data-driven decision-making with the aim of identifying vulnerabilities in the supply chain, such as suppliers with relevant links to regions affected by conflicts or natural disasters (Accenture, 2024).

One of the most common applications of AI has been for vaccine supply chain management often in combination with other Industry 4.0 enablers. **Traditional regression and classification** ML however require prior knowledge to be effective and struggle to handle complex equipment and noisy data. These limitations make applying traditional AI to supply chain monitoring almost impossible, as the latter often consist of short and imbalanced datasets. To tackle these limitations, Jahin et al. recently proposed a novel **hybrid quantum-classical keras neural network** (NN) to overcome the limitations posed by traditional models and the data unavailability. The model, QAmplifyNet, showed increased accuracy, precision, recall and F1 score (the harmonic mean between precision and recall scores).

However, it must be noted that extensive literature review shows that it is the first known instance of quantum-classical NN implementation in the context of supply chain backorder prediction. Jiang et al. (2024) proposed a **deep reinforcement learning model** for solving supply chain management issues, such as inventory management. Utilising an autoregressive architecture, the model can learn and optimise past trajectories, yielding favourable outcomes. Studies suggest that this problem can be overcome by **combining AI and ML with other I4Es**, such as blockchain, IoT and big data (Hu, Xu, & Liu, 2023).

[1] https://www.versed.ai/

Box 5. Considerations from the experts: AI-driven methodologies for supply chain monitoring

Highlights resulting from Experts' consultation

When presented with the list of AI methodologies for supply chain monitoring resulting from the literature review conducted and detailed in the Section above, a few experts from the industry and academia confirmed their knowledge of them, with one having experience in applying Variable Neighborhood Search Model and another with Neighborhood Search Model and Reinforcement Learning Algorithms. However, many others pointed out that they are not currently used AI.Moreover, one stakeholder from US industry provided additional feedback by stating that it is more relevant to train the models in a suitable way for the context where it needs to be applied than considering AI frameworks in a general and agnostic way.

Several stakeholders from diverse backgrounds (industry, academia and government) did not feel confident to establish the most effective methodology to be used in the context of MCM supply chain, pointing out there are too many variables at play. They suggested to 1. *"Look at the market"* as the best methodology is probably the one most widely used and to always 2. *Consider the uncertainty and time constraints of the MCM supply chain context,* as it is often the case for emergency scenarios, when choosing a methodology.

When asked about whether deep learning methodologies could be effective, one stakeholder from academia pointed out to the models need to be trained with large amount of data, which is often unavailable due to data visibility issues: *"sample size matters".* They also provided insight into their current work, stating they use Mixed Integer Programming (MIP) to do inventory management, which works in a similar way to Neighborhood Search Model, but MIP is the better choice since it is an exact algorithm, whereas Neighborhood Search Model is a heuristic one. Another stakeholder from academia confirmed that Neighborhood Search Model is not ideal to use since there is no guarantee of optimality.

When asked about public authorities' needs (US) one stakeholder from the industry highlighted that visibility is essential before any risk mitigation can begin. For instance, the focus is often on identifying suppliers and understanding the risks within the supply chain, especially in terms of country of origin. U.S. regulations, such as those concerning forced labour, are prompting organisations to closely examine whether materials sourced from regions like China's Xinjiang region are being used in their supply chains. A stakeholder from EU government stated that, from a government perspective, the paramount priority is to be able to map the supply chain.

One stakeholder from **government** (**national level**) involved in demand forecasting and supply monitoring both at national level and EU level stated that regarding both areas the level of adoption of AI both at national and EU level is minimal. The same level of adoption (minimal to none) was identified by **two stakeholders from national government** working for centralised health procurements funds in **two different EU countries**. One of them highlighted that the application of AI tools in their everyday work could brought several benefits, for example: to reduce legal and financial issues related to under/over estimation of products and reduce the environmental impact of waste. Finally, a stakeholder from a fourth EU country's national government mentioned that while AI assists in managing and identifying patterns within vast datasets, as they currently track over ten thousand medicines, for instance, the most crucial analytical decisions remain in the hands of humans. One stakeholder from industry pointed out that to carry out effective supply chain monitoring, big amount of data from outside the network (i.e. third parties' delivery, traffic or weather problems) is needed but often not available. Another industry stakeholder presented their use case which consists of a software which integrates Generative AI methodologies to scan billions of news items from around the world every day to get real time data about potential supply chain disruptions.

One stakeholder from industry working on estimating the right supply of medicine at a centralised level (i.e. for the entire country) stated that their flow of information between hospitals, pharmacies and suppliers is currently extremely accurate and transparent. They expressed **concerns that introducing AI tools could disrupt the trust** established among these stakeholders, as many AI-generated outputs may be difficult to explain. One stakeholder from the industry, dealing mainly with supply chain resilience and risk management, provided insights on the methodologies used:

- At first, they implemented natural language processing (NLP) systems to process news data (to get real time insights on potential disruptions of the supply chain). More recently, they transitioned to a generative AI large language model.
- The XGBoost algorithm is used to predict the future prices of critical materials like gold, tin, tungsten, and tantalum.
- To predict potential delays in purchase order deliveries following supply chain disruptions the K-Nearest Neighbors (KNN) algorithm is employed.

Differently, another stakeholder from the industry explained that certain segments of the supply chain have enough data to support AI analysis. However, when it comes to tracking or identifying incoming shortages, the available data is insufficient to power an AI tool effectively. To address this issue, they have requested their partners to report any real shortages where no alternative medicine was provided. Despite these efforts, there are still clear information gaps.

Regarding the use of Industry 4.0 technologies, stakeholders interviewed confirmed their utility to track products across the supply chain to speed up signing of contracts and to connect in real time objects that are in very different locations. A stakeholder from academia mentioned that the supply chain of most products can be monitored using **IoT** and other advanced techniques. For example, in the case of medicines, sensors in the machinery can track the origin of every active pharmaceutical ingredient (API), as well as details such as medicine sizes, production dates, and any batching issues. **Blockchain** technology ensures that this data is recorded in a tamper-proof manner, providing full traceability. Traditional management information systems collect significant amounts of data, such as the source of components, the machines involved in their processing, and the logistics companies responsible for shipping. Once this data is accessible, it allows for further inquiries into the modelling of various scenarios to predict potential outcomes under different conditions, such as increased demand, conflicts, adverse weather, or supply shortages.

The state of the art of AI real-world applications for supply chain monitoring is summarised in Box 8**.**

2.3 Simulation methods for supply chain decision-making

As previously stated, **simulation methods** can be seen as able to overcome some of the limitations of AI, Big Data and ML as being less reliant on historical data and able to determine causality (Nas & Koyuncu, 2019). According to Tordecilla et al. (2021) simulation methodologies have been used in the design of resilience supply chain networks in the context of numerous vulnerability factors of which the most common are demand, capacity, costs and supply shocks.

Simulations are often used together with **optimisation techniques** as they allow testing of the solutions provided by the latter. Specifically, simulations can be used to solve complex problems characterised by high uncertainty by breaking down the problem into small simulations and comparing them (Tolk, 2022).

In the systematic review by Tordecilla et al (2021), more than half of the papers written on simulation-optimisation applied to supply chains only used theoretical data. Of those who used real-world cases, most of them focused on food supply chain (Saif & Elhedhli, 2016), as well as electronic or automotive (Ding, Benyoucef, & Xie, 2008) and healthcare supply chain (2021) Looking at industry applications, however, there are several case studies of simulations run through AnyLogic being applied by industry to solve several supply chain issues. Examples include AnyLogic simulations used to determine the most appropriate production and distribution policies of vaccines, to study the bullwhip effect (amplification of variability in demand caused by the retailer incorrectly forecasting it) for the automotive industry, or to test operational improvements in the technological supply chain (AnyLogic, 2023).

Specifically to healthcare SC, Melnyk et al (2024). discuss the most common applications of simulations have been on **modelling the impact that different disruptive factors** (e.g. COVID-19 cases, policy interventions, increased demand) **would have on the supply chain of MCM** (e.g. vaccines and PPE) **and healthcare services** (e.g. ICU beds capacity). Regarding data needed to apply the different simulations, this has usually been taken from

already available information, such as literature for baseline scenarios on COVID-19 infection patterns and historical data on demand and usage for the particular product examined. There are five simulation methodologies that have been found to be applied to supply chain monitoring and demand forecasting in healthcare settings: **Monte Carlo, Agent-Based Simulation, Discrete Event, System Dynamics, and Hybrid Simulations** (which bring together different methodologies). The choice of which methods to use should be based on a variety of factors, such as the objectives of the decision-makers and the complexity and components of the system (Currie, 2020). Although some problems are seen to be easier modelled using one specific methodology (e.g. ABS for individually oriented problems), most problems could be solved by more than one methodology (Grigoryev, 2018).

Simulations are closely interlinked with AI tools as the former can help test AI solutions by replicating them in real-life scenarios. Simulations can also help fill data gaps needed to run AI through **synthetic data**. For instance, synthetic data created through simulations can be used to verify the efficacy of new ML technologies as well as to train ML within simulated environments, particularly Deep Learning, when historical and real-time data is missing (Peng, et al., 2015); (AnyLogic, 2023).

Table 4.Overview of simulation methodologies

Simulation Methodology	Description & Application	Reference
Monte Carlo	Can be used to study operational risks rather than disruptive risks in the SC. It is also useful for sensitivity analysis in complex systems with multiple uncertain variables, as it provides a comprehensive view of potential outcomes	Ivanov (2020) Begen et al (2024)
Agent Based (ABS)	Most appropriate to study individuals´ behaviours. This can be applied to look at population-level issues	Kasaie et al (2013) Lee et al. (2021) Lorig et al. (2021)
Discrete Event (DES)	To study operational problems as allows for more dynamicity. Can be applied to test high-impact disruptions in the SC.	Osorio et al. (2017)
System Dynamics (SD)	Used for strategic modelling and to study complex systems. Can be applied to study supply chain disruptions.	Ghadge et al. (2022) Shirazi et al. (2021)
Hybrid/system	Coupled multiple simulation models within a single system – ABS for individuals´ behaviours population modelling; DES for process/operational/logistical modelling, SD for biological disease modelling and overall complex system modelling; along with ML and optimization (mixed integer programming).	Camur et al (2023) Lee et al. (2015) Lee et al. (2021)

A. Monte Carlo simulation

Monte Carlo simulation is a computational technique used to model and analyse systems (Mooney, 1997). It is widely used in the supply chain as it allows for complexity and uncertainty, as well as being relatively easy to use (Jellouli & Chatelet, 2001). Monte Carlo has been applied to understand uncertain supply chain scenarios, in **healthcare planning** as well as **inventory management** (Elsayed, Taha, & Hassan, 2022). (Mikhail, El-Beheiry, & Afia, 2019) used this methodology to test the resilience of different supply chain structures towards disruptions However, according to Jabbarzadeh et al (2016) Monte Carlo is not as extensively used alone as much as it is used in conjunction with optimisation techniques such as Integer programming, to test the performance of the developed optimisation models (Klibi & Martel,

2012). By design, Monte Carlo simulation is computationally taxing but tends to produce highly accurate results.

In health emergency context, a recent study by Begen et al (2024) was found to use Monte Carlo simulation, implemented the model through Microsoft Excel, and applied it in a Canadian hospital to support decision-making and ensure appropriate bed capacity in various COVID-19 scenarios.

B. Agent-based Simulation (ABS)

ABS simulations focus on individuals and their behaviours. Its application is somehow more recent than other methods such as SD and DES, as it was limited to academic studies until around 15 years ago (Grigoryev, 2018). When looking at a population, ABS can predict how the behaviours of individuals and their interactions can influence outcomes at a population level (Currie, 2020). It is defined as a **stochastic method,** as accepting a level of randomness and variability in human behaviour. Generally, ABS has been used to understand how behaviours can impact supply chain vulnerabilities (Lorig, Johansson, & Davidsson, 2021). For instance, ABS has been effective in understanding the impact that different levels of information sharing between suppliers and retailers can have on the resilience of the supply chain when considering different demand trends (Tang & Pan, 2014).

Specifically to health supply chains, it has been suggested as an effective method to model pandemics (Epstein, 2009), particularly for disease management (Kasaie, Dowdy, & Kelton, 2013). A recent application of ABS in the context of shortages of vaccine supply during a pandemic scenario utilises ABS simulation, coupled with ML, optimization, DES and compartmental model (system dynamics model); to model different scenarios of disease outbreak and validate the results of a framework that utilises biological characteristics, human behaviour, hospital resources and intervention operations logistics (Lee, et al., 2021). Considering different vulnerabilities in the supply chain, results identified the optimal vaccination strategy to be a mixed one, where some of the high-risk population is given priority for vaccination followed by non-prioritised population. The inclusion of the simulation ultimately allowed to understand the impact that patients´ individual and social behaviours would have on a shortage of vaccines and choose the best strategy according to different scenarios of disease outbreak development. Such information is important to demand forecasting as it provides the timeline on vaccine consumption, and the re-supply timeline.

C. Discrete Event Simulation (DES)

If System Dynamics focus on the system and the dynamic relationship between its entities, Discrete Event Simulation (DES) see entities as individual objects, and model activities according to each individual element from probability distributions (Brailsford & Hilton, 2001). Similarly to ABS, DES is also defined as a **stochastic method**, as it accounts for levels of unpredictability and randomness. It is thus appropriate to solve more detailed **operational problems** and answer specific questions such as identifying vulnerabilities in the production process, resource allocation and comparing medical interventions (Sterman, 2002). Furthermore, DES allows to consider the dynamic behaviours of items, providing dynamic validity of aggregated data provided by AI (Tiacci & Rossi, 2024). In Tordecilla et al (2021) systematic review, DES was found to be the most common type of simulation methodology applied to supply chain disruptions across different contexts. In a healthcare management context, it could be used to look at a hospital with little contact with the outside world, relative low numbers of patients, to compare different scenarios and to obtain quantitative output such as waiting times and resource utilisation (Brailsford & Hilton, 2001).

A review by Vazquez-Serrano et al (2021) looked at 231 papers applying DES modelling in healthcare and concluded that the methodology is being increasingly used in the sector, particularly to allocate resources, improve time and efficiency in healthcare centres.

D. System Dynamics (SD)

System Dynamics consist of a set of different equations that are solved over simulation time (Sterman, 2002). A System Dynamics method can be either deterministic or stochastic, depending on how the model is designed; while it's often considered deterministic in basic applications, the methodology allows for the inclusion of random variables and stochastic elements to represent uncertainty in real-world systems when needed. In a SD simulation, entities are seen as dynamic as,their state changes continuously. As it allows to study complex, dynamic supply chain systems that include dynamic feedback, time delays and time-dependent behaviours, studies tend to use SD at a strategic level to gain insights into the relationship between different entities of a complex system (Sterman, 2002). For this reason, it is considered the perfect tool to utilise when observing disruptions in the supply chain (Petropoulos, 2022). Ghadge, Ivanov & Chaudhuri (2022) have utilised SD simulation to better understand the ripple effect, the impact that a risk has when cascading on all the nodes of the supply chain network. Modelling different scenarios according to four risks, namely disruptions in supply, market demands and transport capacity shortages and all combined; they find that the impacts of disruptions depend on the combination of risks, the origin of the risk in the supply chain network and the type of risk.

In healthcare service management context, SD could be used to analyse a society affected by numerous disease risks or a hospital with a relatively large number of patients and to obtain a qualitative output such as the reason why a specific service is overrunning (Brailsford & Hilton, 2001). SD has also been applied to model the **spread of diseases** (Rong et al., 2021, (Lee, Liu, & Pietz, 2017) and **evaluate public health polices** (Katsaliaki & Mustafee, 2011), particularly related to the COVID-19 pandemic. The traditional model used for applying simulation to pandemic spreading is the SIR model, a three stages model that divides individuals into (1) susceptible, (2) infected and (3) recovered (Kermack & McKendrick, 1927). Shirazi et al. (2021) used the SD methodology to understand vulnerabilities in the downstream part of the supply chain (delivery) during the COVID-19 pandemic. They run a study to understand the best location for plasma collection centres during the COVID-19 pandemic and minimise the total cost and the flow time of plasma supply chain. To do so, they applied a SD methodology to estimate the outbreak of COVID-19 and be able to simulate plasma demand levels. (Azghandi, Griffin, & Jalali, 2018) run the first study to determine the optimal inventory strategy in a pharmaceutical supply chain according to the characteristics of the disruption (product recall). Using SD, they mapped disruptions differing in periodicity, breaths, lengths and size and concluded that inventory strategies which account for features in the disruptions are the ones that reduce the costs related to inventory, holding and shortages.

E. Hybrid simulation

Hybrid simulation (HS) is defined as the utilisation of different simulation techniques within one study. HS have grown in popularity in the past years as thought to overcome the limitations of single models by increasing reliability and accuracy and allowing to capture more details by Lee et al. (2013), (Brailsford & Hilton, 2001); (Mustafee & Fakhimi, 2024). Dos Santos et al (2020) and Kar et al. (2024) conducted a review of different hybrid simulations in healthcare and found **DES and SD** to be the most popular combination in supply chain management, specifically on hospital management issues and disease management. In these studies, hybridization is treated as sequential, and the models remained independent with just small overlapping.

Hybrid simulation has been found to be effective in understanding how to meet unstable demand trends during a disruption like the COVID-19 pandemic, and evaluating the effectiveness of different mitigation interventions related to inventory management and logistics (Camur, et al., 2023). The study by Camur et al. (2023) utilises DES with some elements of ABS (which better represent pandemic trends) to understand realistic demand pattern for oxygen concentrators (OC) during the pandemic, inputting real world data from US

government and research agencies. The model developed by the experts allowed them to understand scenarios based on disruptions on different tiers of the supply chain as well as the impact of change of policies. Results pinpointed inventory policies base on demand data to significantly reduce lead time. In addition, it is claimed to allow the identification of regions and demographics which will be more impacted by a disruption in the supply chain and the rapid testing of different mitigation policies to contain shocks.

AnyLogic is a simulation software which combines different simulation methods, such as DES, SD and ABS (TheAnyLogic Company). AnyLogic has been applied by companies, governments and no-profits (e.g. UK Department for Work and Pensions, American Red Cross) extensively to solve several supply chain issues. For example, it has been used to model Glaxo Smith Kline (GSK) vaccine supply chain (AnyLogic, 2023). Specifically, dynamic simulation and DES were used to model the manufacturing part of the supply chain (including quality control and assurance) and determine optimal production and distribution strategies. AnyLogic has also been used to create *Gojii*, a tool that helps to select the appropriate demand signal and the related inventory policy.

More recently, AnyLogic created Anylogistix to better represent the complexity of the supply chain. For instance, AnyLogistix has been applied to simulate the effects of medicine shortages in a hospital ward and select the right recovery policy (Aldrighetti, Zennaro, Finco, & Battini, 2019); and to understand the impact of disruptions on PPE supply chain during the COVID-19 pandemic (Rong, Wang, Han, & Zhao, 2022). To note, experts have highlighted that many HS studies are still limited to theoretical methodology and lack application, probably because of the large amount of high-detailed data and the expertise required (De Santos et al., 2020).

Table 5. Overview of simulation methodologies in different healthcare supply chain areas

Object of Modelling	Methodology applied	Reference
Inventory management	Monte Carlo, Hybrid (AnyLogistx), system dynamics	Elsayed, Taha and Hassan 2022, Azghandi et al 2018
Hospital management	Monte Carlo, DES	Begen et al 2024, Gabriel et al., 2020,
Blood supply chain (and COVID-19 cases)	System Dynamics, DES	Shirazi et al 202, Osario et al 2017
PPE supply Chain	AnyLogistix (hybrid)	Rong, Wao and Zhao, 2022
Shortages in the pharmaceutical supply chain	AnyLogistix (hybrid), DES	Lozano-Diez et al 2020, Aldrighetti et al 2019
Vaccine and oxygen concentrators demand (and COVID-19 cases)	DES, Hybrid (DES+ABS)	Kamran et al 2022, Camur et al 2022

***Source**: author's elaboration based on literature review findings.*

F. Hybrid modelling: simulations used with other techniques (including AI)

There is now unanimous consensus on the need to combine methodologies to capture the complexity of healthcare supply chain and overcome the limitations that single methodologies pose (Kar et al., 2024). One of the fastest growing research areas consists of combining simulation models with other techniques such as mathematical programming, heuristic algorithms and AI (Lee et al., 2013, 2015, 2022), (Ordu et al. (2020) and (Akhtar, Kraemer, & Gardner, 2019). **DES** is the main simulation method to have been used **in conjunction with other methodologies** to solve healthcare supply chain issues, with the most common application being hospital optimisation, and in particular optimising patient flow and predicting hospital beds´ needs.

One such combination is **simulation-optimisation methods,** which have become popular in healthcare settings as increasing power in decision-making (Lee et al., 2013, 2015) (Ordu et

al., 2021). On one hand, optimisation techniques (mathematical programming and heuristic algorithms) allow for optimal configuration. On the other, simulation methods allow testing of different scenarios with relative outcomes. Interestingly, a systematic review by Tordecilla et al. (2021) on the use of simulation methodologies and optimisation techniques for supply chain networks revealed that optimisation methods are more used than simulation methods, specifically when the purpose of the study is related to disruptions. **Integer linear programming** is one of the most applied optimisation techniques in healthcare supply chain**,** as it allows finding solutions when dealing with integer entities such as transportation units and patients, in contrast to other optimisation models which use continuous values of variables or decision variables (Ali Fallahi et al, 2024). This has been extensively used for pharmaceutical supply chains (Goodarzian et al., 2021), blood supply chains (Gunpinar & Centeno, 2015) and quarantine-based home healthcare (Nabavizadeh et al., 2024). Ordu et al. (2021) developed a simulation-optimisation model using DES aimed at forecasting demand and estimating required bed capacity and staff needs in an English hospital. A study by Osorio et al. (2017) utilises DES to model in conjunction with Integer Optimisation model across three different scenarios of the blood supply chain, from collection to production, storing and distribution. The model results were evaluated using real data from a hospital in Colombia and found to be improving key metrics such as shortages and costs. The model could be used to develop early warnings for vulnerabilities in the blood supply chain and the authors claim that is being considered for application in the integrated central information system of the hospitals‘ district.

Recently, some experts have combined traditional simulations with **real-time feeds** (also called digital twins), creating **Real Time Simulation** (RtS). As this combines historical data (DES) with real-time feeds, it enables short-term decision-making and faster experimentation (Mustafee & Fakhimi, 2024). Although the model has been proposed to be applied to healthcare settings, there are no known applications of it so far.

DES methods have been the most frequently applied to validate AI solutions. Studies by Ortiz-Barrios et al (2023), (2024) applied Random Forest (RF) and DES to predict the likelihood of a COVID-19 patient admission to ICU, thus significantly reducing waiting times. Results were compared to the ones from a Spanish hospital chain and found over 90% accuracy. Once data on the current admissions rate are inputted, the model could be used by ICU managers to provide them with a forecast of the number of patients admitted to ICU in the next week. Nas and Koyunco (2019) applied 10 different ML techniques (e.g. RF, LSMT and decision tree) with the same dataset to predict patients´ arrival rates. These predictions were then inputted into the DES model as a parameter. Results showed that the short-term memory model was the one with the highest accuracy, thanks to which the length of stay was minimised by 7% and the beds of the emergency ward optimised.

Table 6. DES simulation applications with other methodologies

Object of Modelling	Methodology applied	Reference
AI models		
Hospital management (emergency waiting time, bed waiting time)	Random Forest	*Ortiz-Barrios et al 2023/2024*
	Artificial Network	*Shokouh et al 2022*
	ANN	*Zeinali et al., 2015*
	Random Forest, Gradient Boosting and AdaBoost	*Atalan et al., 2022;*
	ANFIS, FFNN, RNN and Ada Boost	*Yousefi et al 2017*
	XGBoost (ML) and SHAP	*Kim et al 2021*

Object of Modelling	Methodology applied	Reference
Optimisation		
Hospital optimisation	Integer program	*Ordu et al 2021*
Blood supply chain	Integer program	*Osario et al 2017*
Pharmaceutical Supply Chain	Integer program, meta-heuristic methods	*Goodarzian et al 2021*
System modelling		
Hospital operations such as patient throughput wait time, queue length, utilisation, resources (labour, equipment, beds etc), disease status and prognosis, treatment outcome and readmission prediction	Integrated system including AI/ML, ABS, DES, Mixed Integer Programming	*Lee et al., 2015*
MCM logistics, mass dispensing operations, medical supply distribution, locations of dispensing facilities, optimal facility staffing and resource allocation, routing of the population, MCM dispensing methods, MCM supply and resupply paths and timeline. Decision support and data management.	Integrated system including AI/ML, ABS, SD, DES, Mixed Integer Programming, cognitive analytics, visualized graph-drawing tools	*Lee et al, 2013, 2015, 2021, 2022.*

***Source**: author's elaboration based on literature review findings.*

AI and DES can be applied by health authorities to increase preparedness against outbreaks and other disruptions. Governments could also implement a dashboard with projected ICU capacity indicators to inform decision-making and more rapidly adapt to disruption (Ortiz-Barrios, et al., 2023). Reviews found that major challenges remain regarding applying simulation techniques: only over 5% of published papers in the field of healthcare-related simulation modelling reported that the model findings had been used in practice (Brailsford & Hilton, 2001); (Katsaliaki & Mustafee, 2011). Barriers to implementation include a lack of resources needed to implement these methods (particularly when it comes to AI) and constraints in representing the complexity and heterogeneity of real-world scenarios (Ortiz-Barrios, et al., 2023). Specifically, regarding AI and simulation applications, for the models to work data need to be detailed and of high quality. Academic experts should work closely with healthcare practitioners to ensure models apply to the real world.

A specific type of hybrid modelling is **systems modelling,** an integrated system using ML, simulation and optimisation techniques which has been shown to result in significant measurable improvement and innovation. This has been used and implemented by (Lee, et al., 2015) which describes an emergency department's (ED) decision support system that couples **machine learning, simulation (SD and DES), and simulation-optimisation (Mixed Integer nonlinear programming)** to improve the ED's timeliness of care, quality of care, and operational efficiency; while reducing avoidable readmissions. The system allows healthcare administrators to globally optimise workflow, considering the uncertainties of incoming patient injuries and diseases and their associated care, thereby significantly reducing patient length of stay (Lee et al., 2015). Optimisation is performed to ensure the best operations and system performance (including throughput, wait time, queue length, and utilisation). The resource allocation (including labour, equipment, and beds), modelled via a nonlinear mixed-integer program, is optimised within each simulation iteration. Machine learning is used to identify discriminatory attributes that can predict whether a patient will return to the ED. The simulator combines processes and operations with real-time feeds. Within the simulation, an individual patient is simulated thoroughly, including medical conditions, arrival times, zones visited, and treatments received. Hence, in addition to modelling the hospital operations, it also characterises an individual by disease type, risk factors, demographics, and payor types; knowledge that the machine-learning analysis uncovers. Upon completion of a patient's treatment and before discharge, the ML classification rule is used within the simulation environment (Lee et al., 2015). Implemented at Grady Memorial Hospital in Atlanta, the system helped reduce length of stay by roughly 33 per cent, resulted in a 28 per cent reduction in ED readmissions, and eliminated more than 32 per cent of the nonurgent-care cases from the ED.

The ED and trauma efficiencies improved throughput by over 16 percent and reduced the number of patients who left without being seen by over 30 percent.

Sponsored by the US Centers for Disease Control and Prevention (CDC), a data-driven informatics decision support system, **RealOpt,** was developed for public health medical preparedness. Considered by some as the singular most comprehensive innovative MCM modelling system built, RealOpt addresses the fundamental challenges in MCM logistics, mass dispensing operations, medical supply distribution, locations of dispensing facilities, optimal facility staffing and resource allocation, routing of the population, MCM dispensing methods, MCM supply and resupply paths and timeline (demand consumption and re-supply schedule). RealOpt combines a heterogeneous suite of modelling techniques: DES for process/operations/logistics, **SD** for disease modelling, **ABS and ML** for human cognitive and behaviour, **Mixed Integer Programming** for resource allocation, facility location, along with **visualisation graph-drawing tools**. The system integrates the modelling modules into a single powerful **decision support and data management suite** platform along with computational (solution) engines, and the **3D geographical spatial information with federal census data and demographic and socioeconomic data**. RealOpt facilitates effective and dynamic operational planning and execution and strategic and tactical decision and policy making. It has enabled the CDC to provide modern tools to local jurisdiction (over 3000 counties) that support dynamic planning for emergencies and that establish a knowledge data bank to provide feedback about the deployment of various techniques and disease characteristics of different viruses. The RealOpt suite has a US user base of over 14,000 public health and emergency directors covering all states, plus many international users. RealOpt has been used in different emergency responses such as seasonal flu vaccination, smallpox and hepatitis vaccine dispensing, H1N1 mass vaccination; fire, flood, earthquake and hurricane responses, COVID-19 diagnostic test and mass vaccination, and hospital re-design for COVID-19 patient care (CDC, 2021), (FEMA, 2024). The system has also been used by numerous hospitals/clinics for optimizing hospital workflow and operations efficiency (de Mesquita, Roush, & Sarkar, 2021). RealOpt allows management and tracking of multiple resource types simultaneously, for example, labour, equipment, beds, financial investment, medical supplies, and countermeasures. In addition, RealOpt includes biodata bank and clinic models of real data for various viral spread patterns and actual operations characteristics. It has been used for compliance by all public health divisions across the 50 states in the US (Lee, et al (2015), (Lee, YF Liu, & Pietz, 2016), (2021), (2023)

Although these systems appear technological complex in nature and require unique expertise to tackle and develop that span multiple modelling domains, the systems appear to be practical and perform well in practice.

Box 6. Considerations from the experts: simulation models for supply chain

Highlights resulting from Experts' consultation

Experts from different backgrounds raised the importance of simulation methodologies to be able to test different scenarios and effectively respond to health emergencies. A government expert raised this as a priority for public health institutions, particularly to understand the impact of disease evolvement in different contexts.

Three academic experts mentioned having experience in applying simulation methodologies and described them as long-established tools used when problems are too complex to be solved with traditional mathematical formulas. They indicated that simulation and optimisation techniques are extensively used in inventory management and optimisation. When discussing the best methodology to use in MCM supply chain however, they indicated the lack of one single methodology. Monte Carlo was described as a popular methodology as easy to use, with applications in inventory management during the pandemic, however lengthy to run computationally. According to one industry stakeholder, Monte Carlo simulations will play a significant role in the future in assessing risks, demonstrating the value that simulation will bring to optimisation of the supply chain. However, he also stressed how currently, the focus is on fundamental applications.

ABS was described to be indicated for micro-level (e.g. hospitals), but that if used at a macro level could potentially also bring benefits. DES was confirmed to mainly being used for hospital management. Mixed Integer Program Modelling has been successfully used by one of the academic experts for inventory management of blood supply chain. When discussing simulation methods in conjunction with AI, one academic expert outlined their potential benefits but highlighted how they are rarely used together because of the amount of data and expertise needed to apply both in conjunction. Another academic expert confirmed that simulation, AI and optimisation combined have been implemented successfully in large healthcare systems with measurable improvement in ED's timeliness of care, quality of care, prognosis prediction, and operational efficiency; while reducing avoidable readmissions. RealOpt developer mentioned that the CDC system was developed without any data, and the team quickly performed hospital drills and participated in events to establish good estimates and gather real data on various operations and human behaviour patterns. The expert emphasised that developers should not be deterred by the lack of data. Rather, they should focus on building realistic models that work in real life and perform rapid prototyping to establish good data for running the models. Comparing data needs for simulations versus AI, one academic expert highlighted how simulations might require less granular data than AI, although simulations used in emergency scenarios (with limited historical data) are not able to give a precise prediction but rather provide a baseline scenario. To tackle the limitations of data availability, an industry expert proposed Deep Q algorithm, a technique that combines Reinforcement Learning with Deep Neural Network Models to run several simulations of different scenarios and then recommend a specific policy able to optimise a certain outcome based on a pre-established goal.

2.4 Limitations and challenges to implementation

A. Data availability and visibility

Data scarcity and lack of **visibility** in MCM supply chains are fundamental challenges hindering the application of AI methodologies and simulation methods. ML, for instance, require a high amount of data to be trained and evaluated, while for simulations a common challenge is presented in the validation of the simulation code. MCM supply chains, given their complexity, often lack visibility, not just at the upstream level (e.g. product manufacturing) but also considering ancillary supplies and all other components that may affect the product downstream, and thus relevant data may not be available.

The lack of adequate data is even more critical during health emergencies since historical comparable information is scarce and data collection may be hindered by time and resources constraints. Indeed, some key data elements can only be collected during the course of the emergency, and it becomes important to have a database able to adapt over time as the event evolves (National Academies of Sciences, 2017). Data needs may also highly differ depending on the specific emergency situation and the MCM considered, as well as on the stakeholder perspective (e.g. industry, organisation at international or national level or local hospitals).

As the adage "you cannot manage what you cannot see" suggests, limited data visibility hinders the extent to which stakeholders can access timely information about supply chain operations, hindering supply chain management and the application of the discussed methodologies (MacCarthy, Ahmed, & Demirel, 2022). Moreover, data scarcity erodes trust and collaboration among stakeholders (Ioannou & Demirel, 2022).

B. Data integrity and quality

Poor data quality has been mentioned as one of the major barriers to the implementation of AI tools and simulation methods, both for industries and healthcare institutions (Cannas et al., 2024). Since the success of any AI model depends on the quality of the data it is trained on, poor data can lead to unreliable outcomes (colloquially expressed as "Garbage in – garbage out").

In real-world scenarios, there may be many quality and integrity issues, such as inconsistencies in data collection, incomplete databases or noisy data (Zhao, et al., 2024).

Health records, for instance, are usually a structured and reliable data source, but they can vary significantly across regions and healthcare providers due to differences in the data entry methods, terminologies used and granularity, leading to strong inconsistencies. Moreover, inadequate modality to collect such data can generate incomplete data (i.e. missing data or errors in data entry), which is a pervasive issue in many datasets and further complicates the applicability of AI-driven methods (Zhao, et al., 2024).

Noisy data (i.e. data containing considerable irrelevant or extraneous information) can arise from various sources such as monitoring of social media platforms or news. AI algorithms are designed to manage high amounts of data and filter relevant information. However, authentic signals can be lost in too noisy databases, leading to false alarms or missed detections.

To ensure the integrity of AI models, it is essential to implement robust data preprocessing techniques and real-time validation methods. These processes help distinguish genuine epidemiological trends from noise and safeguard the accuracy of predictions. As models adapt to new data in real-time, validating the accuracy of incoming data is critical since anomalies, outliers, or incorrect data can significantly skew forecasts (Zhao, et al., 2024).

C. Privacy and security

AI-tools are vulnerable to different security threats (Hu, et al., 2021). While AI technologies offer immense potential for demand forecasting and supply monitoring; its reliance on vast amounts of data, often containing sensitive information, raises significant security concerns. These are different throughout the whole process, from the stage of data collection, data processing, model training, inference and final integration. For instance, the data collection stage is susceptible to threats related to data biases, fake data and data breaches (Dilmaghani, et al., 2019). These threats can have severe consequences for organisations and restrain them from using these new technologies (Raimundo & Rosário, 2021). In addition, data privacy concerns disincentive data sharing and the implementation of AI systems in an ethical way.

There are however numerous countermeasures which can be taken to mitigate security issues and allow harnessing the benefits of AI while safeguarding sensitive data and anonymisation (see Conclusion for some suggestions) (Hu, et al., 2021)

D. Human and financial resources and infrastructure

Healthcare institutions and other low-resource settings often lack the latest knowledge and expertise needed to effectively apply AI and simulation methods (Motamedi, et al., 2024).

AI implementations need interdisciplinary collaboration between data scientists, supply chain experts, epidemiologists and domain experts to develop robust data integration frameworks. This requires not only technical expertise for data integration but also specific theoretical knowledge to achieve a deep understanding of patterns and characteristics of different data types. However, in real-case scenarios this level of inter-disciplinarity is often not available, hindering the successful implementation of innovative tools (Zhao, et al., 2024).

Furthermore, financial resources are often insufficient to invest in methods that are considered innovative and with uncertain efficacy (Vázquez-Serrano, Peimbert-García, & Cárdenas-Barrón, 2021). For instance, costs of implementation, energy consumption and operations are considered a major challenge in applying a smart vaccine supply chain system (e.g., the cost to install and maintain IoT devices, the transaction cost of blockchain monitoring, and the computational cost of ML models for forecasting) (Hu, Xu, & Liu, 2023)

Deep learning models are characterised by high computational demands, also a major limitation for implementation in real-world scenarios (Zhao, et al., 2024). Training and deploying these models demand substantial resources, including processing power, memory and time. Similarly, complex detailed simulations that account for multiple variables can also be computationally intensive, especially when running multiple iterations to improve accuracy

and reliability. To face this high computational demand, organisations need to have access to complex infrastructure e.g. high-performance computing clusters or specialised hardware. Aside from being costly, utilising these technologies is also time intensive: models need days to train or hours to generate a prediction, which might not be practical to inform rapid interventions. Finally, the significant consumption of energy, especially for models adopted on a large scale, increases costs but also contributes to environmental impact.

E. Integration & interpretability

An additional barrier to implementation can derive from the lack of data and systems integration. To model complex scenarios, data from different sources should be integrated into an evolving response structure (National Academies of Sciences, 2017). Public health databases have been developed for decades and several electronic health records are already being regularly collected. However, these databases currently lack harmonisation and common standards, which can contribute to increasing difficulties in data integration. As these systems have not been designed to inform AI models, which are dynamic and data-intensive; it is difficult to integrate the latter into these systems, and compatibility issues may arise. For instance, AI methodologies such as deep learning or reinforcement learning require specific computational environments, software libraries and data structures, which highly differ from traditional databases and software platforms currently used in many health systems (Zhao, et al., 2024).

Interpretability and **applicability** of the results are other key factors needed for the successful implementation of AI models in supply chain forecasting and monitoring. However, interpretability is one of the most criticised areas in ML forecasting models, since the model's inner working is often considered opaque (so-called “black box” phenomenon) and hard to understand, limiting its wider adoption. Models evolving through real-time adjustments further contribute to increasing complexity and reducing interpretability (Zhao, et al., 2024). When decisions based on disease prediction may have significant public health implications, this inability to understand and explain model outcomes can be a significant limitation to evidence-based decision-making. For these reasons, interpretability is a key area in the new wave of ML research (Fildes, 2022).

The **generalisability** of the results is also important for a wider practical adoption. Overfitting of deep learning models may occur when a model trained on limited or noisy data learns the noise in the data rather than the underlying pattern. In these cases, the model will provide high accuracy on the training data but poor generalisation to new unseen ones and, therefore, low applicability.

F. Cultural and organisational resistance

The application of new technologies is often met with scepticism, as they are perceived to be complex to understand, requiring structural changes in management and exposing the data at risk of security breaches (Bauer, 2024)

Consistently, difficulties in change management are one of the main barriers to AI implementation identified by the literature (Hangl, Behrens, & Krause, 2022). This is because the integration of AI models into an organisation's management and decision-making system generates changes in the workflow and business process, which normally cause internal resistance (Hangl, Behrens, & Krause, 2022).

Furthermore, AI systems are seen as ´black box´ and as requiring human oversight (involvement of individuals to ensure values such as dignity and autonomy are respected). This can lead to scepticism, particularly when AI-driven decisions suggest unexpected actions or conflict with other sources of information.

Finally, ethical considerations play a crucial role in the implementation of AI. AI models rely on large datasets often containing sensitive information, raising concerns around consent and

transparency. For instance, traditional epidemiological studies involve explicit informed consent to increase participants' awareness. Clear guidelines on data collection, storage, and usage of personal or sensitive data are crucial. Increasing transparency on data sources, methodologies, and potential risks involved can enhance trust and facilitate the adoption of AI into the organisation. However, the need to improve transparency should be balanced with privacy and security measures.

Box 7. Considerations from the experts: limitations and challenges to implementation

Highlights resulting from Experts' consultation

When discussing barriers to the implementation of AI tools and simulation methodologies, **lack of visibility in the supply chain and data quality** emerged as prominent issues. On the one hand, some experts underlined these issues as particularly prominent in the healthcare field, especially during emergency situations with scarce historical data to train AI models. On the other, at least one expert from each of the three fields suggested that institutions should not wait to obtain complete data visibility but start building solutions. For instance, they could consider data repurposing of already available information or focus on increasing data quality through dialogue and cooperation. Similarly, two stakeholders from the public sector underlined that data availability is not the main obstacle to implementing AI as countries have access to abundant historical data thanks to regulations (e.g. mandatory reporting). There also remains an issue regarding dealing with a too large amount of data and the ability to extract meaningful information. Although some experts highlight that generative AI could support in summarising such wide information, it remains a challenge to decipher quality data from all sorts of synthetic or non-relevant data. Nevertheless, all experts agreed that data quality can be improved (e.g., the structure of databases, frequency of data gathering, level of details at key stages of manufacturing processes). An industry stakeholder explained how its company solved the issue of data visibility by relying on another provider to access information and buying data on pharmacies' stock levels to fill data gaps across the supply chain.

One of the barriers hindering better data quantity and quality is limited **cooperation between EU countries and across fields of expertise** (e.g. industry and government). Regarding the first, initiatives to improve data exchange and visibility of the European member states' supply chain would be highly beneficial (i.e. Nordic pharmaceutical forum) and should include discussions on the implementation of AI methodologies. However, cooperating in implementing AI systems could be hindered by **differences among Member States** (e.g., diverse and often non-interoperable sources, differences in reporting standards, terminology, logistics and infrastructures). Notably, there is currently no consistency on the definition of "shortage" of drugs at the European level. For example, Denmark has a list of 30 deficient drugs, while Italy has 3,800, even though the market is the same. The difference is in the approach: in Italy, a shortage is defined by regulatory criteria, including any risk of non-supply, while in Denmark the direct impact on patients is considered.

When implementing monitoring systems, experts often experience reluctance from suppliers to share data, which undermines the willingness to start the process. Very often, this is rooted in issues regarding navigating business-to-business relationships (i.e. developing effective ways of approaching different stakeholders by demonstrating the benefits they can get from sharing data and implementing AI systems). Interestingly, stakeholders that were based in smaller countries characterised by simpler and centralised health supply networks (e.g. a limited number of wholesalers) found it easier to map their supply chain.

Most stakeholders agreed that **data sharing and security concerns** are also among the biggest obstacles. Public entities must comply with privacy regulations and are concerned about protecting the safety and anonymity of data to be provided to AI, especially when the software is developed by a third party. At the same time, manufacturers and suppliers may be reluctant to share detailed information on their supply chain to protect highly sensitive and confidential data. It is essential to demonstrate that data collection systems are secure (e.g. sharing information about software security through third-party audits or certifications like ISO 27001, ISO 27701, and SOC 2 Type 2 audits and GDPR compliance). Academic interviewees mentioned that these risks can be easily reduced with different techniques such as blockchain and simulated data. However, two academic experts warned against the overreliance on artificially generated data, which can lead to unreliable results. Federated learning was suggested by many stakeholders as a solution to increase security since it allows training a model without data exchange.

Finally, further barriers mentioned by stakeholders referred to significant **resistance to change** from institutions and industry. This is not unusual and is being observed within healthcare and other industry sector. The public sector faces a cultural challenge, with a slower pace of innovation compared to the private sector also driven by **bureaucratic complexity**, which delays the adoption of innovative techniques and disincentives startups to pursue business-to-government opportunities for collaboration. An industry stakeholder also expressed their reluctance to apply AI due to perceiving this as a ´black box´, thus raising concerns related to trust and accountability. Similarly, academic experts mentioned public authority struggles to trust the results of AI models, as often seemingly contradictory. This outlines the importance of knowledge translation to provide compelling evidence through understandable language.

3. Discussion and Conclusion

Results showed that research and application of AI tools and simulation methods in supply chain forecasting and monitoring are steadily increasing. The COVID-19 pandemic provided the drive to accelerate research on these tools, as traditional mathematical methods were shown to be insufficient to handle the scale of supply chain disruptions. However, this is a very recent research field and there is no consensus yet on the most suitable methodologies to be applied in the context of health emergencies. Implementations of AI and simulation methodologies in real-case scenarios are still limited, as not many organizations have the skilled and computational resources to adopt AI forecasting methods successfully (Fildes, 2022). At the moment, AI-tools are mostly offered and used by start-ups and consultancy companies to solve supply chain issues during normal times to gain competitive advantage. Box 8 summarised key insights and examples of implementations in the public and private sectors. However, a growing number of real-world use cases are expected to be published in the next years.

Focusing on **demand forecasting,** the literature review and consultations highlighted that this is an extended and rapidly evolving field of research. Many traditional statistical models (such as ARIMA) are well established and continue to be applied to demand forecasting. At the same time, AI solutions (i.e. ML and deep learning) are increasingly explored by experts, often in combination with traditional approaches to overcome models´ limitations, speed up the forecasting process and increase prediction accuracy. AI support is especially critical in contexts characterised by high complexity or high degree of uncertainty, such as during health emergencies. Many different models have been used in the literature for demand forecasting across sectors, including in the MCM field. Experts are comparing their performance to support stakeholders in the selection of the most appropriate model. However, the choice of which model to use depends on many factors such as model assumptions and data characteristics, data quantity and quality and predictors selected. For instance, XGBoost is currently a cost-effective tool applied in real-world scenarios, as it provides accurate forecasting in studies focusing on blood demand or medicine shortages; but it has some limitations when applied to datasets with imbalanced and mixed groups and, therefore, is not always considered a suitable methodology. Overall, ensemble approaches seem to be among the most cutting-edge solutions for many cases of MCM demand forecasting since they combine different models to improve prediction capacity. Moreover, federated learning emerged as an effective solution to protect privacy while collecting a high amount of data, which is an essential requirement to train machine learning for accurate predictions.

On the supply side, the literature and the consultations conducted showed that AI-driven methodologies can support **supply chain monitoring and inform planning,** mainly by analysing data collected through other Industry 4.0 technologies such as the IoT, blockchain, and Big Data analytics. AI can help map the whole supply chain and consequently support the accurate and fast identification of potential disruptions and vulnerabilities. These technologies,

used in conjunction, can enhance transparency, traceability, and operational efficiency, particularly in the context of MCM.

Several digital technologies companies have emerged in the past years offering platforms that, with the use of AI and ML tools, can help companies optimising their SC mapping, monitoring disruptions, and consequently respond and prevent bottlenecks. Methodologies used include NLP, Neural Networks and Deep Learning. However, services marketed as "AI mapping for supply chain visibility" are also offered in cases where the actual involvement of AI is very limited. This practice represents a sort of "AI washing," since AI involvement is exaggerated for marketing purposes only. The digital transformation of supply chain management through AI and other advanced technologies offers significant benefits, particularly in improving visibility and decision-making processes. However, the effective application of these tools still depends on a large volume of robust data input (often lacking in healthcare systems and in emergency situations), collaboration across stakeholders to obtain data and significant financial resources.

Finally, **simulation methods** can significantly improve decision-making related to MCM supply chain, as well as the effective management of resources during health emergency scenarios. The literature review showed simulation methodologies have been applied for some time by industries (e.g. medicines and vaccines manufacturers, big retailers, and automotive industries) to optimise supply chains through software using hybrid methodologies such as AnyLogic. In the public healthcare sector, system modelling and RealOpt have been used by US to improve hospitals decision making and optimise emergency response. Consultations highlighted the need for healthcare institutions to test different scenarios of MCM supply chains. Furthermore, there is evidence that applying simulations can support solving supply management issues. Application of simulation methods to system-level interventions (both national and at a policy level) and health emergencies is increasing, with applications in the US showing they can be effectively used to test the results of different policy interventions during health emergency contexts with unstable demand patterns. EU health institutions should explore their implementation in conjunction with AI and optimisation techniques in the context of different MCM demand and shortage scenarios, to manage resources in hospital settings, for inventory optimisation and to test the impact that different policy options could have on the spread of diseases.

To address critiques and lack of trust caused by the limitations of the different methodologies, EU institutions could be looking at bringing all methodologies together by applying **system modelling** and **RealOpt** to emergency scenarios manage resources in hospital settings, for inventory optimisation and test the impact that different policy options could have on the spread of diseases by Lee et al (2022).

Box 8. Focus on practical applications of AI-supported methodologies and simulation methods for supply chain monitoring and demand forecasting in real-world scenarios

Real-world applications of AI methodologies and simulation methods

Private sector

Results showed that applications of AI to increase supply resilience are still rare and mainly used in the **data collection and transparency phase**. Only a few players are already exploring the **full potential of AI** in improving resilience. For instance, some are running multiple scenario analyses to assess the business impact of potential risks and then taking targeted actions to mitigate them. Other technologies, such as **IoT and blockchain**, are more frequently applied, especially by multinational companies.

Information on specific methodologies applied is rarely publicly available since manufacturers are reluctant to share details about their supply chain and the AI methodologies used. Some major retailers, such as Amazon and Walmart, have made substantial investments in data science capabilities to leverage the latest advancements in ML forecasting; but the majority still rely on less advanced techniques or depend on external software vendors and consultants to meet their

forecasting needs. Similarly, in the healthcare sector, some major **wholesalers** active in the EU are working on implementing AI systems to improve monitoring of the market and avoid excessive stock.

Consultancy and service companies are playing a critical role in AI implementation since they specialise in supporting clients to better define needs and build customised solutions, starting from high-quality databases and strong data infrastructure. Some of the services currently implemented by such companies are:

- **Demand forecasting,** offered by companies such as i) *Coupa AI-powered* demand modelling, which can recognise and learn from complex trends and demand patterns to identify the impact on supply chain **(Coupa, 2024)**; ii) *Anaplan PlanIQ,* which leverage advanced statistical and ML forecast algorithms to make accurate predictions **(Anaplan, 2021)** or iii) *RELEX ML* tool to model consumer behaviour for more accurate forecasts **(RELEX, 2024)**.
- **Monitoring of shocks** through analysis of news that can impact the supply chain through natural language processing (NLP) systems or generative AI;
- **Prediction of prices** of critical materials, provided by *Resilinc* (Resilinc, 2024)
- **Estimate of potential delays** in purchase order delivery after a supply chain disruption currently offered by *Resilinc*.
- **Multi-tier mapping** through Hybrid AI and supplier validation, e.g. *Resilinc* (Resilinc, 2024)
- **Safety stock optimisations** and **inventory management** through AI, probabilistic planning and Monte Carlo simulations (e.g., *RELEX*).
- **Simulation modelling** through *AnyLogic* software (AnyLogic, 2023)
- **Risk management** through interactive platforms, from companies such as *Everstream* (Everstream, 2024) and *Altana* (Altana, 2024)

Public sector

Public organisations are in an **earlier stage of the implementation process** compared to private industries. In many cases, AI-tools application is now starting to be discussed as a potential area of investment.

The experts' consultations highlighted few cases of government applying AI-driven methodologies for demand forecasting and/or simulation methods across different sectors, mainly energy, defence and healthcare. Outside of the EU, RealOpt has broad dissemination across all levels of public health emergency responders and hospitals across the United States. The US government is also contracting specialised service companies and academia. For example, during the COVID-19 pandemic, *Resilinc* assisted the US Strategic National Stockpile with **tracking the manufacturing locations** and **sourcing raw materials** for N95 masks. At the time, this was a crucial early step in understanding supply chain vulnerabilities. A US public entity in the defence sector, also applied AI to **supplier identification and supply chain risk** analysis. The **Canadian Blood Services** is closely collaborating with a group of researchers to explore the performance of different AI tools in improving **demand forecasting** accuracy. The UK government is also contracting the company Altana AI in mapping their national supply chain. It is not however clear into what extent AI will be applied.

Within the EU, this Study found no cases of current AI implementation, and few cases of public entities working to prepare for AI implementation. According to a ongoing survey conducted by CHESSMAN among EU Member States, only one is using some form of AI for demand forecasting. In Norway, a public entity is working to explore the possible implementation of AI tools focusing on defining the need and improving the quality of data. The Italian Medicine Agency (AIFA) is currently conducting a pilot to implement AI methodologies for medicine shortages, leveraging on their experience on pharmaceutical crime and countering illicit drug trafficking. EU institutions are funding many research projects that can provide useful insights on AI tools for forecasting (e.g. ReMAP project on predicting aircraft maintenance, CHESSMEN initiative on medicines shortages) or monitoring (e.g. ECO4CO project for supporting healthcare institutions in monitoring, management and forecasting activities through AI, social media analytics, geospatial analysis and satellite imaging) as well as to understand how to mitigate current challenges (e.g. EU-HIP on systems interoperability).

In the context of MCM supply chains, the adoption of **AI and simulation-based** tools can enhance EU's crisis preparedness and response capabilities by enabling a more accurate prediction of trends and increasing visibility compared to traditional tools. These methods can facilitate the identification of new trends in demand and provide a correct and fast estimation of the supply needed to avoid disruptions. By better predicting demand, they also allow to reduce risks of legal issues related to over/underestimation and encourage more sustainable practices by reducing waste. Simulation methods allow testing the impact of different real-world scenarios such as interventions and disruptions, enabling decision-makers to identify the suited response for different contexts. They can be used both for emergency preparedness (e.g. to understand SC criticalities) as well as for emergency response (e.g. to reallocate resources and test the effectiveness of policies).

However, for AI and simulation methods to be successfully applied, **strategic policy actions** are needed. The information gathered through the literature and the consultations with experts allowed to identify the major limitations and challenges (see Section 3.4) and develop a series of suggested actions that EU institutions could implement to create an adequate environment for the implementation of AI-tools.

To start, policy-makers should engage in a collaborative exercise to clearly **define the needs** behind AI-tools applications, if possible focusing on a specific category of MCM and developing a tailored analysis of the issues and benefits expected from the adoption of new tools. This step should not be underestimated since the operational questions developed will drive data needs, collection strategy and models to be applied. Once this has been clarified, activities should focus on understanding and mapping current data viability, quality and gaps. Concerning data already available, it is essential to work on solving potential access issues (e.g., interoperability between different systems) and start exploring the potential uses of this information. Lack of full data visibility should not stop the initial application of AI methodologies and simulation methods. A plethora of data is already available, especially after the COVID-19 pandemic, and some data gaps could also be filled by data repurposing. Collaborations with academia can be helpful since some scholars are specifically focusing their research on data cleaning, integration of different data systems and data pre-processing.

In parallel, significant efforts should be dedicated to improving data availability and visibility, as well as data integrity and quality. EU institutions should aim at **mapping the end-to-end MCM supply chain** to be able to exploit the full potential of AI tools. Data collection and mapping exercises should particularly focus on the upstream part of the SC (e.g. data on manufacturers and suppliers) as this is where most gaps are present, and bottlenecks are created. Data collection can be aided by incentivising the use of Industry 4.0 enablers such as IoT and automated monitoring platforms with AI incorporated.

Improving data access is however not possible without strong business-government relationships**.** Public authorities should invest in **collaboration and capacity-building initiatives** across interdisciplinary teams with experts from computer science, medicine, law and policy**.** Industry, government and academia often work separately, hindering access to data and the successful application of these technologies in the real world. Academia should be encouraged to improve the generalisability and applicability of the discussed technologies, explaining their use with compelling evidence and guiding institutions in their application (Bauer, 2024). Industry should be incentivised to improve its end-to-end SC visibility and monitoring and demand forecasting through all Industry 4.0 Technologies. Spaces for information sharing and collaboration across areas should be established, as well as start incentivising industry to share data and expertise by providing incentives and tools to reduce privacy and security concerns.

Indeed, issues around data privacy and security play a significant role in hindering the implementation of these new technologies. EU institutions should start **identifying legal aspects**, such as the need for legislative changes; and **explore methodologies for data safeguarding,** such as enhancing cybersecurity and restrictions for access across systems

and put in place strong risk management strategies (Bauer, 2024)**.** Education and training for AI professionals on ethical practices and data privacy laws are also important to create a culture of responsibility and awareness. Several countermeasures can be applied to different stages of AI application to mitigate security threats. Data security protection strategies can reduce issues related to data collection (Hu, et al., 2021). Technologies such as blockchain and federated learning can mitigate issues surrounding anonymisation by decentralising data. The latter provides a solution to leverage sensitive data while preserving privacy, as it allows AI models to learn from decentralised data sources without the need to transfer the data itself, thereby lowering the risk of privacy violations. Data minimisation strategies can also contribute to ensuring that only the data strictly necessary for a specific task is collected and processed, minimising the potential for misuse. To address issues surrounding data safeguarding (and in parallel achieve standardisation), a trusted centre within the EU could aggregate data, ensuring anonymity by stripping sensitive details and maintaining confidentiality. Once data quantity and quality have been improved, cutting-edge AI methodology can be applied to further improve the control over the supply chain by identifying bottlenecks and preventing potential disruptions.

Finally, public authorities should **boost investment in research and development,** including training and upskilling on AI and other Industry 4.0 Enablers technologies, such as blockchain and hybrid simulations. Increasing expertise would also help to address the **resistance** that is currently present in institutions regarding the use of AI and the risks that this poses. Given the pace at which these technologies are developing, upskilling should be continuous to maintain competencies up-to-date. Furthermore, to tackle barriers to application related to the high-computational demand of deep learning models adequate infrastructures are needed. Cloud-based solutions and investment in energy-efficient solutions can also contribute to reduce implementation issues.

To conclude, by addressing the current barriers that hinder the application of AI in MCM supply chain monitoring and forecasting, the EU can harness the potential of AI and simulation methods to ensure a resilient MCM supply chain and, ultimately, adequate access to healthcare for European citizens.

Bibliography

Abdollahiasl, A., Dinarvand, R., Nikfar, S., & Jaberidoost, M. (2013). Pharmaceutical supply chain risks: A systematic review. *DARU Journal of Pharmaceutical Sciences*.

Abeyratne, S. A., & Monfared, R. P. (2016). Blockchain ready manufacturing supply chain using distributed ledger. *International Journal of Research in Engineering and Technology*.

Accenture. (2024). Supply chain networks in the age of generative AI: Turning promise into performance. Retrieved from https://www.accenture.com/content/dam/accenture/final/accenture-com/document-2/Accenture-Supply-Chain-Networks-In-The-Age-Of-Generative-AI.pdf

Adulyasak, Y., Benomar, O., Chaouachi, A., Cohen, M. C., & Khern-Am-Nuai, W. (2023). Using AI to detect panic buying and improve products distribution amid pandemic. *AI & society*, 1-30.

Akhtar, M., Kraemer, M., & Gardner, L. M. (2019). A dynamic neural network model for predicting risk of Zika in real time. *BMC medicine*, 17, 1-16.

Aldarraji, Vega-Márquez, Pontes, Mahmood, & Riquelme. (2024). Addressing energy challenges in Iraq: Forecasting power supply and demand using artificial intelligence models. *Heliyon*. doi:10.1016/j.heliyon.2024.e25821

Aldrighetti, R., Zennaro, I., Finco, S., & Battini, D. (2019). Healthcare supply chain simulation with disruption considerations: A case study from Northern Italy. *Global Journal of Flexible Systems Management, 20(Suppl 1)*, 81-102.

Alotaibi, S., & Mehmood, R. (2018). Big Data Enabled Healthcare Supply Chain Management: Opportunities and Challenges. *SCITA*.

Altana. (2024). *Public Sector - Case Studies & Deep Dives.* Retrieved from: https://altana.ai/solutions/public-sector.

Anaplan. (2021). *Plan IQ*. Retrieved from https://www.anaplan.com/content/dam/anaplan/assets/documents/datasheet/anaplan-plan-iq-datasheet.pdf

AnyLogic. (2023). Business Simulation Case Studies-AnyLogic.

Ardabili, S. F.-K. (2020). COVID-19 Outbreak Prediction with Machine Learning. *SSRN Electronic Journal.* doi:https://doi.org/10.2139/ssrn.3580188

Arora, & Grigas. (2018). Importance of Supply Chain Management in Healthcare of Third World Countries. *International Journal of Supply and Operations Management, 5(1), 101-106*.

Azarafza, M. A. (2020). COVID-19 Infection Forecasting based on Deep Learning in Iran. doi:https://doi.org/10.1101/2020.05.16.20104182

Azghandi, R., Griffin, J., & Jalali, M. S. (2018). Minimization of drug shortages in pharmaceutical supply chains: A simulation-based analysis of drug recall patterns and inventory policies. *Complexity, 2018(1), 6348413.*

Balasubramanian, S., Shukla, V., Islam, N., Upadhyay, A., & Duong, L. (2023). Applying artificial intelligence in healthcare: lessons from the COVID-19 pandemic. *International Journal of Production Research*, 1-34.

Barratt, M., & Oke, A. (2007). Antecedents of supply chain visibility in retail supply chains: a resource-based theory perspective. *Journal of operations management*, 25(6), 1217-1233.

Bathula, A., Gupta, S. K., Merugu, S., Saba, L., Khanna, N. N., Laird, J. R., & Suri, J. S. (2024). Blockchain, artificial intelligence, and healthcare: the tripod of future—a narrative review. *Artificial Intelligence Review*, 57(9), 238.

Bauer, E. (2024). *Potential application of emerging technologies in public procurement.* TALLINN UNIVERSITY OF TECHNOLOGY.

Begen, M. A., Rodrigues, F. F., Rice, T., & Zaric, G. S. (2024). A forecasting tool for a hospital to plan inbound transfers of COVID-19 patients from other regions. *BMC Public Health,* , 24(1), 505.

Beldek, T. K. (2020). Supply chain management in healthcare: a literature review. In Proceedings of the International Symposium for Production Research 2019. *Springer International Publishing.*

Belgodère, D, P. B., MC, J., V, R., W, J., V, R., . . . J, A. (2023). Patient and public involvement in the benefit-risk assessment and decision concerning health products: position of the Scientific Advisory Board of the French National Agency for Medicines and Health Products Safety (ANSM). *BMJ Glob Health*, 8 (5).

Blanchard, D. (2021). *Supply chain management best practices.* John Wiley & Sons.

Borges, D., & Nascimento, M. C. (2022). COVID-19 ICU demand forecasting: A two-stage Prophet-LSTM approach. *Applied Soft Computing, 125, 109181*. doi:https://doi.org/10.1016/j.asoc.2022.109181

Boston Consulting Group, &. W. (2022). *Boston Consulting Group, & Wellcome Trust. (2022).* Retrieved from https://cms.wellcome.org/sites/default/files/2022-04/understanding-the-antibiotic-manufacturing-ecosystem-2022.pdf

Brailsford, S. C., & Hilton, N. A. (2001). A comparison of discrete event simulation and system dynamics for modelling health care systems.

Brintrup, A., Kosasih, E., Schaffer, P., Zheng, G., Demirel, G., & MacCarthy, B. L. (2023). Digital supply chain surveillance using artificial intelligence: definitions, opportunities and risks. *International Journal of Production Research*.

Camur, M. C., Thanos, A. E., Yund, W., Tseng, C. Y., White, C. C., & Iakovou, E. (2023). An integrated system dynamics and discrete event supply chain simulation framework for supply chain resilience with non-stationary pandemic demand. *In 2023 Winter Simulation Conference (WSC)* , (pp. 1617-1628).

Cannas, V. G., Ciano, M. P., Saltalamacchia, M., & Secchi, R. (2024). Artificial intelligence in supply chain and operations management: a multiple case study research. *International Journal of Production Research*, 62(9), 3333-3360.

Cao, S. J. (2023). Exploring blockchain-based traceability for food supply chain sustainability: Towards a better way of sustainability communication with consumers. *Procedia Computer Science*.

CDC, C. f. (2021). *The Division of State and Local Readiness (DSLR) Friday Update.* Retrieved from https://www.cdc.gov/orr/readiness/resources/00_docs/2021.05.28_DSLR-Friday-Update_508Compliant.pdf

Chimmula, V. K. (2020). Time series forecasting of COVID-19 transmission in Canada using LSTM networks. *Chaos, Solitons & Fractals*, 35, 109864. doi:https://doi.org/10.1016/j.chaos.2020.109864

Christopher, M. &. (2004). Mitigating supply chain risk through improved confidence. *International journal of physical distribution & logistics management*, 34(5), 388-396.

Christopher, M., & Peck, H. (2004). Building the resilient supply chain. *International Journal of Logistics*.

Coupa. (2024). *Demand Disruption Checklist*. Retrieved from https://get.coupa.com/rs/950-OLU-185/images/Demand-Disruption-Checklist.pdf

Coustasse, A., Tomblin, S., & Slack, C. (2013). Impact of Radio-Frequency Identification (RFID) Technologies on the Hospital Supply Chain: A Literature Review. *Perspectives in Health Information Management*.

Currie, C. S. (2020). How simulation modelling can help reduce the impact of COVID-19. . *Journal of Simulation*, 14(2), 83-97.

Dallas, M. P., Horner, R., & Li, L. (2021). The mutual constraints of states and global value chains during COVID-19: The case of personal protective equipment. *World Development*, 139, 105324. .

De Giovanni, P. (2019). Digital supply chain through dynamic inventory and smart contracts. *Mathematics*, 7(12), 1235.

de Mesquita, P. J., Roush, R., & Sarkar, J. (2021). County-Level Planning for Efficient Distribution of Emergency Medical Countermeasures with RealOpt Software. *Health security,*, 19(5), 532-540.

Desai, MUG, K., S, B., A, C., P, N., M, H., . . . B, L. (2019). Real-time Epidemic Forecasting: Challenges and Opportunities. Health Secur. *Health Secur.* doi:10.1089/hs.2019.0022

Dey, R., & Salem, F. M. (2017). Gate-variants of Gated Recurrent Unit (GRU) neural networks. *IEEE 60th International Midwest Symposium on Circuits and Systems (MWSCAS), Boston, MA, USA,*. doi:10.1109/MWSCAS.2017.8053243

Dilmaghani, S., Brust, M. R., Danoy, G., Cassagnes, N., Pecero, J., & B. P. (2019). Privacy and security of big data in AI systems: A research and standards perspective. *In 2019 IEEE international conference on big data (big data)* , 5737-5743.

Ding, H., Benyoucef, L., & Xie, X. (2008). Stochastic multi-objective production-distribution network design using simulation-based optimization. *International Journal of Production Research,* , 47(2), 479–505. .

dos Santos, V. H., Kotiadis, K., & Scaparra, M. P. (2020). A review of hybrid simulation in healthcare. *In 2020 winter simulation conference (wsc)*, (pp. 1004-1015).

Duan, L., & Xiong, Y. (2015). Big data analytics and business analytics. . *Journal of Management Analytics,*, 2(1), 1-21.

Elsayed, N., Taha, R., & Hassan, M. (2022). Measuring Blood Supply Chain Performance Using Monte-Carlo Simulation. *IFAC-PapersOnLine*, 55(10), 2011–2017.

EMA. (2021). *Reflection Paper Forecasting Demand of Medicinal Products in the EU EEA.* Retrieved from https://www.ema.europa.eu/en/documents/other/reflection-paper-forecasting-demand-medicinal-products-eueea_en.pdf

EMA. (2024). *European Shortages Monitoring Platform- European Medicines Agency* . Retrieved from https://www.ema.europa.eu/en/human-regulatory-overview/post-authorisation/medicine-shortages-availability-issues/european-shortages-monitoring-platform

ENISA. (2021). *Threat Landscape for Supply Chain Attacks.* ENISA (European Union Agency for Supply Chain Attacks).

ENISA. (2024). *Threat landscape .* ENISA (Euopean Agency Union for CyberSecurity).

Epstein, J. M. (2009). Modelling to contain pandemics. *Nature, 460(7256),*, 687-687.

Erevelles, S., Fukawa, N., & Swayne, L. (2016). Big Data consumer analytics and the transformation of marketing. *Journal of business research*, 69(2), 897-904.

European Commission. (2023). *Commission steps up actions to address critical shortages of medicines and strengthen security of supply in the EU*. Retrieved from Addressing critical shortages of medicines in the EU.

European Commission. (2023). *HERA | Funding and opportunities*. Retrieved from https://health.ec.europa.eu/health-emergency-preparedness-and-response-hera/funding-and-opportunities_en#:~:text=HERA%20Invest%20is%20a%20financing,the%20HERA%20Work%20Plan%202023%20

Everstream. (2024). *Actionable intelligence- Everstream Analytics. .* Retrieved from https://www.everstream.ai/solutions/actionable-intelligence/.

Fanoodi, B. M. (2019). Reducing demand uncertainty in the platelet supply chain through artificial neural networks and ARIMA models. *Computers in Biology and Medicine, 113, 103415.* doi:https://doi.org/10.1016/j.compbiomed.2019.103415

FEMA, F. E. (2024). *Planning, Decision Support, and Modeling Resources for Chemical Incidents* . The U.S. Department of Homeland Security.

Fildes, R. K. (2022). Post-script—Retail forecasting: Research and practice. *International Journal of Forecasting, 38(4), 1319–1324*. doi:https://doi.org/10.1016/j.ijforecast.2021.09.012

Finkenstadt, J. D., & Handfield, R. (2021). Blurry vision: Supply chain visibility for personal protective equipment during COVID-19. *Journal of Purchasing and Supply Management* .

Gao, Y. G. (2023). Vaccine supply chain coordination using blockchain and artificial intelligence technologies. *Computers & Industrial Engineering, 175, 108885*. doi:https://doi.org/10.1016/j.cie.2022.108885

Garba, A., Dwivedi, A. D., Kamal, M., Srivastava, G., Tariq, M., Hasan, M. A., & Chen, Z. (2021). A digital rights management system based on a scalable blockchain. . *Peer-to-Peer Networking and Applications*, 14, 2665-2680.

GEP. (2024). *A Deep Dive Into Upstream and Downstream Supply Chain*. Retrieved from GEP: https://www.gep.com/blog/strategy/upstream-and-downstream-supply-chain-a-deep-dive

Gezgin, E., Huang, X., Prakash, S., & Idelfonso, S. (2017). *Digital transformation: Raising supply-chain performance to new levels.* McKinsey & Company. Retrieved from https://www.mckinsey.com/capabilities/operations/our-insights/digital-transformation-raising-supply-chain-performance-to-new-levels

Ghadge, A. E., Ivanov, D., & Chaudhuri, A. (2022). Visualisation of ripple effect in supply chains under long-term, simultaneous disruptions: a system dynamics approach. . *International Journal of Production Research,* , 60(20), 6173-6186.

Goic, M., Bozanic-Leal, M., Badal, M., & Basso, L. J. (2021). COVID-19: Short-term forecast of ICU beds in times of crisis. *PLOS ONE*. doi: https://doi.org/10.1371/journal.pone.0245272

Grand View Research. (2024). *Artificial Intelligence Market Size, Share & Trends Analysis Report By Solution, By Technology (Deep Learning, Machine Learning, NLP, Machine Vision, Generative AI), By Function, By End-use, By Region, And Segment Forecasts, 2024 - 2030.* Grand View Research.

Grigoryev, I. (2018). *AnyLogic in Three Days: Modelling and Simulation Textbook.* AnyLogic.

Gunpinar, S., & Centeno, G. (2015). Stochastic integer programming models for reducing wastages and shortages of blood products at hospitals. . *Computers & Operations Research*, 54, 129-141.

Guo, L., Fang, W., Zhao, Q., & Wang, X. (2021). The hybrid PROPHET-SVR approach for forecasting product time series demand with seasonality. *Computers & Industrial Engineering, 161, 107598.* doi:https://doi.org/10.1016/j.cie.2021.107598

Hamway, S. (2024). *Supply Chain Agility Index.* Miami: Nucleus Research.

Hangl, J., Behrens, V. J., & Krause, S. (2022). Barriers, Drivers, and Social Considerations for AI Adoption in Supply Chain Management: A Tertiary Study. *Logistics, 6(3), 63.* doi:https://doi.org/10.3390/logistics6030063

Hariharan, R., Sundberg, J., Gallino, G., Schmidt, A., Arenth, D., Sra, S., & Fels, B. (2020). An Interpretable Predictive Model of Vaccine Utilization for Tanzania. *Frontiers in Artificial Intelligence*. doi:https://doi.org/10.3389/frai.2020.559617

Harper, P. R., & Shahani, A. K. (2003). A decision support system for the care of HIV and AIDS patients in India. *European Journal of Operational Research*, 147(1), 187-197.

Harris-Wilson, K. (2023). *Exploring the influence of synthetic data and diverse methodologies on spare parts demand forecasting.* Doctoral dissertation, Erasmus University Rotterdam.

Hayden, J., & Shillingford, D. (2024). *AI in Supply Chain Risk Management: what every supply chain executive should know about AI and Machine learning. .* Accessible on request from https://www.everstream.ai/special-reports/ai-supply-chain-risk-management/.

HERA. (2023). *Annual Work Plan of the Health Emergency, Preparedness and Response.* Health Emerency Response Authority (HERA).

Hu, H., Xu, J., & Liu, M. (2023). Vaccine supply chain management: An intelligent system utilizing blockchain, IoT and machine learning. *Journal of Business Research*.

Hu, Y., Kuang, W., Qin, Z., Li, K., Zhang, J., Gao, Y., & Li, K. (2021). Artificial intelligence security: Threats and countermeasures. *ACM Computing Surveys (CSUR), 55(1), 1-3*.

Huang, Y. X. (2020). Medical service demand forecasting using a hybrid model based on ARIMA and self-adaptive filtering method. . *BMC Med Inform Decis Mak*, https://doi.org/10.1186/s12911-020-01256-1.

Hyder, A., Buckeridge, D. L., & Leung, B. (2013). Predictive validation of an influenza spread model. *PloS one*.

IBM. (2019). *Supply chain planning and analytics*. Retrieved from https://www.ibm.com/products/planning-analytics/supply-chain-planning?utm_content=SRCWW&p1=Search&p4=43700068108624254&p5=p&p9=58700007558656354&gad_source=1&gclid=Cj0KCQjwxsm3BhDrARIsAMtVz6O0y8b6MsJk1ZQg1u5pQ1QKVWa0LyrJR91SrrKORFwdJAB_6ka63dEaAgcgEALw_wc

IBM. (2024). *What is inventory management?* Retrieved from IBM: https://www.ibm.com/topics/inventory-management

IBM. (2024). *What is supply chain optimization?* Retrieved from IBM: https://www.ibm.com/topics/supply-chain-optimization#:~:text=IBM-,What%20is%20supply%20chain%20optimization%3F,performance%20in%20a%20supply%20network.

Imai, C., Armstrong, B., Chalabi, Z., Mangtani, P., & Hashizume, M. T. (2015). Time series regression model for infectious disease and weather. *Environmental research*, 142, 319-327.

Ioannou, I., & Demirel, G. (2022). Blockchain and supply chain finance: a critical literature review at the intersection of operations, finance and law. . *Journal of Banking and Financial Technology, 6(1),* , 83-107.

Ivanov, D. (2020). Predicting the impacts of epidemic outbreaks on global supply chains: A simulation-based analysis on the coronavirus outbreak (COVID-19/SARS-CoV-2) case. . *Transportation Research Part E: Logistics and Transportation Review,* , 136, 101922.

Iwendi, C. B. (2020). COVID-19 Patient Health Prediction Using Boosted Random Forest Algorithm. *Frontiers in Public Health*. doi:https://doi.org/10.3389/fpubh.2020.00357

Jabbarzadeh, A., Fahimnia, B., Sheu, J. B., & Moghadam, H. S. (2016). Designing a supply chain resilient to major disruptions and supply/demand interruptions. *Transportation Research Part B: Methodological*, 94, 121-149.

Jackson, I., Ivanov, D., Dolgui, A., & Namdar, J. (2024). Generative artificial intelligence in supply chain and operations management: acapability-based framework for analysis and implementation. *INTERNATIONAL JOURNAL OF PRODUCTION RESEARCH*.

Jellouli, O., & Chatelet, E. (2001). Monte Carlo simulation and genetic algorithm for optimising supply chain management in a stochastic environment. *IEEE International Conference on Systems, Man and Cybernetics. e-Systems and e-Man.*

Jiang, J., Liu, X., Ren, T., Wang, Q., & Zheng, Y. e. (2024). Deep Reinforcement Learning for Solving Management Problems: Towards A Large Management Model. *arxiv*.

Kairouz, P., McMahan, H. B., Avent, B., Bellet, A., Bennis, M., Bhagoji, A. N., . . . Gardner, J. (2021). Advances and Open Problems in Federated Learning. *arXiv:1912.04977v3*.

Kamran, M. A. (2023). A new vaccine supply chain network under COVID-19 conditions considering system dynamic: Artificial intelligence algorithms. . *Socio-Economic Planning Sciences*, , 85, 101378.

Kar, E., Fakhimi, M., Turner, C., & Eldabi, T. (2024). Hybrid simulation in healthcare: a systematic exploration of models, applications, and emerging trends. . *Journal of Simulation*, 1-19.

Kasaie, P., Dowdy, D. W., & Kelton, W. D. (2013). An agent-based simulation of a tuberculosis epidemic: understanding the timing of transmission. *In 2013 Winter Simulations Conference (WSC)* , (pp. 2227-2238).

Katsaliaki, K., & Mustafee, N. (2011). Applications of simulation within the healthcare context. Journal of the operational research society, . 62(8), 1431-1451.

Kazmin, D., Nakaya, H. I., Lee, E. K., Johnson, M. J., Van Der Most, R., Van Den Berg, R. A., & Pulendran, B. (2017). Systems analysis of protective immune responses to RTS, S malaria vaccination in humans. *Proceedings of the National Academy of Science*, 114(9), 2425-2430.

Kermack, W. O., & McKendrick, A. G. (1927). A contribution to the mathematical theory of epidemics. *Proceedings of the royal society of london. Series A, Containing papers of a mathematical and physical character*, (pp. 115(772), 700-721.).

Khaldi, R. E. (2017). Artificial Neural Network Based Approach for Blood Demand Forecasting: Fez Transfusion Blood Center Case Study. *Proceedings of the 2nd International Conference on Big Data, Cloud and Applications, 1–6*. doi:https://doi.org/10.1145/3090354.3090415

Klibi, W., & Martel, A. (2012). Modeling approaches for the design of resilient supply networks under disruptions. *International Journal of Production Economics*, 135(2), 882-898.

Koç, E., & Türkoğlu, M. (2022). Forecasting of medical equipment demand and outbreak spreading based on deep long short-term memory network: the COVID-19 pandemic in Turkey. *Signal, Image and Video Processing, 16(3), 613–621.* doi:https://doi.org/10.1007/s11760-020-01847-5

Kosasih, E., & Brintrup, A. (2022). Towards Digital Supply Chain Risk Surveillance. *IFAC PapersOnLine*.

Kumar, R., & Tripathi, R. (2019). Traceability of counterfeit medicine supply chain through Blockchain. *11th international conference on communication systems & networks (COMSNETS) IEEE.*, 568-570.

Kwon, H. J., Park, S., Park, Y. H., Baik, S. M., Park, D. J., & . (2024). Development of blood demand prediction model using artificial intelligence based on national public big data. *DIGITAL HEALTH*. doi:https://doi.org/10.1177/20552076231224245

Lal, A. A. (2022). Pandemic preparedness and response: exploring the role of universal health coverage within the global health security architecture. *The Lancet Global Health*.

Lee, E. K., Atallah, H. Y., Wright, M. D., Post, E. T., Thomas IV, C., Wu, D. T., & Haley Jr, L. L. (2015). Transforming hospital emergency department workflow and patient care. *Interfaces*, 45(1), 58-82.

Lee, E. K., Li, Z. L., Liu, Y. K., & LeDuc, J. (2021). Strategies for vaccine prioritization and mass dispensing. *Vaccines (Basel)*. doi:10.3390/vaccines9050506

Lee, E. K., Liu, Y. F., Debboun, M., Shah, U. A., & Nakaya, H. I. (2022). A General-Purpose Modeling Framework for Analyzing Infectious Disease Outbreaks.

Lee, E. K., Liu, Y., & Pietz, F. H. (2017). A compartmental model for Zika virus with dynamic human and vector populations., (pp. Vol. 2016, p. 743).

Lee, E. K., Pietz, F., Benecke, B., Mason, J., & Burel, G. (2013). Advancing public health and medical preparedness with operations research. *Interfaces*, 43(1), 79-98.

Lee, E. K., YF Liu, & Pietz, F. (2016). A Compartmental Model for Zika Virus with Dynamic Human and Vector Populations. *AMIA Annu Symp Proc*, (pp. 743–752).

Lee, E., Yuan, F., Mann, B. J., & Egan, B. (2023). Handling Imbalanced and Poorly Separated Data: a Multi-Stage Multi-Group Machine Learning Approach. *In 2023 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*, (pp. 4336-43).

Li, N., Arnold, D. M., Down, D. G., Barty, R., Blake, J., Chiang, F., . . . Heddle, N. M. (2022). From demand forecasting to inventory ordering decisions for red blood cells through integrating machine learning, statistical modelling, and inventory optimization. *Transfusion*, 62(1), 87–99. doi:https://doi.org/10.1111/trf.16739

Li, N., Pham, T., Cheng, C., McElfresh, D. C., Metcalf, R. A., Russell, W. A., . . . Goel, R. (2023). Blood Demand Forecasting and Supply Management: An Analytical Assessment of Key Studies Utilizing Novel Computational Techniques. *Transfusion Medicine Reviews,*. doi:https://doi.org/10.1016/j.tmrv.2023.150768

Li, Q., Wen, Z., Wu, Z., Hu, S., Wang, N., Li, Y., . . . He, B. (2021). A Survey on Federated Learning Systems: Vision, Hype and Reality for Data Privacy and Protection. *arXiv:1907.09693v7*.

Li, T., Sahu, A. K., Talwalkar, A., Smith, V., ., & . (2020). Federated Learning: Challenges, Methods, and Future Direc. *IEEE Signal Processing Magazine*, arXiv:1908.07873v1.

Long, P., Lu, L., Chen, Q., Chen, Y., Li, C., & and Luo, X. (2023). Intelligent selection of healthcare supply chain mode – an applied research based on artificial intelligence. *Frontiers Public Health*.

Looi, M. K. (2024). The European healthcare workforce crisis: how bad is it? *bmj*, 384.

Lorig, F., Johansson, E., & Davidsson, P. (2021). Agent-based social simulation of the COVID-19 pandemic: A systematic review. *Journal of Artificial Societies and Social Simulation*, 24(3).

MacCarthy, B. L., Ahmed, W. A., & Demirel, G. (2022). Mapping the supply chain: Why, what and how? *International Journal of Production Economics, 250, 108688*.

Mah Muam, P., Skalna, I., & Muzam, J. (2022). Natural Language Processing and Artificial Intelligence for Enterprise Management in the Era of Industry 4.0. *Applied Sciences*.

Manavalan, E., & Jayakrishna, K. (2019). A review of Internet of Things (IoT) embedded sustainable supply chain for industry 4.0 requirements. . *Computers & industrial engineering*, 127, 925-953.

Mediavilla, M. A., Dietrich, F., & Palm, D. (2022). Review and analysis of artificial intelligence methods for demand forecasting in supply chain management. *Procedia CIRP*. doi:https://doi.org/10.1016/j.procir.2022.05.119

Mele, F. D., Musulin, E., & Puigjaner, L. (2005). Supply chain monitoring: a statistical approach. *Computer Aided Chemical Engineering*.

Melnyk, S. A., Thürer, M., Blome, C., Schoenherr, T., & Gold, S. (2024). (Re)-discovering simulation as a critical element of OM/SCM research: call for research. *International Journal of Operations and Production Management,*, 44(7), 1376–1389.

Mikhail, M., El-Beheiry, M., & Afia, N. (2019). Investigating resilient supply chain design determinants using Monte Carlo simulation. *8th international conference on industrial technology and management (ICITM). IEEE.*, (pp. 27-31).

Miljković, N. P. (2024). Results of EAHP's 2023 shortages survey. *European Journal of Hospital Pharmacy*.

Mooney, C. Z. (1997). *Monte carlo simulation .* Sage.

Motamedi, M., Dawson, J., Li, N., Down, D. G., Heddle, N. M., & . (2024). Demand forecasting for platelet usage: From univariate time series to multivariable models. *PLOS ONE*. doi:https://doi.org/10.1371/journal.pone.0297391

Mulopulos, S. (2023). *Supply Chain Resilience Policy: Theory, Practice, and Action.* Fulbright New Zealand.

Mustafee, N., & Fakhimi, M. (2024). Towards an Integrative Taxonomical Framework for Hybrid Simulation and Hybrid Modelling. . *In Hybrid Modeling and Simulation: Conceptualizations, Methods and Applications. Cham: Springer Nature Switzerland.*, (pp. 3-22).

Myrzashova, R., Alsamhi, S. H., Shvetsov, A. V., Hawbani, A., Wei, X., & . (2023). Blockchain Meets Federated Learning in Healthcare: A Systematic Review With Challenges and Opportunities,. *IEEE Internet of Things Journal*, vol. 10, no. 16, pp. 14418-14437,.

Nakamoto, S. (2008). Bitcoin: A Peer-to-Peer Electronic Cash System.

Nas, S., & Koyuncu, M. (2019). Emergency department capacity planning: a recurrent neural network and simulation approach. *Computational and mathematical methods in medicine*, 4359719.

National Academies of Sciences, E. a. (2017). Building a National Capability to Monitor and Assess Medical Countermeasure Use During a Public Health Emergency: Going Beyond the Last Mile: Proceedings of a Workshop (M. Boname, T. Wizemann, & J. Snair, Eds.). *National Academies Press*. doi:https://doi.org/10.17226/24912

National Academies of Sciences, Engineering, and Medicine. (2022). *Future Planning for the Public Health Emergency Preparedness Enterprise: Lessons Learned from the COVID-19 Pandemic: Proceedings of a Workshop.* National Academic Press.

Nguyen, D. C., Ding, M., & Pathirana, P. N. (2021). Blockchain and AI-based solutions to combat coronavirus (COVID-19)-like epidemics: A survey. *IEEE Access*.

Norwegian Institute of Public Health. (2024). *EU interoperability with HERA's IT platform (EU-HIP).* NIPH.

Norwood Bailey, F., & Peel, D. (2021). Supply chain mapping to prepare for future pandemics. *Applied Economics Perspectives and Policy*.

NVIDIA. (n.d.). *XGBoost*. Retrieved from NVIDIA: https://www.nvidia.com/en-us/glossary/xgboost/

OECD. (2022). *OECD Framework for the Classification of AI Systems .* Paris: OECD Publishing.

OECD. (2023). *Artificial intelligence*. Retrieved from https://www.oecd.org/digital/artificial-intelligence/

Olawade, B. W. (2023). Using artificial intelligence to improve public health: a narrative review. *Frontiers Public Health*.

Omar, I. A., Jayaraman, R., Debe, M. S., Salah, K., & Yaqoob, I. (2021). Automating Procurement Contracts in the Healthcare Supply Chain Using Blockchain Smart Contracts. *IEEE Access*.

Ordu, M., Demir, E., Tofallis, C., & Gunal, M. M. (2021). A novel healthcare resource allocation decision support tool: A forecasting-simulation-optimization approach. . *Journal of the operational research society*, 72(3), 485-500.

Ortiz-Barrios, M., Arias-Fonseca, S., Ishizaka, A., Barbati, M., Avendaño-Collante, B., & Navarro-Jiménez, E. (2023). Artificial intelligence and discrete-event simulation for capacity management of intensive care units during the Covid-19 pandemic: A case study. *J Bus Res.* doi: 10.1016/j.jbusres.2023.113806

Ortiz-Barrios, M., Ishizaka, A., Barbati., M., Arias-Fonseca, S., Khan, J., Gul, M., . . . Pérez-Aguilar, A. (2024). Integrating discrete-event simulation and artificial intelligence for shortening bed waiting times in hospita.

Page, M. J., McKenzie, J. E., Bossuyt, P. M., Boutron, I., Hoffmann, T. C., Mulrow, C. D., . . . McGuinnes. (2021). The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ, 372*(n71).

Pall, R., Gauthier, Y., Auer, S., Mowaswes, W., ., & . (2023). Predicting drug shortages using pharmacy data and machine learning. *Health Care Management Science*, 26(3), 395–411. doi:https://doi.org/10.1007/s10729-022-09627-y

Panzitta, M., Ponti, M., Bruno, G., Cois, G., D'Arpino, A., Minghetti, P., . . . Ricci, M. (2017). The strategic relevance of manufacturing technology: An overall quality concept to promote innovation preventing drug shortage. *Int J Pharm*, 516(1-2):144-157.

Parbat, D., & Chakraborty, M. (2020). A python based support vector regression model for prediction of COVID19 cases in India. . *Chaos, Solitons & Fractals,* , 138, 109942.

Peng, X., Sun, B., Ali, K., & Saenko, K. (2015). Learning deep object detectors from 3d models. *In Proceedings of the IEEE international conference on computer vision* , (pp. 1278-1286).

Perazzoli, S., de Santana Neto, J., & de Menezes, M. (2022). Systematic analysis of constellation-based techniques by using Natural Language Processing. *Technol. Forecast. Soc. Chang.*

Petropoulos, F. M. (2022). COVID-19: Forecasting confirmed cases and deaths with a simple time series model. *International Journal of Forecasting*. doi:https://doi.org/10.1016/j.ijforecast.2020.11.010

Phuong, J. M. (2019). The impacts of medication shortages on patient outcomes: A scoping review. *PLOS ONE*. doi: https://doi.org/10.1371/journal.pone.0215837

Piccolomini, E., & Zama, F. (2020). Monitoring Italian COVID-19 spread by a forced SEIRD model. *PloS one*, 15(8), e0237417.

Pundir, A. K., Jagannath, J. D., & Ganapathy, L. (2019). Improving supply chain visibility using IoT-internet of things. . *9th annual computing and communication workshop and conference (ccwc) IEEE.*, (pp. (pp. 0156-0162).).

Querec, T. D., Akondy, R. S., Lee, E. K., Cao, W., Nakaya, H. I., Teuwen, D., & Pulendran, B. (2009). Systems biology approach predicts immunogenicity of the yellow fever vaccine in humans. *Nature immunology*, 10(1), 116-125.

Raimundo, & Rosário. (2021). The Impact of Artificial Intelligence on Data System Security: A Literature Review. Sensors (Basel).

Rathipriya, R., Rahman, A. A., Dhamodharavadhani, S., Meero, A., Yoganandan, G., & . (2023). Demand forecasting model for time-series pharmaceutical data using shallow and deep neural network model. *Neural Comput Appl.*, 35(2):1945-1957. doi:10.1007/s00521-022-07889-9

RELEX. (2024). *Unified supply chain for pharmaceutical distributors and wholesalers*. Retrieved from https://www.relexsolutions.com/industries/wholesale-distribution/pharmaceutical/

Ren, L., Jia, Z., Laili, Y., & Huang, D. (2023). Deep Learning for Time-Series Prediction in IIoT: Progress, Challenges, and Prospects. *IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS*.

Resilinc. (2024). *Multi-tier mapping: AI + Supplier validated data*. Retrieved from https://resource.resilinc.com/rs/863-OTG-034/images/Resilinc%20Multi%20Tier%20Mapping%20Data%20Sheet%202024.pdf?_gl=1*ydpbff*_gcl_au*NTAxNTU1OC4xNzI1MjcwNzEx*_ga*OTU5MjY4Nzc5LjE3MjU0Mzk3ODM.*_ga_B0E7GYBYS9*MTcyOTc2NDgwMC4xMC4xLjE3Mjk3NjUyNjguNjAuMC4xMzUxN

Riahi, Y. S. (2021). Artificial intelligence applications in supply chain: A descriptive bibliometric analysis and future research directions. . *Expert Systems with Applications*, 173, 114702.

Román-Portabales A, L.-N. M.-A. (2021). Systematic Review of Electricity Demand Forecast Using ANN-Based Machine Learning Algorithms. *Sensors (Basel)*. doi:10.3390/s21134544

Rong, W., Wang, P., Han, Z., & Zhao, W. (2022). An epidemic spreading simulation and emergency management based on system dynamics: a case study of China's university community. . *Complexity, 2022(1), 9164404.*

Rudin, C. C. (2022). Interpretable machine learning: Fundamental principles and 10 grand challenges. *Statistic Surveys*, 16, 1-85.

Saberi, S., Kouhizadeh, M., Sarkis, J., Shen, L., ., & . (2019). Blockchain technology and its relationships to sustainable supply chain management. *International journal of production research*, 57(7), 2117-2135.

Saif, A., & Elhedhli, S. (2016). Cold supply chain design with environmental considerations: A simulation-optimization approach. *European Journal of Operational Research,.*, 251(1), 274-287.

Saisridhar, P., Thuerer, M., & Avittathur, B. (2024). Assessing supply chain responsiveness, resilience and robustness (Triple-R) by computer simulation: a systematic review of the literature. *International Journal of Production Research*, 62(4), 1458-1488.

Savachkin, A., & Uribe, A. (2012). Dynamic redistribution of mitigation resources during influenza pandemics. *Socio-Economic Planning Sciences*, 46(1), 33-45.

Schilling, M., Rickmann, L., Hutschenreuter, G., Spreckelsen, C., ., & . (2022). Reduction of Platelet Outdating and Shortage by Forecasting Demand With Statistical Learning and Deep Neural Networks: Modeling Study. *JMIR Medical Informatics, 10(2), e29978.* doi:https://doi.org/10.2196/29978

Schöpper, H., & Kersten, W. (2021). Using Natural Language Processing for Supply Chain Mapping: A Systematic Review of Current Approaches. *5th International Conference on Computational Linguistics and Intelligent Systems.* Kharkiv, Ukraine: CEUR Workshop Proceedings.

Senna, P., Reis, A., Marujo, L. G., Ferro de Guimarães, J. C., Severo, E. A., & dos Santos, A. C. (2023). The influence of supply chain risk management in healthcare supply chains performance. *Production Planning & Control*, 35(12), 1368-1383.

Serdarasan, S., & Tanyas, M. (2012). Dealing with Complexity in the Supply Chain: The Effect of Supply Chain Management Initiatives. *SSRN*.

Shafer, S. M., & Smunt, T. (2004). Empirical simulation studies in operations management: context, trends, and research opportunities. *Journal of Operations Management*, 22 (345-354).

Shafiekhani, S., Namdar, P., & Rafiei, S. (2022). A COVID-19 forecasting system for hospital needs using ANFIS and LSTM models: A graphical user interface unit. *DIGITAL HEALTH, 8, 205520762210850.* doi:https://doi.org/10.1177/20552076221085057

Shirazi, H. K. (2021). A stochastic bi-objective simulation–optimization model for plasma supply chain in case of COVID-19 outbreak. . *Applied Soft Computing*, 112, 107725.

Singh, A. A., & Shiny, I. D. (2024). Inventory Management and Sales Prediction using XGBoost. *Proceedings of the 3rd International Conference on Applied Artificial Intelligence and Computing (ICAAIC-2024).* IEEE.

Solamain, B., & Cohen, I. (2024). *Research Handbook on Health, AI and the Law.*

Soyiri, I. N., & Reidpath, D. D. (2013). An overview of health forecasting. *Environmental Health and Preventive Medicine*. doi:https://doi.org/10.1007/s12199-012-0294-6

Stanley, S. (2019). Healthcare supply chain cybersecurity: A growing concern. *Becker's Hospital Review*.

Stergiou, K. D., Minopoulos, G. M., Memos, V. A., Stergiou, C. L., Koidou, M. P., & Psannis, K. E. (2022). A Machine Learning-Based Model for Epidemic Forecasting and Faster Drug Discovery. *Applied Sciences, 12(21), 10766.* doi:https://doi.org/10.3390/app122110766

Sterman, J. (. (2002). System Dynamics: systems thinking and modeling for a complex world.

Subramanian. (2021). Effective Demand Forecasting in Health Supply Chains: Emerging Trend, Enablers, and Blockers. *Logistics*. Retrieved from https://doi.org/10.3390/logistics5010012

Tang. (2005). Perspectives in supply chain risk management: a review. *Perspectives in Supply Chain Risk Management: A Review*.

Tang, Z., & Pan, Y. (2014). Agent-based supply chain management modeling and simulation. *IEEE International Conference on Information and Automation (ICIA)*, (pp. 1255-1259).

Tiacci, L., & Rossi, A. (2024). A discrete event simulator to implement deep reinforcement learning for the dynamic flexible job shop scheduling problem. . *Simulation Modelling Practice and Theory,* , 134, 102948.

Tolk, A. (. (2022). imulation-based optimization: Implications of complex adaptive systems and deep uncertainty. *information* , 13(10), 469.

Tordecilla, R. D., Juan, A. A., Montoya-Torres, J. R., Quintero-Araujo, C. L., & Panadero, J. (2021). Simulation-optimization methods for designing and assessing resilient supply chain networks under uncertainty scenarios: A review. . *Simulation modelling practice and theory*, 106, 102166.

Tseng, Y. J., & Shih, Y. L. (2020). Developing epidemic forecasting models to assist disease surveillance for influenza with electronic health records. . *International Journal of Computers and Applications*, 42(6), 616-621.

Valluri, A., North, M. J., & Macal, C. M. (2009). Reinforcement learning in supply chains. *International Journal of Neural Systems*.

Vázquez-Serrano, J. I., Peimbert-García, R. E., & Cárdenas-Barrón, L. E. (2021). Discrete-event simulation modeling in healthcare: a comprehensive review. *International journal of environmental research and public health*, 18(22), 12262.

Waller, M. A., & Fawcett, S. E. (2013). Data science, predictive analytics, and big data: a revolution that will transform supply chain design and management. . *Journal of Business logistics,* , 34(2), 77-84.

Wamba, S. F., Queiroz, M. M., & Trinchera, L. (2020). Dynamics between blockchain adoption determinants and supply chain performance: An empirical investigation. *International Journal of Production Economics*, 229, 107791.

Wang, Q., & Ingham, N. (2008). A DISCRETE EVENT MODELLING APPROACH FOR SUPPLY CHAIN SIMULATION. *International Journal of Simulation Modelling (IJSIMM),*, 7(3).

Wei, H., Li, N., Wu, J., Zhou, J., Drew, S., ., & . (2023). Federated Blood Supply Chain Demand Forecasting: A Case Study.

WHO. (2024). Surveillance in emergencies.

Williams, B., Roh, J., Tokar, T., Swink, M., ., ., & . (2013). Leveraging Supply Chain Visibility for Responsiveness: The Moderating Role of Internal Integration. *Journal of Operations*.

Wollek, S., Singaravelu, S. S., & National Academies of Sciences, E. a. (2022). Coordinating Surveillance and Medical Countermeasure Response. *Future Planning for the Public Health Emergency Preparedness Enterprise*.

Wong, W. P. (2023). Digitalization enhancement in the pharmaceutical supply network using a supply chain risk management approach. *Scientific Reports*.

Wright, J. (2023). Healthcare cybersecurity and cybercrime supply chain risk management. *Health Economics and Management Review*.

Yan, B., Tang, X., Liu, B., Wang, J., Zhou, Y., Zheng, G., . . . Tu, W. (2020). An Improved Method for the Fitting and Prediction of the Number of COVID-19 Confirmed Cases Based on LSTM.

Younis, H., & Wuni, I. Y. (2023). Application of industry 4.0 enablers in supply chain management: Scientometric analysis and critical review. *Helyion*.

Zandavi, S. M., Rashidi, T. H., & Vafaee, F. (2020). Forecasting the Spread of Covid-19 Under Control Scenarios Using LSTM and Dynamic Behavioral Models.

Zhang, W., & Li, X. (2024). A data-driven combined prediction method for the demand for intensive care unit healthcare resources in public health emergencies. *BMC Health Services Research, 24(1), 477.* doi:https://doi.org/10.1186/s12913-024-10955-8

Zhao, A. P., Li, S., Cao, Z., Hu, P. J.-H., Wang, J., Xiang, Y., . . . Lu, X. (2024). AI for science: Predicting infectious diseases. *Journal of Safety Science and Resilience*. doi:https://doi.org/10.1016/j.jnlssr.2024.02.002

Zheng, G., Kong, L., & Brintrup, A. (2023). Federated machine learning for privacy preserving, collective supply chain risk prediction. *nternational Journal of Production Research*. doi:https://doi.org/10.1080/00207543.2022.2164628

Zheng, Y., Liu, L., Shi, V., Huang, W., & Liao, J. (2022). A Resilience Analysis of a Medical Mask Supply Chain during the COVID-19 Pandemic: A Simulation Modeling Approach. *International Journal of Environmental Research and Public Health*, 19 (13).

Zhong, J., Hu, X., Alghamdi, O., Elattar, S., Al Sulaie, S., & . (2023). XGBoost with Q-learning for complex data processing in business logistics management. *Information Processing and Management*.

Zivkovic M, B. N.-T. (2021). COVID-19 cases prediction by using hybrid machine learning and beetle antennae search approach. *Sustain Cities Soc.* doi:doi: 10.1016/j.scs.2020.102669.

Annex 1: Methodology

This section outlines the methodology followed in this report for the review of the literature on the use of AI-supported methodologies and simulation methods for demand forecasting and supply monitoring. Moreover, this section also describes the methodology used to gather additional inputs on the topic from key stakeholders, covering from the selection of stakeholders up to the analysis of the inputs gathered.

Scientific literature review

The focus of the systematic literature review was to analyse the literature reporting on how further innovation in software aspects, especially AI-based tools, can improve the monitoring of the MCM supply chain in the EU, increase the accuracy of forecasting and guide evidence-based decision-making. An extensive literature search was conducted following the PRISMA guidelines and using the academic database PubMed. The box below presents the search strings used for this systematic literature review. The search strings were developed based on the key research questions of the study and following several discussions among the study team. The strings were tested to make sure valid and reliable outcomes were obtained. Studies published within the last 10 years (2015-2024) were considered, and all eligible studies had to be open source, the full text available, and be written in English.

Box 9. Research questions and query strings.

Titles and abstracts of articles published between 2015 and 2024 were searched for the following terms in PubMed:

Research question #1: What AI-driven methodology and simulation methods can be used for demand forecasting of MCM in the EU?

- Search string: (“AI” OR “artificial intelligence*” OR “ML” OR “Machine Learning” OR “Deep Learning” OR “LLM” OR “large language model*” OR “predictive analytic*” OR “simulation method*” OR “simulation model*” OR “predictive method*”) AND (“demand forecast*” OR “demand prediction*” OR “demand surge*”)
- Results: 86
- Screening questions:
  - Are the results relevant for HERA to forecast demand of critical MCMs, estimate disease outbreak trajectories, and to study the impact of disruptive events?
  - Are the results useful for planning future public health interventions?

Research question #2: What AI-driven methodology and simulation methods can be used for supply chain monitoring of MCM in the EU?

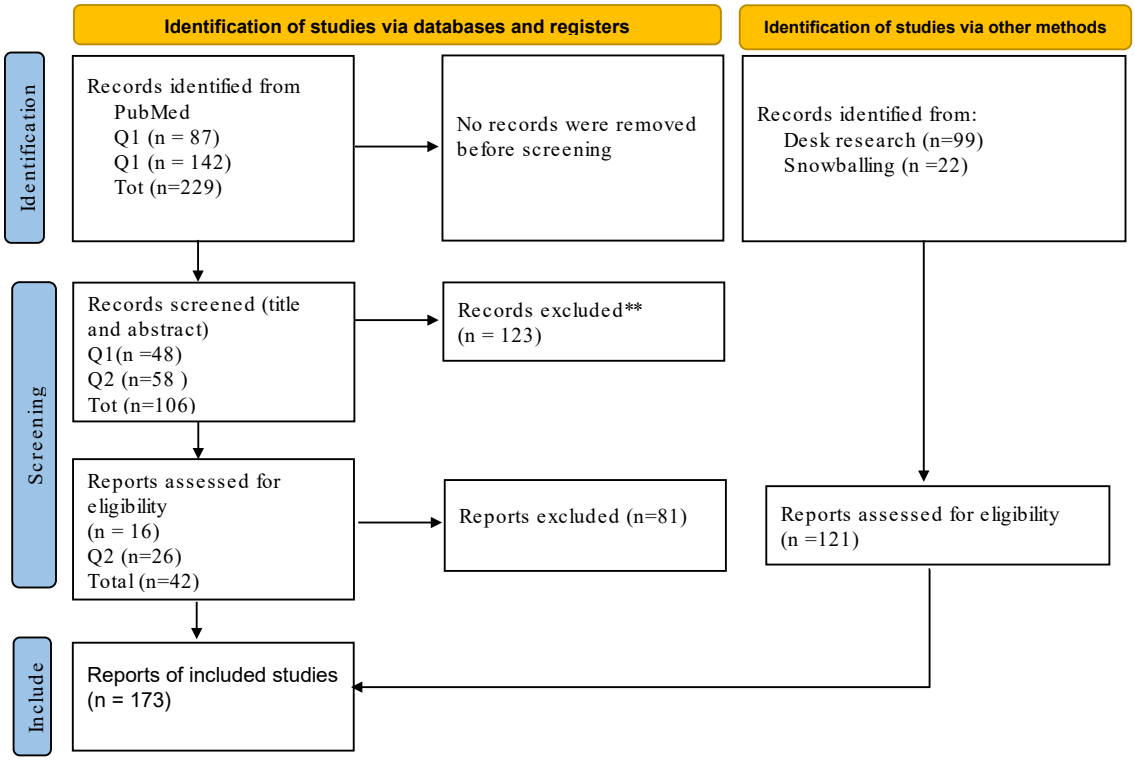
- Search string: (“AI” OR “artificial intelligence*” OR “ML” OR “Machine Learning” OR “Deep Learning” OR “LLM” OR “large language model*” OR “predictive analytic*” OR “simulation method*” OR “simulation model*” OR “predictive method*”) AND (“supply chain*” OR “supply network*”) AND (“monitor*” OR “forecast*” OR “track*” OR “surveillance*” OR “optimization” OR “resilience”)
- Results: 142
- Screening questions:
  - Are the results relevant for HERA to monitoring MCM, map the supply chain of critical MCMs and identify actual/potential bottlenecks?
  - Are the results relevant for HERA to prevent or mitigate critical medicine shortages in the EU?

Search hits were downloaded, resulting in one file prepared for the screening of the search hits (n=228). Citations were screened by title and abstract to see if they explored 1) AI-supported methodologies used for demand forecasting, 2) AI-supported methodologies for supply monitoring, or 3) simulation methods for comparing different scenarios, optimising supply chains and driving public health intervention. Based on these criteria, 106 publications were selected for further full-text review.

A quick full text review was conducted before potentially eligible reports were further assessed. Studies were excluded if they were: 1) not peer-reviewed, 2) outside of the research scope, 3) non-relevant to the research questions. A total of 42 studies were selected for data extraction. Reasons for exclusion for the other studies were recorded. The study team manually added 131 references identified outside of the database search, which emerged from checking the bibliography of relevant publications (snowballing approach) or through desk research. A single reviewer extracted data from each included report. A second reviewer checked extractions and appraisals for accuracy and completeness. A narrative synthesis was performed.

The total number of records after each screening round was documented using the PRISMA flow diagram template (see Figure below).

Figure 3. PRISMA flow diagram summarising the records of the literature review



***Source**: authors' elaboration based on Page et al. (2021)*

Consultation activities

The consultation activities were carried out for two months and aimed to achieve three primary objectives:

- **Validate and bridge research** gaps: To corroborate and fill the research gaps identified in existing literature by integrating insights from experts across academia, public health laboratories, industry, and government.
- **Validate AI methodologies and simulation methods:** To assess the validity and feasibility of AI supported methodologies identified in the literature for demand forecasting and supply chain monitoring as well as simulation methods for supply chain optimisation.

To meet these objectives, the consultations included in-depth one-on-one interviews (60 minutes long). Prior to the consultations, three distinct interview guidelines were developed, each tailored to a specific stakeholder category (academia, industry, and government). These guidelines were reviewed and validated by HERA, ensuring comprehensive coverage of key topics while allowing flexibility for individual expertise.

Each guideline included the following sections:

- Introduction and consent for recording and minute-taking.
- Background questions on the interviewee's profile and experience in the field.
- Questions regarding the use of AI methodologies and simulation methods for demand forecasting and supply chain monitoring, considering applicability to MCM supply chain.
- Questions exploring implementation challenges (with a focus on public authorities' perspective), research bottlenecks and priorities.
- Open-ended questions for additional insights, uncovered areas, or recommendations for further consultation.

A comprehensive stakeholder mapping exercise was conducted at the outset of the study (and lasted six months), identifying key experts across the three stakeholder categories. This mapping was validated by HERA and informed the subsequent engagement activities, which included targeted emails to request participation in interviews.

The consultation strategy envisioned a target number of participants for each category:

- Academia (N = 5)
- Industry (N = 5)
- Government (N =5)

During the two-month consultation period, many stakeholders were included in the engagement activities, after a thorough review of their expertise to achieve the target. The table below summarises the final number of stakeholders interviewed for each category.

Table 7. *Consultation activities.*

	Interview (N)
Academia	3
Industry	5
Government	7

Ultimately, **3** stakeholders from academia, **5** from industry, and **7** from government were consulted, providing invaluable feedback and input.

Interviews followed a semi-structured format, allowing for adaptation based on the interviewee's specific expertise. Preliminary materials, including interview guidelines and tables of AI methodologies and simulation methods identified in the literature, were shared with interviewees in advance. Following each interview, detailed minutes were shared for review and approval before incorporating the feedback into the study.

The valuable insights gathered from the consultation activities are presented at the end of each relevant results section (Box 2-5). These findings collectively contributed to the final discussion in section 3.

GETTING IN TOUCH WITH THE EU

In person

All over the European Union there are hundreds of Europe Direct information centres. You can find the address of the centre nearest you at: https://europa.eu/european-union/contact_en

On the phone or by email

Europe Direct is a service that answers your questions about the European Union. You can contact this service:

– by freephone: 00 800 6 7 8 9 10 11 (certain operators may charge for these calls),

– at the following standard number: +32 22999696, or

– by email via: https://europa.eu/european-union/contact_en

FINDING INFORMATION ABOUT THE EU

Online

Information about the European Union in all the official languages of the EU is available on the Europa website at: https://europa.eu/european-union/index_en

EU publications

You can download or order free and priced EU publications from: https://op.europa.eu/en/publications. Multiple copies of free publications may be obtained by contacting Europe Direct or your local information centre (see https://europa.eu/european-union/contact_en).

EU law and related documents

For access to legal information from the EU, including all EU law since 1952 in all the official language versions, go to EUR-Lex at: http://eur-lex.europa.eu

Open data from the EU

The EU Open Data Portal (http://data.europa.eu/euodp/en) provides access to datasets from the EU. Data can be downloaded and reused for free, for both commercial and non-commercial purposes.



intellera consulting

Predi[t
*By